



中华人民共和国国家标准

GB/T 39852—2021

物品电子编码 标签数据转换

Electronic product code—Tag data translation

2021-03-09 发布

2021-10-01 实施

国家市场监督管理总局 发布
国家标准化管理委员会

目 次

前言 III

1 范围 1

2 规范性引用文件 1

3 术语和定义 1

4 缩略语和符号 1

5 EPC 系统与标签数据 2

 5.1 EPC 系统 2

 5.2 标签数据 3

6 标签数据转换 6

 6.1 标签数据转换概述 6

 6.2 转换过程 7

7 TDT 标记文件 9

 7.1 TDT 标记文件概述 9

 7.2 TDT 标记文件的定义方式和附加要求 10

 7.3 TDT 标记文件的元素和属性 21

8 标签数据转换算法 27

 8.1 概述 27

 8.2 功能算法和应用程序接口 (API) 27

附录 A(规范性附录) TDT 标记文件 33

 A.1 SGTIN-64 33

 A.2 SGTIN-96 45

 A.3 SGTIN-198 56

 A.4 SSCC-64 67

 A.5 SSCC-96 77

 A.6 SGLN-64 86

 A.7 SGLN-96 98

 A.8 SGLN-195 109

 A.9 GDTI-96 119

 A.10 GDTI-113 130

 A.11 GRAI-64 141

 A.12 GRAI-96 152

 A.13 GRAI-170 163

 A.14 GIAI-64 175

GB/T 39852—2021

A.15	GIAI-96	183
A.16	GIAI-202	192
A.17	GSRN-96	200
A.18	GID-96	208
参考文献		211

前 言

本标准按照 GB/T 1.1—2009 给出的规则起草。

本标准由全国物流信息管理标准化技术委员会(SAC/TC 267)提出并归口。

本标准起草单位:中国物品编码中心、北京仁聚智汇科技发展有限公司、深圳市标准技术研究院。

本标准主要起草人:李志敏、罗艳、施煜、李媛红、张彪、鄢若韞、易晓珊、胡敏、董腾、王隆。



物品电子编码 标签数据转换

1 范围

本标准规定了物品电子编码系统与标签数据、标签数据转换、标签数据转换标记文件和标签数据转换算法的方法和要求。

本标准适用于物品电子编码系统信息交换与传输过程中的标签数据转换与处理。

2 规范性引用文件

下列文件对于本文件的应用是必不可少的。凡是注日期的引用文件,仅注日期的版本适用于本文件。凡是不注日期的引用文件,其最新版本(包括所有的修改单)适用于本文件。

GB/T 12905 条码术语

ISO/IEC 15962 信息技术 项目管理用射频识别(RFID) 数据协议:数据编码规则和逻辑存储功能[Information technology—Radio frequency identification (RFID) for item management—Data protocol: data encoding rules and logical memory functions]

ISO/IEC 19762 信息技术 自动识别与数据采集 统一词汇[Information technology—Automatic identification and data capture (AIDC) techniques—Harmonized vocabulary]

GS1 EPC 标签数据标准(GS1 EPC Tag Data Standard)

GS1 通用规范(GS1 General specifications)



3 术语和定义

GB/T 12905、ISO/IEC 19762 界定的术语和定义适用于本文件。

4 缩略语和符号

下列缩略语和符号适用于本文件。

ABNF:扩展巴科斯范式(Augmented Backus-Naur Form)

AI:应用标识符(Application Identifier)

ALE:应用层事件(Application Level Event)

DNS:域名解析系统(Domain Name System)

EPC:物品电子编码(Electronic Product Code)

EPCIS:物品电子编码信息服务(Electronic Product Code Information Services)

GDD:全球数据字典(Global Data Dictionary)

GDSN:全球数据同步网络(Global Data Synchronization Network)

GPC:全球产品分类(Global Product Classification)

GS1:国际物品编码协会(Global Standard)

LLRP:底层读写器协议(Low Level Reader Protocol)

NAPTR:名称授权指针(Naming Authority Pointer)

ONS:对象名称解析服务(Object Naming Service)

- RFID:射频识别(Radio Frequency Identification)
- TDS:标签数据标准(Tag Data Standard)
- TDT:标签数据转换(Tag Data Translation)
- URI:统一资源标识符(Uniform Resource Identifier)
- URN:统一资源名称(Uniform Resource Name)
- XML:可扩展置标语言(Extensible Markup Language)
- ..._{hex}:十六进制表示
- (...)_{bin}:二进制表示

5 EPC 系统与标签数据

5.1 EPC 系统

5.1.1 EPC 系统组成

EPC 是 GS1 物品编码的电子代码形式。EPC 系统是 GS1 系统的组成部分,是在信息网络基础上,以 EPC 或 GS1 物品编码的其他形式(如条码、OCR 字符)代码,作为物品标识以及信息传递或检索关键字,利用 RFID 技术和其他自动识别与数据采集技术,实现物品信息的自动识别与采集、检索、查询和传递,从而达到物与信息网络互联、物与物互联的标准化服务网络。

EPC 系统由标识与数据采集层、EPC 信息系统层、应用层、EPC 网络服务层四部分组成,其架构如图 1 所示。

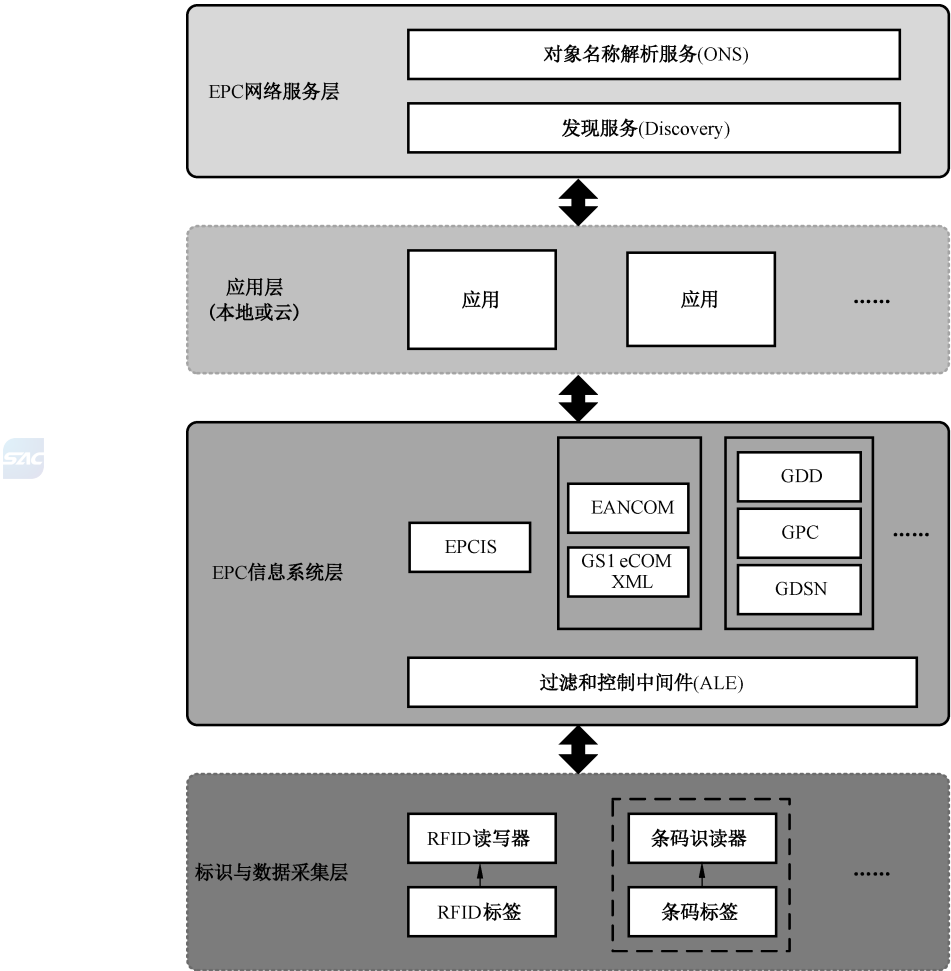


图 1 EPC 系统架构图

5.1.2 标识与数据采集层

标识与数据采集层实现物品标识信息的自动采集,由标识和读写器构成。基于 RFID 技术的标识与数据采集层由 RFID 标签和识读器组成,其他技术不在本标准中规定。

RFID 标签是装载物品电子编码(EPC)的自动识别载体,通常附着于标识物品上,是物品全生命周期的唯一标识,由天线和芯片组成。EPC 在 RFID 标签内以二进制形式存储。

RFID 读写器是读取或者读写 RFID 标签存储信息的设备。读写器可内嵌底层读写器协议(LLRP),控制读写器读取原始标签数据,通过中间件与信息系统交换信息。

5.1.3 EPC 信息系统层

EPC 信息系统层为应用层提供数据过滤以及数据共享服务,包括实现过滤功能的应用层事件(ALE)接口,实现数据共享服务的数据同步、GS1 EANCOM 和 GS1 eCOM XML 标准,实现物品事件信息服务的 EPCIS 标准等。

注 1: GS1 EANCOM 是由 GS1 开发的,应用于商业流通领域的电子数据交换规范。

注 2: GS1 eCOM XML 是由 GS1 开发的基于 XML 语言的电子数据交换规范。

5.1.4 应用层

应用层是指用户根据需求采用 EPC 或 GS1 物品编码的其他形式代码为关键字搭建的各种应用。

5.1.5 EPC 网络服务层

EPC 网络服务层提供在互联网上寻找具体应用、具体对象的网络服务,包括对象名称解析服务(ONS)和发现服务(Discovery)等,其中,对象名称解析服务(ONS)是通过唯一的物品电子代码对实体对象进行识别与发现的网络服务模式。

5.2 标签数据

5.2.1 标签数据级别

标签数据是指 EPC 系统中,从 RFID 标签到对象名称解析服务(ONS)的各种格式的数据,包括七种格式:

- 二进制格式(BINARY);
- 标签编码 URI 格式(TAG_ENCODING);
- 纯标识 URI 格式(PURE_IDENTITY);
- GS1 文本数据字段格式(LEGACY);
- GS1 应用标识符字符串格式(LEGACY_AI);
- GS1 数据传输字符串格式(ELEMENT_STRING);
- ONS 域名格式(ONS_HOSTNAME)。

其中,二进制格式、标签编码 URI 格式和纯标识 URI 格式是 RFID 标签编码的三种格式,GS1 文本字段、GS1 应用标识符格式和 GS1 数据传输字符串格式格式是在 EPC 系统和用户应用系统中存储、传输 GS1 数据的格式,ONS 域名格式是标签数据在对象名称解析服务(ONS)中发起查询和检索使用的格式。

图 2 给出了 EPC 系统中的七种标签数据级别,以及编码过程和解码过程的方向。标签数据的格式自上而下级别降低,RFID 标签的二进制存储格式为最低级别。标签数据按照级别从高级别格式向低级别格式进行标签数据转换的过程称之为标签数据转换的编码过程(向着二进制格式的方向);反之,将标签数据按照级别从低级别格式向高级别格式进行标签数据转换的过程称之为标签数据转换的解码过程(远离二进制格式的方向)。



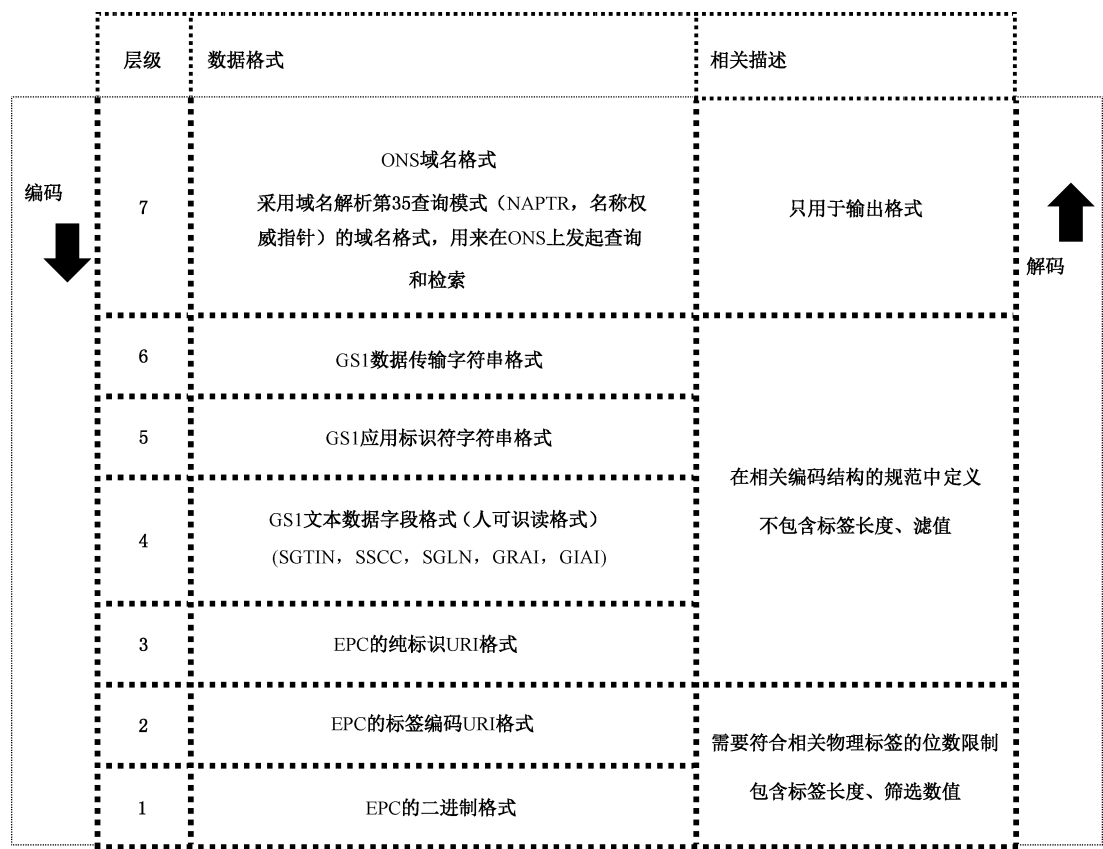


图 2 标签数据的格式层级

5.2.2 二进制格式



二进制格式是 EPC 存储在 RFID 标签的 EPC/UII 存储区中的格式，应在 EPC 系统的标识和数据采集层使用，由下列部分组成：

- a) 标头(header)，表示具体的编码结构和编码结构的位长，一般为 8 位二进制（也有特例，如 SGTIN-64 是 2 位）；
- b) 滤值，一般为 3 位二进制，用来表示不同的包装层级或者在应用中快速筛选和过滤读取的 RFID 标签；
- c) 其他字段，如分区、厂商识别代码、商品项目代码和唯一序列号等。

示例：

GTIN 为“06954606200019”，序列号为“20181109001”，采用 SGTIN-96 编码结构时，其二进制格式为：

00110000 001 100 100001001010011000001001110 000000000000000001
00010010110010111000110100100100001001

注：滤值为 1，GS1 厂商识别代码长度为 8。

其中：

标头(header)： 00110000
滤值： 001
分区数值： 100
GS1 厂商识别代码：100001001010011000001001110
商品项目代码： 000000000000000001
序列号： 00010010110010111000110100100100001001

5.2.3 标签编码 URI 格式

标签编码 URI 格式是 URN 字头与数据段表示的标签编码。标签编码 URI 格式与存储在物理标签中的 EPC 的二进制格式进行一一对应,包含编码结构、位长信息(对于固定长度的 EPC 以及滤值等)。标签编码 URI 格式示例如下。

示例:

5.2.2 示例中数据的标签编码 URI 格式为:

```
urn:epc:tag:sgtin-96:1.69546062.00001.20181109001
```

5.2.4 纯标识 URI 格式

纯标识 URI 格式是 URN 字头与数据段表示的 EPC 标识。在 EPC 信息服务中存储和传输 EPC 代码时,默认采用纯标识 URI 格式。纯标识 URI 格式可避免应用软件受到诸如标签位长、滤值的影响,使采用不同位长标签进行编码的同一个标识对象得到相同的、唯一的纯标识 URI 格式的 EPC 代码数据,纯标识 URI 格式示例如下。

示例:

5.2.2 示例中数据的纯标识 URI 格式为:

```
urn:epc:id:sgtin:69546062.00001.20181109001
```

5.2.5 GS1 文本数据字段格式

GS1 文本数据字段格式是 EPC 系统应用层使用的一种表示 GS1 数据的格式,对于不符合 GS1 规范的用户应用此格式。GS1 文本数据字段格式构成如下:

- 每个数据段采用“名称=值”的形式表示;
- 前后两个数据段之间用“;”(英文分号)分割;
- AI 用表示其含义的英文名称替换。

示例:

5.2.2 示例中数据的 GS1 文本数据字段格式为:

```
gtin=06954606200019;serial=20181109001
```

5.2.6 GS1 应用标识符字符串格式

GS1 应用标识符字符串是由一个或多个应用标识符及其标识的数据段连接在一起的可供人工识读的字符串格式,其中,应用标识符(AI)需要用括弧(英文括弧)包裹,GS1 应用标识符的规则见 GS1 通用规范。

示例:

5.2.2 示例中数据的 GS1 应用标识符字符串格式为:

```
(01)06954606200019(21)20181109001
```

5.2.7 GS1 数据传输字符串格式

GS1 数据传输字符串格式基于 GS1 应用标识符,是 GS1 系统中传输、通信的标准数据格式。GS1 数据传输字符串格式在 GS1 系统的各个应用、服务和系统之间传递 GS1 数据时使用,也在 EPC 系统应用层使用。GS1 数据传输字符串格式由若干段 GS1 应用标识符与数据首尾相接所组成,并且:

- 每个数据段由 GS1 应用标识符起始(注意:GS1 应用标识符不加括弧),后面紧跟该应用标识符指示的数据;
- 对于前后两段数据,前段 GS1 应用标识符指示的数据为定长时不加<FNC1>字符;当前段 GS1 应用标识符指示的数据是变长时,在前段末尾与后段前须添加<FNC1>字符;若变长的数据段处于最后位置时,字符串末尾不加<FNC1>字符。

注：〈FNC1〉字符的 ASCII 值为 29，(1D)_{hex}。

示例：

5.2.2 示例中数据的 GS1 数据传输字符串格式为：

01069546062000192120181109001

其中：

GTIN(AI 01): 01 06954606200019

序列号(AI 21):21 20181109001

5.2.8 ONS 域名格式

ONS 域名格式是一个符合 DNS 域名解析第 35 查询模式的格式。该格式的标签数据用来在 ONS 中发起查询和检索时使用。该格式只作为输出格式。

示例：

5.2.2 示例中数据的 ONS 域名格式为：

00001.69546062.sgtin.id.onsepc.com

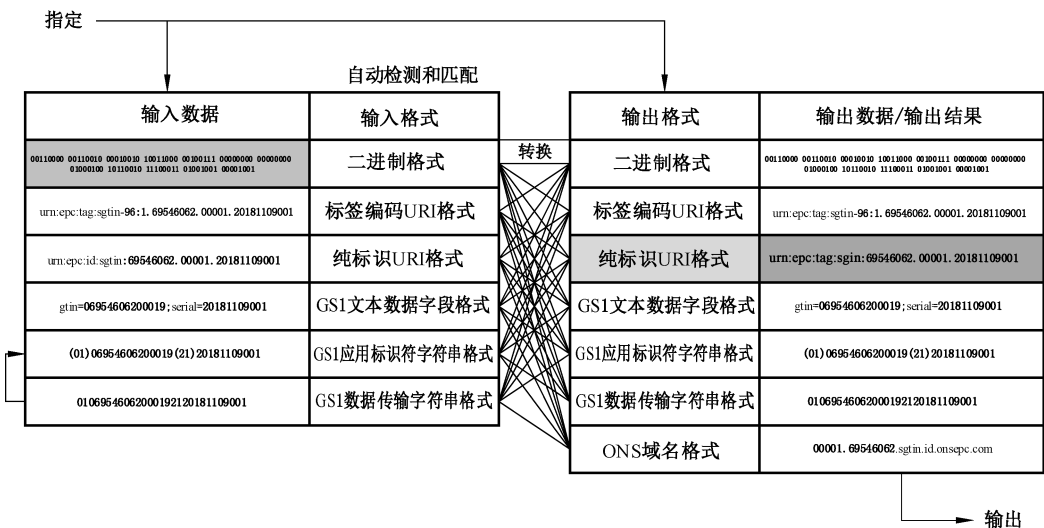
注 1：DNS 域名解析第 35 查询模式，是指名称权限指针(NAPTR)记录类型，NAPTR 记录是 Internet 的域名系统中的一种资源记录。NAPTR 记录与服务记录(SRV)的组合允许链接多个记录以形成复杂的重写规则，其产生新的域标签或统一的资源标识符(URI)。

注 2：由于对象名称解析服务(ONS)存在不同的版本，有的版本支持序列号查询，有的版本不支持，所以从其他格式转换为 ONS 域名格式依照采用的对象名称解析服务(ONS)的版本可能会舍弃部分信息(如序列号部分)。

6 标签数据转换

6.1 标签数据转换概述

EPC 的编码结构包括 SGTIN、SSCC、SGLN、GRAI、GIAI、GSRN、GDTI 和 GID 等，每一种编码结构都具有图 2 所示的七种格式。标签数据转换是指 EPC 编码在不同格式间的相互转换。标签数据转换是指同一个编码结构的不同格式之间的相互转换，不能跨编码结构进行。转换可以跨层级进行。标签数据转换过程的输入与输出如图 3 所示。



注：“指定”表示输入数据是给定的，输出格式是由用户或者应用确定。

图 3 标签数据转换过程的输入与输出

不同格式的编码转换时数据段并非完全双向对应。例如，图 2 中，格式的第 1 级(二进制格式)和第 2 级(标签编码 URI 格式)之间存在一一对应关系，而第 3 级到第 6 级格式无编码位长和滤值信息，是通过将纯标识 URI 格式添加、转换，或组合必要的辅助文本信息构成。第 7 级格式用于对象名称解析服

务发起查询和检索,该格式不要求和 EPC 编码的任一格式完全对应,因此只作为输出格式,不作为输入格式参与编码过程。

6.2 转换过程

6.2.1 转换环节

标签数据转换过程是自动识别输入的 EPC 编码,并将该编码格式转为需要的输出格式。转换过程分为输入、转换、输出三个环节,如图 4 所示。

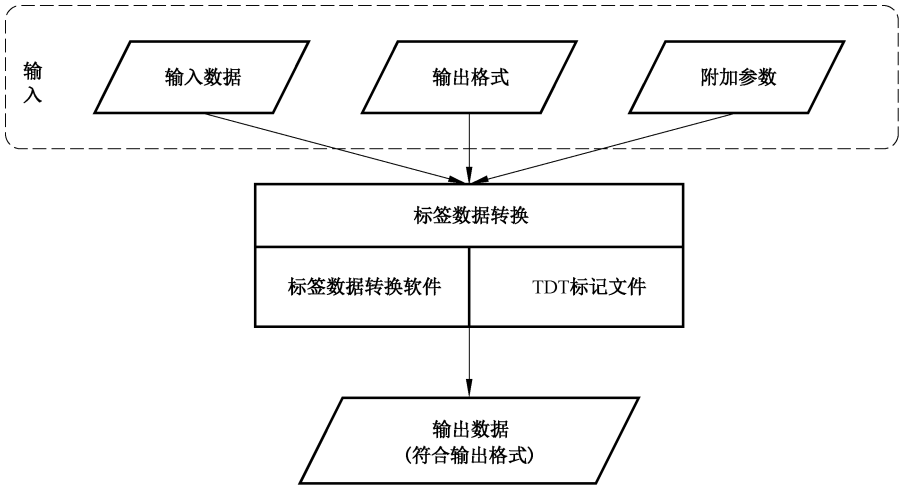


图 4 标签数据转换过程

6.2.2 输入

6.2.2.1 概述

输入环节包括已知的输入数据、目标输出格式要求以及其他附加参数。

6.2.2.2 输入数据

输入数据是由用户输入的或是由外部应用接口输入的,其采用的格式定义为输入格式。输入格式是 EPC 其中格式中除 ONS 格式的任何一种格式。标签数据转换过程应能通过输入数据,自动检测、判断以及匹配输入格式。

6.2.2.3 输出格式

输出格式是由用户或者应用确定。

操作过程中,有些格式较其他格式包含的信息多。如,标签二进制格式和标签编码 URI 格式除包含 EPC 代码数据,还包含 EPC 代码存储到 RFID 标签中所需的位长和滤值。当这两个格式作为输入格式向其他格式转换时会舍弃这些信息;当 ONS 域名作为输出格式时,任何格式向 ONS 域名格式转换会舍弃序列号信息。且 ONS 域名只能作为输出格式。

6.2.2.4 附加参数

转换过程需要输入用于确定输入值含义及转换输出格式的参数,比如,GS1 文本数据字段格式包含了 EPC 代码的所有基本信息(厂商、商品项目代码和序列号等),但是该格式并没有明确指明厂商识别代码和类别代码之间的边界,所以在实际转换中就需要厂商识别代码的长度来作为附加参数参与到整个转换过程中来。这种参数称之为附加参数。

编码方向和解码方向标签数据转换过程及附加参数的参与情况分别见图 5a)和图 5b)。从图 5 可



以看出,编码过程需要附加参数;解码过程除了某些情况之外(比如 64 位编码)一般不需要附加参数。

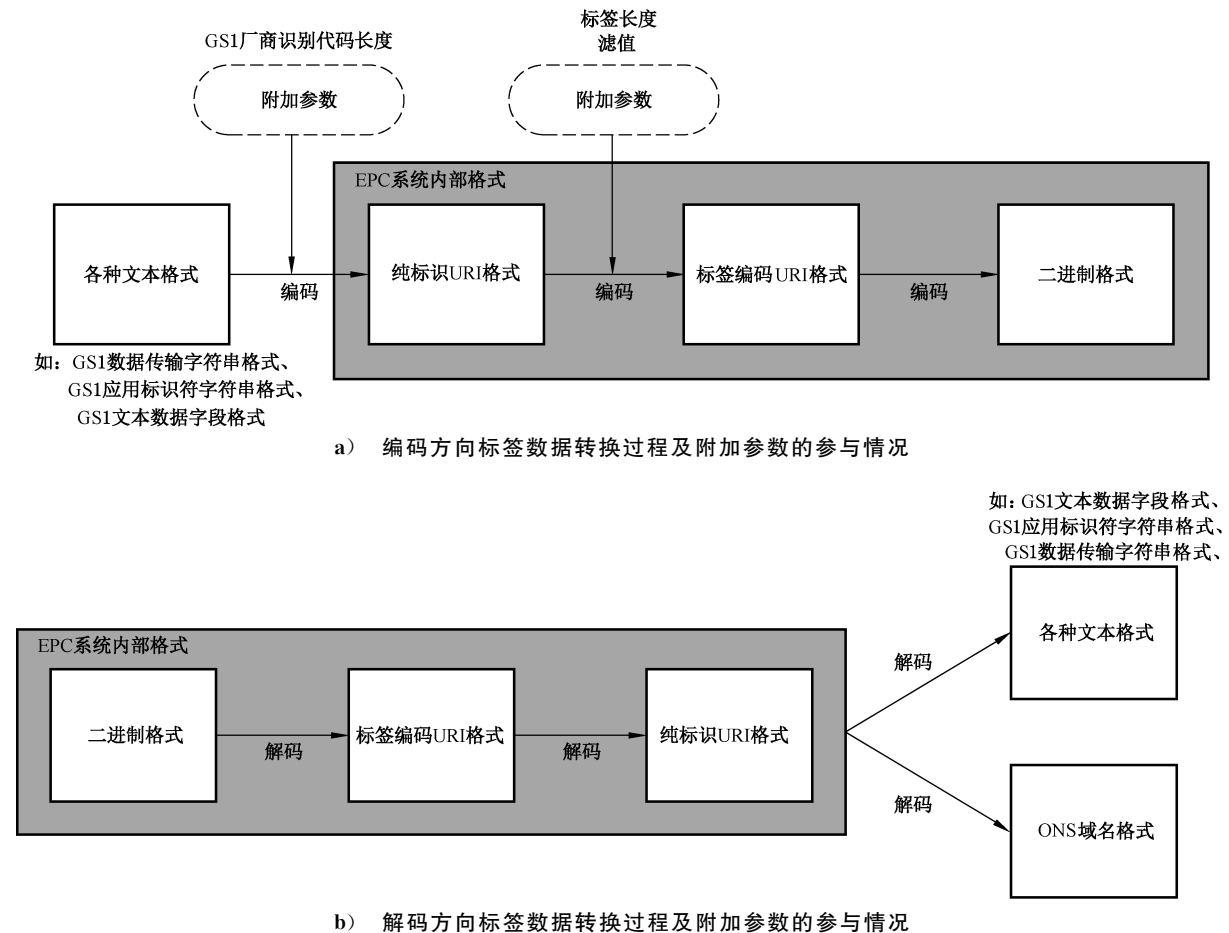


图 5 标签数据转换的编码和解码过程中附加参数参与情况

附加参数可以分为两类:

- 解析参数:解释输入数据含义的参数,比如参数 `gs1companyprefixlength`,就是指示 GS1 文本数据字段数据或者用 GS1 应用标识符分割的两种 GS1 数据字符串中的厂商识别代码长度的参数,用于提取厂商识别代码以供转换使用;
- 格式参数:补充输出结果中需要而输入数据中没有的,且与格式有关的说明参数,比如滤值 `filter` 和标签长度 `tagLength`。

附加参数在 TDT 标记文件中定义,TDT 文件要求见第 7 章。

6.2.3 转换

转换包含 TDT 标记文件和标签数据转换算法两个部分,其中:

- a) TDT 标记文件是用来定义 EPC 各种编码结构(如 SGTIN-64、SSCC-96、SGTIN-96、SGLN-195、GRAI-96、GID-96 等)以及在每种编码结构内部七种格式之间的转换规则。关于 TDT 标记文件的详细规定见第 7 章,具体标记文件见附录 A。
- b) 标签数据转换算法是实现标签数据转换的程序机制。包括维护功能,即通过 TDT 标记文件读取各种编码结构的转换规则;转换功能,即根据具体的输入数据、附加参数和输出格式完成转换功能的程序动作。标签数据转换算法的详细规定见第 8 章。转换过程分为 5 步:
 - 1) 通过标头匹配,判断输入格式和编码结构;
 - 2) 进行编码结构和输入格式检查,并将输入数据分解成相应的字段;

- 3) 经过必要的计算(字符串操作、二进制到十进制/字母数据转换、填充等)将字段数据从输入格式转换为输出格式中相应字段所需要的格式;
 - 4) 使用必要的规则中定义的操作计算和派生出输出中必要的其他字段的值;
 - 5) 将相应的字段和常数值按输出格式进行组合,并将结果作为输出数据进行输出。
- 基本步骤如图 6 所示。

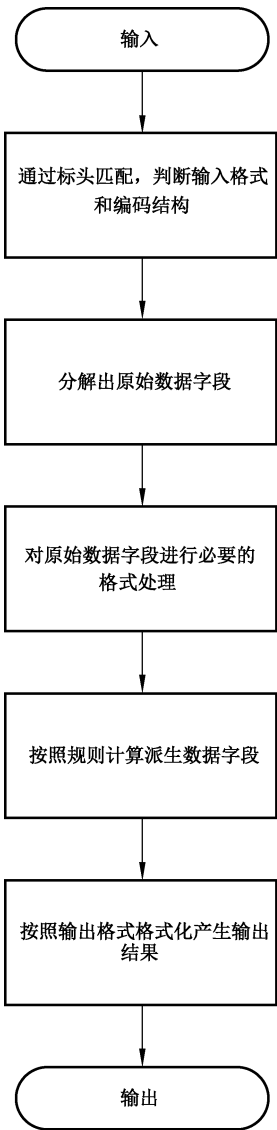


图 6 标签数据转换的基本步骤

6.2.4 输出

输出是通过标签数据转换过程,按输出格式要求得到输出数据的环节。

7 TDT 标记文件

7.1 TDT 标记文件概述

TDT 标记文件是一组定义了特定编码结构中如何在不同级别标签数据格式之间进行编码和解码规则的标记语言文件,采用 XML 格式进行描述。

TDT 标记文件的标记定义方式和相关规则按 7.2 给出的规定。TDT 标记文件中的各种元素和属

性说明按 7.3 给出的规定。本标准规定的 TDT 标记文件共 18 个,详见附录 A。

7.2 TDT 标记文件的定义方式和附加要求

7.2.1 TDT 标记文件的定义方式

TDT 标记文件采用 XML 格式,定义了根元素、编码结构元素、格式级别元素、选项元素、字段元素和规则元素共 6 个元素,各元素之间的相互关系采用 UML 类图描述,如图 7 所示。TDT 标记文件有关附加要求应符合 7.2.2~7.2.13 给出的细节。

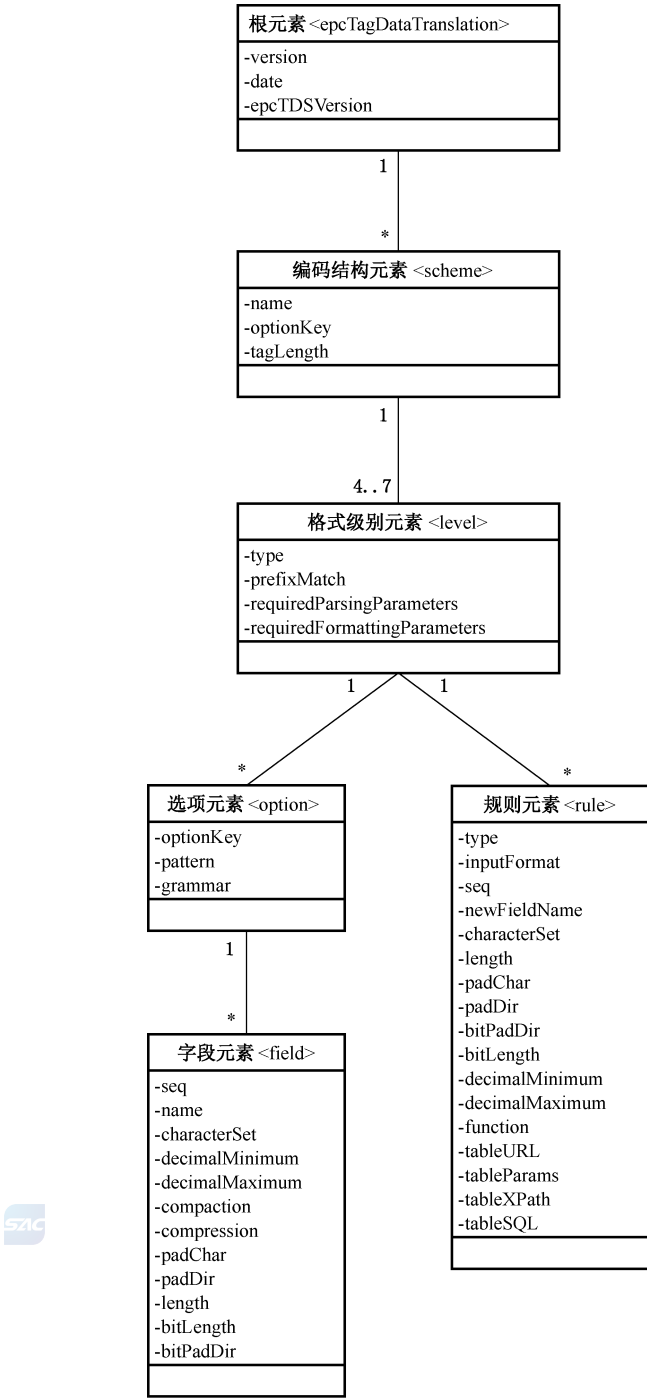


图 7 TDT 标记文件中各种元素的关系

7.2.2 输入数据和输出数据的定义方式

标签数据转换分别使用正则表达式和扩展巴科斯范式 ABNF 来定义不同级别 EPC 格式的转换规则。正则表达式主要是用来匹配输入数据,并从其中分割出不同的数据字段(位、数字、字母的组合)。扩展巴科斯范式 ABNF 主要用来定义输出数据的格式化规则,即如何通过原始的数据字段、派生的数据字段和常数值组合出标签数据转换过程的输出结果。

正则表达式采用符合 Perl 语言的正则表达式语法,并且需要支持否定式的零宽断言(zero-length negative lookahead)。

注: TDT 标记语言中的正则表达式并非仅仅符合 XSD 规范的正则表达式,因为其不支持否定式的零宽断言(zero-length negative lookahead)。典型编程语言如 Perl、Java、C#、.Net 的正则表达式库一般都支持否定式的零宽断言。

按照 ABNF 格式所定义的 grammar 属性描述了如何将转换过程中所获得的各个字段以及相关固定的常数值组合在一起最终构成输出数据。在 grammar 属性中,固定值用单引号包裹,没有用单引号包裹的代表数据字段(原始的或者派生的)的名称。这些数据字段在转换过程中需要使用实际获得的值进行替换。grammar 属性的每个字段(固定值字段和数据字段)之间使用空格分割,在替换组合完成后需要删除。

在选项元素<option>的子元素字段元素<field>中还包含其他相关的约束条件和格式转换规则,用于检查或者校验 EPC 是否存在错误。

上述定义模式相对复杂,但增加了未来的扩展性,特别是对 URI 格式,每个元素独立定义,方便检查,且可在未来使用更长存储位数标签时便于扩展规则。

7.2.3 输入格式的判断

标签数据转换过程应能自动根据输入数据(字符串)判断其编码结构和具体的格式。其他附加参数,如,标签长度可通过输入数据获得(如输入二进制格式或标签编码 URI 格式,格式本身带有标签长度);或由外部输入及用户选择。另外,在有些格式之间(如:在纯标识 URI 格式、GS1 应用标识字符串等格式、GS1 数据传输字符串之间)转换不需要知道标签长度,此时在 TDT 标记文件中任选一个同编码格式的标记文件就可完成上述转换过程。

标签数据转换过程允许的编码结构和格式很多,为了减小计算量提高效率,TDT 标记文件的所有格式的定义都包含前缀检查(prefixMatch)属性。查找输入格式时,先使用该前缀检查属性跳过不符合的项,然后使用正则表达式进行匹配检查排除不符合项,对于正则表达式完全匹配的项,就可根据匹配结果提取字段进行下一步转换步骤。

7.2.4 输出格式的名称缩略语

TDT 标记文件中输出格式及各种格式缩略语如下:

- BINARY:二进制格式;
- TAG_ENCODING:标签编码 URI 格式;
- PURE_IDENTITY:纯标识 URI 格式;
- LEGACY:GS1 文本数据字段格式;
- LEGACY_AI:GS1 应用标识符字符串格式(GS1 供人识别字符串格式,AI 有括弧);
- ELEMENT_STRING:GS1 数据传输字符串格式(AI 没有括弧,变长 AI 数据段用<FNC1>分割);
- ONS_HOSTNAME:ONS 域名格式。

本标准还对每种 EPC 编码结构规定了仅用于输出的 ONS 域名格式,该格式符合 DNS 域名解析第

35 查询模式(NAPTR,名称权威指针),用于发起 ONS 的查询和搜索,该格式只允许作为输出格式,不在输入格式自动匹配之列。

7.2.5 附加参数

附加参数一般在编码过程中使用而不在解码过程中使用。如图 5 b)所示,解码过程从二进制格式开始,经历标签编码 URI 格式、纯标识 URI 格式,最后转换为 GS1 文本数据字段格式、GS1 应用标识符字符串格式或 GS1 数据传输字符串格式,这个过程中可能会丢失一些信息(如标签长度、滤值、GS1 厂商识别代码的长度等)。在解码过程中需要附加参数的情况只有在 64 位标签时,才会需要使用 GS1 厂商识别代码与 GS1 厂商识别代码索引之间的转换表作为附加参数。除了上面这种情况,整个解码过程不需要附加参数的参与。

编码过程有些情况下需要如下附加参数,才能解析输入字符串:

- GS1 厂商识别代码的长度;
- 实际的标签长度;
- 滤值(比如表示包装层级:单品级、箱品级还是托盘级)。

标签数据转换算法应考虑这些附加参数输入和获取的方式。编程可采用在输入参数中引入“名称-值”对的查找表或关联数组(如 Dictionary、Hashtable)方式,或采用适合的正则表达式直接从输入的文本字符串提取的方式等。

GTIN 和 GLN 两类 GS1 标识在 EPC 应用时需提供一个系列号作为补充。此情况下,系列号不应采用附加参数的方式传输到标签数据转换过程,应将该系列号作为输入数据的一部分。如果输入采用的是 GS1 文本数据字段字符串格式,在 GTIN 或 GLN 后添加一个“;serial=(系列号)”;如果采用的是 GS1 应用标识符字符串或 GS1 数据传输字符串,GTIN 时添加应用标识符 21(系列号),GLN 时添加应用标识符 254(系列号)。其他编码结构(如 SSCC、GRAI、GIAI、GDTI、GSRN),本身就是系列化编码,不需要再添加系列号部分。示例见表 1。8.2 规定了标签数据转换软件的应用程序接口,其中输入参数仅允许:输入数据、附加参数、输出格式,不需要考虑个性值和中间变量。

表 1 各 GS1 标识关键字转换为标签数据转换输入数据的示例

编码结构	GS1 文本数据字段格式	GS1 应用标识符字符串格式
SGTIN	gtin=00037000302414;serial=10419703	(01)00037000302414(21)10419703
SSCC	sscc=000370003024147856	(00)000370003024147856
SGLN	gln=0003700030241;serial=1041970	(414)0003700030241(254)1041970
GRAI	grai=00037000302414274877906943	(8003)00037000302414274877906943
GIAI	giai=00370003024149267890123	(8004)00370003024149267890123
GSRN	gsrn=061414123456789012	(8018)061414123456789012
GDTI	gdti=0073796100001	(253)0073796100001
GID	generalmanager=5;objectclass=17;serial=23	无相应的应用标识符字符串的表现形式

在标签数据转换的过程中,若附加参数的值与从输入数据中解析出来的值相冲突,以输入数据中解析出来的值为准。比如,如果输入的滤值 filter 被设置成 5,而从输入数据中获得的滤值 filter 是 3,则在后面操作时以 3 作为滤值 filter 的取值。

考虑到编程语言的特异性,本标准附加参数也应以字符串形式传输到标签数据转换程序之中。如采用“key=value”(名称=值)对的形式,两个参数设置之间用“;”(英文分号)分割。这可在需要时简便的将参数转换为名称-值对查找表(如 Dictionary、Hashtable),且能兼容不同编程语言之间的差异性。

7.2.6 字段数据校验和字段数据的转换与派生

在字段元素<field>和规则元素<rule>包含了对该字段数据进行校验(正则表达式)和转换与派生的规则(内建函数和操作)。如,字段所有字符是否在指定的(数字或者字母)范围内、字符长度是否符合要求(校验);数据字段转换结构的长度、如何使用填充字符等(转换和派生)。TDT 标记文件通过此方式和约束来保证未来的扩展性。

7.2.7 对数值型字段的约束和检查要求

对于数值型字段超界的情况,应对字段具体数值进行约束和检查要求。

示例:

SSCC 的系列号部分最大允许 10 位十进制数,允许的取值是 1-9,999,999,999,而其二进制编码是不多于 34 位的二进制数,34 位二进制数可以表示的数值范围是 0-17,179,869,183,超出了上面的范围,所以我们需要在二进制格式中加入一个规则检查 SSCC 的系列号部分是否在合理的取值范围内。

为了避免上述问题,标签数据转换中对于表示数字或者数字组合的字段使用 decimalMinimum 属性表示最小可取数值,decimalMaximum 属性表示最大可取数值;表示数字字母组合的字段不需要上述属性。一旦规定了 decimalMinimum 和 decimalMaximum,标签数据转换软件就需要将该字段使用合理的算法(二进制转十进制、文本转数值等)转换为数值(十进制),然后进行如下检查:

$$\text{decimalMinimum} \leq \text{field} \leq \text{decimalMaximum}.$$

一旦超界,需要返回错误或者抛出异常。

7.2.8 对文本型字段的约束和检查要求

对于文本型字段,在字段标记<field>会规定 characterSet 属性来约束该字段字符的范围。该属性使用正则表达式的方式来定义。在正则表达式中,方括弧“[...]”内是允许的字符、星号“*”代表上述字符范围内的字符可以出现零次到多次。在检查时建议添加“^”表示匹配字段的起始位置、“\$”匹配字段的结束位置。例如“^[0-7]*\$”表示整个字符串由数字字符 0-7 的组合组成,不允许其他字符出现。

示例:

[01]* - 仅允许数字字符 0 和 1;

[0-7]* - 仅允许数字字符 0 到 7;

[0-9]* - 仅允许数字字符 0 到 9;

[0-9 A-Z\]* - 仅允许数字字符 0 到 9、空格(ASCII 码 32)、大写字母 A 到 Z 和连字符“-”。

注:此处 9 与 A 之间有一个空格。

characterSet 允许在标签数据转换的过程中检查字段的字符是否在允许的范围内,保证 EPC 数据的正确传输。但有特例,如:GRAI 的系列号部分允许字母和数字,但是在 GRAI-96 和 GRAI-64 编码结构中系列号仅允许存储数字(这是由于位不够造成的),此时如果将字母数字系列号的 GRAI 传进标签转换过程应该报错,可是如果这时候该系列号正好全是数字字符又能够转换就不会产生错误。因此在设计具体软件时,应做好必要的需求分析和充分的软件测试与检查工作。

另外,一个 GRAI 可以采用两种不同的方式进行编码:全数字型的 GRAI-96 和字母数字型的 GRAI-170。如果一个字段包含 compaction 属性,那么其就应按照字母数字型字符串转换为二进制编码的形式进行编码;如果不包含 compaction 属性,就应考虑按照整数或者整型字符串转换为二进制编码的形式进行编码。而且如果设置了 padChar 属性,除了上述规则外,还应按照位数要求和填充规则,使用 padChar 属性中规定的填充字符对其转换结果进行填充。

如果既设置了 compaction 属性,又设置了 characterSet 属性,不能因为有 compaction 属性就仅对字段做字母数字检查,仍应也按照 characterSet 属性的设置来进行检查。

7.2.9 字段的填充

7.2.9.1 填充字符(padChar)和填充方向(padDir)

padChar 属性和 padDir 属性只适用于非二进制格式。

padChar 属性规定的是填充所使用的字符。本标准允许的填充字符包括数字字符 0——ASCII 码 48(30_{hex})、空格——ASCII 码 32(20_{hex})和其他用户指定的字符。

padDir 属性指明填充字符填充的方向,有两个取值:LEFT(左侧填充)和 RIGHT(右侧填充)。

length 属性指明该字段数据的字符长度。

length 属性、padDir 属性和 padChar 属性出现在字段元素<field>和规则元素<rule>之中。填充规则是使用 padChar 属性规定的填充字符和 padDir 属性所规定的填充方向按照 length 属性所规定的长度进行填充。对于字段元素<field>上述三个属性要么同时设置,要么同时省略。对于规则标记<rule>,即使定义了 length 属性,不要求同时定义 padDir 属性和 padChar 属性;因为规则元素里定义的函数的返回结果可能并不需要进一步的填充,设置的 length 属性近用字符长度是否符合要求。

为了避免不必要的重复定义,标签数据转换过程的校验步骤、解析步骤、规则执行步骤和结果构造应遵循以下规则:

- a) 如果在一个格式级别的<level>元素中的不同选项元素<option>中有一个字段元素<field>(原始数据字段)包含有 padChar 属性、padDir 属性和 length 属性的定义,那么在同一个格式级别的<level>元素内部的所有不同选项元素<option>之间的相同字段元素<field>(原始数据字段)都应该使用相同的 padChar 属性、padDir 属性和 length 属性进行填充。如果在一个格式级别的<level>元素中有一个规则元素<rule>(派生数据字段)包含有 padChar 属性、padDir 属性和 length 属性的定义,那么在同一个格式级别的<level>元素内部的所有相同规则元素<rule>(派生数据字段)都应该使用相同的 padChar 属性、padDir 属性和 length 属性进行填充。
- b) 如果在一个标签编码 URI 格式的格式级别元素<level>下的一个字段元素<field>或者一个规则元素<rule>包含有 length 属性、padDir 属性和 padChar 属性,那么标签编码 URI 格式以上层级的所有格式(如纯标识 URI 格式、GS1 数据传输字符串格式、文本字段格式、ONS 域名格式等)的格式级别元素<level>所对应的该字段(原始的或者派生的)都应该使用相同的 padChar 属性、padDir 属性和 length 属性进行填充。
- c) 如果一个二进制格式的格式级别元素<level>下的一个字段标记<field>或者一个规则标记<rule>包含有 length 属性、padDir 属性和 padChar 属性,且标签编码 URI 格式的格式级别元素<level>下没有定义 padDir 属性和 padChar 属性,这表示从其他格式编码转换为二进制格式时,需要在转换之前将非二进制格式的数据按照填充规则进行填充,之后完成二进制转换;反之,如果从二进制格式解码为其他格式时,应该将二进制转换之后的文本结果按照与这个填充规则相反的方式将添加的填充字符去除得到转换结果(若转换到的格式有其他填充规定,还需要按照这个规则做进一步的填充)。这里,对于任意一个 EPC 代码结构,如果二进制和非二进制格式的 padChar 属性和 padDir 属性规定完全相同,就不必再在二进制格式中规定 padChar 属性和 padDir 属性了。如果在同一个 TDT 定义文件中,对二进制格式和非二进制格式都定义了完全相同的 padChar 属性和 padDir 属性,二进制格式中的定义会被忽略。

示例:

填充:GS1 厂商识别代码除了二进制格式之外,所有其他格式的填充字符是数字字符 0,填充方向是 LEFT(padChar="0",padDir="LEFT"),在这些格式中前置 0 也是编码非常重要的组成部分。

7.2.9.2 二进制填充方向(bitPadDir)和二进制长度(bitlength)

对于二进制格式(<level>元素属性 type 的值为 BINARY,即 type="BINARY"),其下的字段元素

⟨field⟩或者规则元素⟨rule⟩中可能会定义二进制填充方向(bitPadDir)和二进制长度(bitLength)属性。二进制填充不需要填充字符的定义,仅允许填充二进制位 0。二进制填充方向(bitPadDir)可缺省,如果定义了 bitPadDir,bitPadDir 的取值应为 LEFT(左侧填充)或 RIGHT(右侧填充),代表二进制填充的方向。二进制长度(bitlength)代表二进制数据的预期长度。

二进制填充并不总是向左侧填充,对于数值型的数据一般是在左侧填充,比如 SGTIN-96 的系列号部分的二进制填充方向为左侧(bitPadDir=“LEFT”);对于字母数字型数据一般是在右侧填充,比如 GRAI-170 的系列号部分的二进制填充方向就是右侧(bitPadDir=“RIGHT”)。在解码时,应考虑二进制填充问题,特别是字母数字型数据二进制全 0 字符很可能是二进制填充造成的,需要在转换时去掉二进制填充位或者忽略掉二进制全 0 字符。

7.2.9.3 填充流程

图 8 是从非二进制格式向二进制格式进行编码时是否进行填充字段(或者剔除填充字符)的判断规则流程图。

图 9 是从二进制格式向非二进制格式进行解码时是否进行填充字段(或者剔除填充字符)的判断规则流程图。

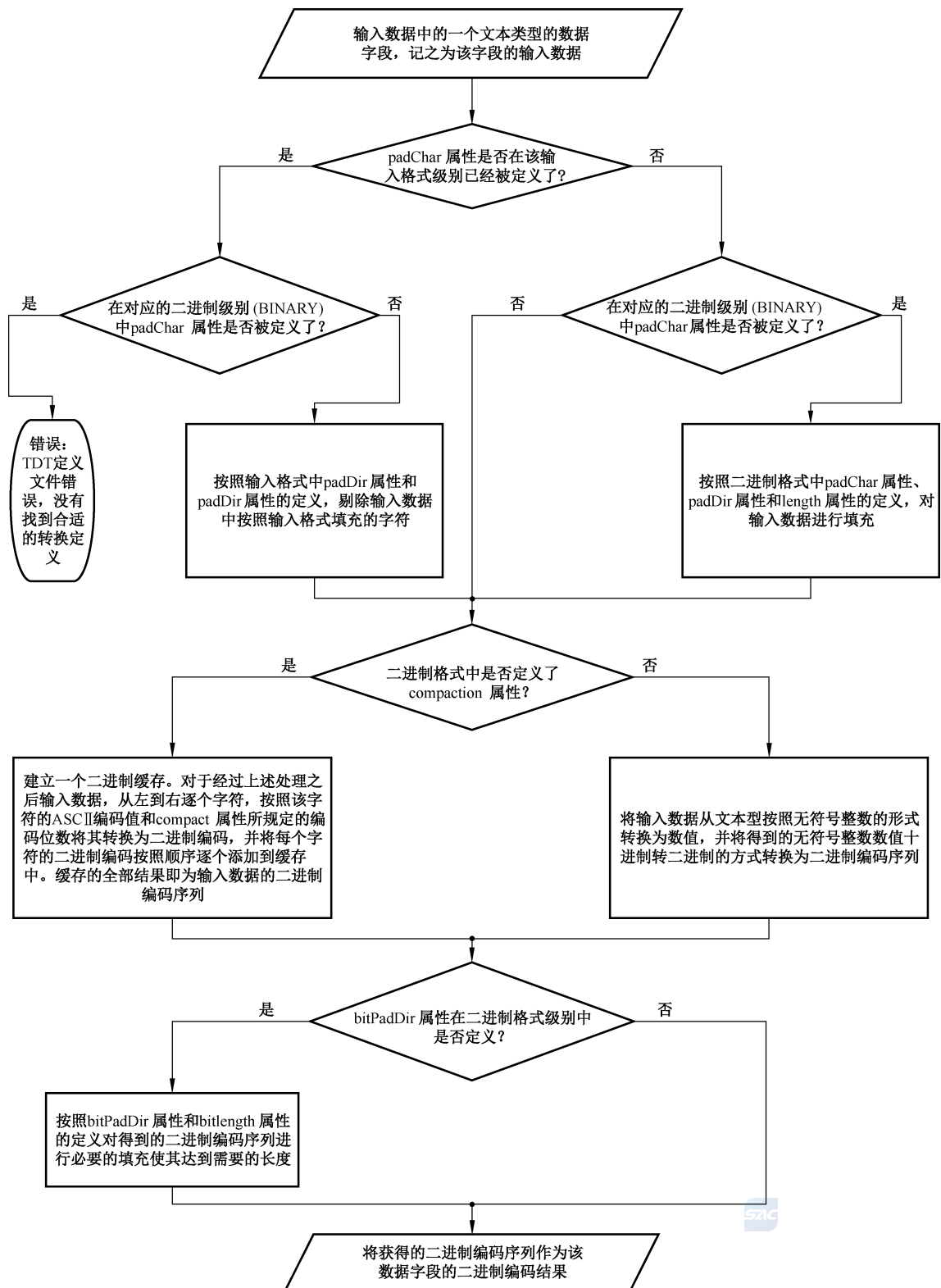


图 8 编码过程填充流程

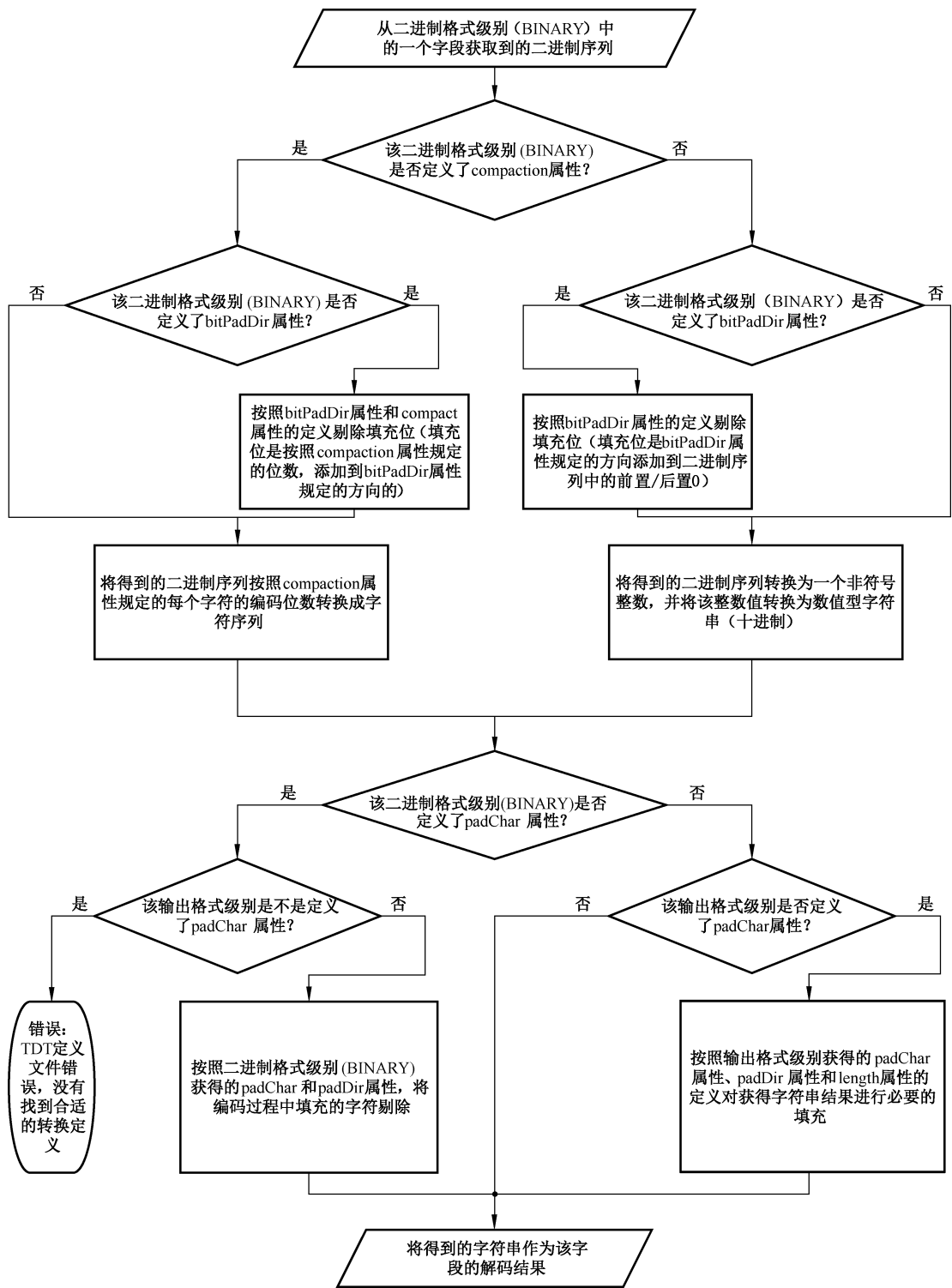


图 9 解码过程填充流程

示例：

对于 96 位的 SGTIN 编码 (SGTIN-96), TDT 定义文件中字段 `gs1companyprefix`, 在选项 7 (optionKey = "7") 下, 在非二进制格式的定义中长度属性值为 7, 填充字符是数字字符 0 (padChar = "0")、左侧填充 (padDir = "LEFT"); 同样的选项二进制格式的定义二进制长度 (bitLength = "24")、二进制填充方向 (bitPadDir = "LEFT"), padDir 属性和 padChar 属性未设置。这代表当解码时, 一个 24 位的二进制 (00000000 10010000 10001000)_{bin} 的 `gs1companyprefix` 字段:

- a) 先剔除掉二进制序列前面所有的前置 0, 转换为 (10010000 10001000)bin;
- b) 将上面的结果转换为十进制整数 37000, 并转换为 4 位十进制字符串“37000”;
- c) 之后按照非二进制格式转换规则左侧填充数字字符 0 补足 7 位, 即“0037000”。

转换过程中可能会产生一些特殊的结果。比如: 对于 SGLN, 当 gs1companyprefix 长度是 12 位时, 地址编码(locationref)是长度为 0 的字符串, URI 结果因会出现两个连续的点“..”, 如“urn:epc:id:sgln:123456789012..12345”。这样的结果是正确的, 不可以在两个点之间加数字 0, 因为数字 0 是 1 个字符长度而非 0 个字符长度。

7.2.10 字段的编码和压缩

当字符型(非数值型)字符串在二进制格式中进行编码时, 该字段定义为一个文本字段。一个文本字段标记(field)中可能包含以下两个属性:

- compaction 属性(字段编码属性);
- compression 属性(字段压缩属性)。

如果没有设置 compaction 属性代表该字段是整数型或者全数字型字符串, 这时采用的是十进制到二进制转换的方式进行编码。如果设置了 compaction 属性, 代表的是该字段是文本字段, 每个字符应该按照其 ASCII 编码值以及该属性设置的位数长度转换为二进制进行编码。compaction 属性可以选取的值为: 5 位(5-bit)、6 位(6-bit)、7 位(7-bit) 和 8 位(8-bit), 这些取值所代表的具体编码方法见 ISO 15962。

compression 属性是预留的属性, 作为压缩属性, 指明字段数据编码的压缩方式(如 zip/gzip 压缩或者 Huffman 编码压缩等)。

7.2.11 TDT 标记文件中使用的一些字段名称及其含义

TDT 标记文件中字段名称的格式没有强制要求, 但根据 XML 语法建议采用小写字母数字组合, 不使用空格和连字符。在不同的编码格式定义之间也没有保留字, 这意味着两个不同的编码格式可能有使用相同名称的字段, 但这两个字段的意义和属性可能都不一样。每一个编码格式的定义都是独立的一个名称空间, 互相不发生关系。表 2 给出了一些本标准规定的 TDT 标记文件中公用的字段名称及其含义。

表 2 字段名称及其含义

名称	含义
Filter	滤值, 取值范围十进制数字 0~7
Serial	系列号, 十进制或者字母数字
gs1companyprefix	GS1 厂商识别代码
gs1companyprefixlength	GS1 厂商识别代码的长度, 即字符个数, 十进制, 比如 gs1companyprefix = "0037000" → gs1companyprefixlength = "7"
tagLength	EPC 标识的编码位长, 64/96/256/等等
gs1companyprefixindex	一个整数型的索引数字, 用来查找实际的 GS1 厂商识别代码, 仅用于 64 位标签
Itemref	厂商内部给一类对象或者一个 SKU 所指定的编码, 用于 GTIN
locationref	厂商内部给一个地址所指定的编码, 用于 GLN
Assetref	一个资产的系列编码(数字或者字母数字), 用于 GIAI
Serialref	一个系列编码(数字或者字母数字), 例如用于 SSCC
serviceref	一个厂商对一个服务所指定的编码, 用于 GSRN
documenttype	一个厂商对一个类文档或者资料所指定的编码, 用于 GDTI

7.2.12 规则元素<rule>和派生字段

在构造输出格式时,一些必备的字段不能够直接通过输入格式模式匹配获得,因此需要一系列的规则来通过已知的数据字段派生出输出结果需要的其他字段。在 TDT 标记文件中采用规则标记<rule>来定义上述转换、派生规则。我们将通过规则标记<rule>所规定的操作产生的新的数据字段成为派生字段或者派生数据字段。

标签数据转换过程又可分为如表 3 所示的两个主要的阶段。

表 3 数据转换的两个阶段

阶段	说明
提取阶段(EXTRACT)	输入数据经过解析获得相关数据字段(称之为原始数据字段/输入数据字段)后进行的一系列操作
格式化阶段(FORMAT)	对原始数据字段、派生数据字段使用相应的格式化语法构造输出结果的一系列操作

标签数据转换过程中所规定的各种规则标记<rule>,只适用于该规则定义所处的格式级别内(<level>)。规则标记<rule>按照其 type 属性分别作用于表 3 所示的两个阶段。解析阶段(EXTRACT)的规则仅对输入格式级别有用,格式化阶段(FORMAT)的规则仅对输出格式级别有用。一个格式级别内,规则可以没有也可以有多条,当有多条规则并存时,规则标记<rule>的 seq 属性规定了规则的执行顺序,从小到大依次执行。

示例 1:

典型的转换和派生是将纯标识 URI 格式相应字段,转换为 GS1 应用标识符字符串格式或 GS1 数据传输字符串格式中 GTIN 字段的解码过程中的解析阶段(EXTRACT)。要将输入数据中提取出的厂商识别代码(companyprefix)、类别代码(itemref)进行派生。

- a) GS1 包装指示符(indicator digit)是类别代码(itemref)的首位数字,将其提取出来并将剩余的部分记为新的类别编码(itemref remainder);
- b) 构造 GTIN 除校验位的部分(gtin prefix);
- c) 计算 GS1 校验位(check digit)。

计算过程如下(里面使用了核心函数按 7.2.13 的规定):

- 1) indicator digit = SUBSTR(itemref,0,1);
- 2) itemref remainder = SUBSTR(itemref,1);
- 3) gtin prefix = CONCAT(indicator digit,companyprefix,itemref remainder);
- 4) check digit = GS1CHECKSUM(gtin prefix)。

示例 2:

编码过程的格式化阶段(FORMAT),即从 GS1 数据传输字符串格式或 GS1 应用标识符字符串格式转换为纯标识 URI 格式,是从 GTIN 字段(gtin)开始,且需要厂商识别代码长度(gslcompanyprefixlength)作为附加参数,派生过程如下:

- a) gtin prefix remainder=SUBSTR(gtin,1,12);
- b) indicator digit=SUBSTR(gtin,0,1) itemref remainder=SUBSTR(gtin prefix remainder,gslcompanyprefixlength);
- c) itemref=CONCAT(indicator digit,itemref remainder);
- d) gslcompanyprefix=SUBSTR(gtin prefix remainder,0,gslcompanyprefixlength)。

7.2.13 核心函数



标签数据转换的编码和解码过程中需要用到表 4 所示的核心函数。

表 4 核心函数

核心函数	说明
SUBSTR(string, offset)	从<offset>位置开始获得子字符串,<offset>取值从 0 开始
SUBSTR (string, offset, length)	从<offset>位置开始获得长度为<length>的子字符串,<offset>取值从 0 开始
CONCAT(string1, string2, string3,...)	将字符串首尾相连组合成一个新的字符串
LENGTH(string)	字符串的长度
GS1CHECKSUM(string)	通过输入的数字字符串计算 GS1 校验位
TABLELOOKUP (inval, tablename, incol, outcol)	从表<tablename>中的<incol>列查找输入值<inval>的位置,并返回<outcol>列同行的值作为输出 请注意表的 URI 是通过另一个参数 tableURL 设置的,相应 XPath 参数或者 SQL 参数分别由 tableXPath 和 tableSQL 定义
add(String, int)	将输入的字符串转换为数值与输入的整数相加,将结果转换为字符串输出
multiply(String, int)	将输入的字符串转换为数值与输入的整数相乘,将结果转换为字符串输出
divide(String, int)	将输入的字符串转换为数值与输入的整数做除法,将商转换为字符串输出
subtract(String,int)	将输入的字符串转换为数值与输入的整数相减,将结果转换为字符串输出
mod(String, int)	将输入的字符串转换为数值与输入的整数做除法,将余数转换为字符串输出(即模运算)

为了完整实现标签数据转换功能,标签数据转换软件需要根据使用的编程语言提供上述函数的对应实现版本,可以是语言中已经支持的函数、也可以是自定义函数或者方法。

计算各个 GS1 编码结构校验位的核心函数(GS1CHECKSUM)的算法见 GS1 通用规范。

很多的编程语言(Java、C++、C#、Visual Basic、Perl、Python)对同一函数的实现方式可能有区别。如:提取子字符串函数 SUBSTR,有的编程语言是子字符串长度(length)参数,有的编程语言是子字符串结束位置(endIndex)参数,见表 5。

表 5 不同编程语言中 SUBSTR 函数的不同形式

编程语言	核心函数 SUBSTR(string,offset,length)的具体实现	说明
C++(STL)	string.substr(offset, length);	
C#	String.Substring(offset, length);	
Visual Basic	String.Substring(offset,length)	
Java	Java.lang.String String.substring(beginIndex, endIndex)	beginIndex = offset endIndex = offset+length
Perl	substr(\$ stringvariable, \$ offset, \$ length);	
Python	String[start:end]	start = offset end = offset+length

如果规则标记<rule>使用到了 TABLELOOKUP 函数,该标记应具有下列属性:

- tableURL 属性;
- tableXPath 属性或者 tableSQL 属性。

TABLELOOKUP 函数表示从一个表格中查找数据。该表格的格式可以是 XML 格式、逗号分隔格式(CSV)、制表符分割格式(TSV),或者标签数据转换软件任一存储表格的格式。由于具体表格的格式可能多种多样,tableURL 属性和 tableParams 属性以及与之相对应的 tableXPath 属性或者 tableSQL 属性的设置仅代表进行 TABLELOOKUP 操作时的逻辑查找方式,而不是具体的代码,且与具体的表格存储和传输格式无关(具体实现由转换软件完成)。

对于 XML 格式存储的表格或者类似 XML 格式存储的表格,规则标记<rule>需要具有 tableURL 属性、tableXPath 属性和 tableParams 属性。tableParams 属性是一个逗号分隔的字符串(一串逗号分割的表格参数),这字符串可由无引号包含的前面步骤得到的字段名或者全是数字或者单引号包含的文本构成的常数值组成。在 tableXPath 出现的“\$1”代表了 tableParams 属性定义的第一个表格参数。tableXPath 在传递到 XML DOM 解析器之前需要使用 tableParams 的定义、字段的值、常数对 tableXPath 进行相应的替换和处理。

当有多个参数出现在 tableParams 属性中,\$2 代表第二个参数,\$3 代表第三个参数,依次类推。

对于逗号分隔格式(CSV)、制表符分割格式(TSV)或者其他可以转换为类似关系型数据库表格形式的表格,规则标记<rule>需要具有 tableURL 属性、tableSQL 属性和 tableParams 属性。tableParams 属性的定义与采用 tableXPath 时的方式类似,其中变量仍用 \$1、\$2、……标识,查询前需将 tableSQL 进行必要的替换。

示例:

GS1 厂商识别代码索引(companyprefixindex)在 64 位标签或者 64 位编码结构中使用,用来代替 GS1 厂商识别代码(companyprefix)。在 <https://www.gs1.org/standards/bc-epc-interop> 上可以找到 GS1 厂商识别代码索引(companyprefixindex)和 GS1 厂商识别代码(companyprefix)的对应表,该表格以 XML 格式提供。

对于 XML 格式的对应表,查找的规则标记<rule>的各个属性为:

——解码过程:

- tableURL 属性和 function 属性设置如下:
tableURL="https://www.gs1.org/standards/bc-epc-interop"
function="TABLELOOKUP(companyprefixindex,'GEPC64Table',companyprefixindex,companyprefixindex)"
- tableXPath 和 tableParams 属性设置如下:
tableXPath="/GEPC64Table/entry[@index='\$1']/@companyPrefix"
tableParams="companyprefixindex";

——编码过程:

- tableURL 属性和 function 属性设置如下:
tableURL="https://www.gs1.org/standards/bc-epc-interop"
function="TABLELOOKUP(companyprefix,'GEPC64Table',companyprefix,companyprefixindex)"
- tableXPath 和 tableParams 属性设置如下:
tableXPath="/GEPC64Table/entry[@companyPrefix='\$1']/@index"
tableParams="companyprefix"。

7.3 TDT 标记文件的元素和属性

7.3.1 根元素的属性和子元素

7.3.1.1 根元素

TDT 定义文件使用元素<epcTagDataTranslation>作为标记文件的根元素。

7.3.1.2 根元素属性

根元素属性见表 6。

表 6 根元素属性

属性名	说明	示例值
Version	TDT 定义文件的版本号	1.6
Date	创建时间戳	2011-01-18T11:33Z
epcTDSVersion	使用的 TDS 版本号	1.6

7.3.1.3 根元素的子元素

根元素子元素的名称及描述见表 7。

表 7 根元素的子元素

子元素名称	描述
Scheme	请见 7.3.2 编码结构标记<scheme>

7.3.2 编码结构元素的属性和子元素

7.3.2.1 编码结构元素

TDT 定义文件的编码结构元素为<scheme>,对于 TDS 中定义的所有标识/编码结构来说,编码结构元素<scheme>规定了在数据转换过程中的编码和解码细节,包含有下列编码结构/标识的转换规则: GDTI-96、GDTI-113、GIAI-64、GIAI-96、GIAI-202、GID-96、GRAI-64、GRAI-96、GRAI-170、GSRN-96、SGLN-64、SGLN-96、SGLN-195、SGTIN-64、SGTIN-96、SGTIN-198、SSCC-64、SSCC-96。

7.3.2.2 编码结构元素的属性

编码结构元素的属性名称、说明及示例值见表 8。

表 8 编码结构元素的属性

属性名称	描述	事例值
Name	编码结构的名称	GDTI-96,GDTI-113,GIAI-64,GIAI-96, GIAI-202, GID-96, GRAI-64, GRAI-96, GRAI-170, GSRN-96, SGLN-64,SGLN-96,SGLN-195,SGTIN-64, SGTIN-96, SGTIN-198,SSCC-64,SSCC-96
optionKey	适用于该编码结构的众多选项所代表的参数名称。数据格式的转换应在具有相同 optionKey 属性的<option>元素之间进行	gs1companyprefixlength

表 8（续）

属性名称	描述	事例值
tagLength	该属性表示 EPC 代码结构的二进制编码长度，即在 Gen2 RFID 标签中内存 EPC/UII 区域 20 h 位置开始的位数。对于变长的 EPC 代码，tagLength 属性不需要定义；对于固定长度的 EPC 代码需要定义该属性值	64、96 或者更大数值

7.3.2.3 编码结构元素的子元素

编码结构元素的子元素名称及描述见表 9。

表 9 编码结构元素的子元素

子元素名称	描述
Level	在该编码结构下，各种级别格式编码和解码规则的定义标记

7.3.3 格式级别元素的属性和子元素

7.3.3.1 格式级别元素

TDT 定义文件的格式级别元素为<level>，该元素提供了每一个格式级别的相应定义，包括前缀检查等。该元素内部包含有嵌套的选项元素<option>和一系列规则元素<rule>，选项元素<option>包含了具体的正则表达式匹配规则和扩展巴科斯范式 ABNF 语法规则的输出结果格式化规则，规则元素<rule>用来计算相应的派生字段。

7.3.3.2 格式级别元素的属性

格式级别元素的属性名称、说明及示例值见表 10。

表 10 格式级别元素的属性

属性名称	描述	示例值
Type	表示格式的级别	BINARY TAG_ENCODING PURE_IDENTITY LEGACY LEGACY_AI ELEMENT_STRING ONS_HOSTNAME
prefixMatch	编码解码过程中用到的前缀，用于匹配	00001010 uri:epc:tag:sscc-64 uri:epc:id:sscc sscc= (00)

表 10（续）

属性名称	描述	示例值
requiredParsingParameters	在解析输入数据过程中使用的各种附加参数(以逗号分隔)	gs1companyprefixlength
requiredFormattingParameters	在格式化输出数据过程中使用的各种附加参数(以逗号分隔)	filter,tagLength

7.3.3.3 格式级别元素的子元素

格式级别元素的子元素名称及描述见表 11。

表 11 格式级别元素的子元素

子元素名称	描述
Option	包含正则表达式匹配规则和结果格式化语法
Rule	包含计算或者获取派生字段的计算规则

7.3.4 选项元素的属性和子元素

7.3.4.1 选项元素

TDT 定义文件的选项元素为<option>。

7.3.4.2 选项元素的属性

选项元素的属性名称、描述及示例值见表 12。

表 12 选项元素的属性

属性名称	属性描述	示例值
optionKey	表示一个编码结构元素<scheme>内部存在的不同选项情况。这些不同选项用optionKey属性的值加以区分。对于变长的EPC,optionKey属性可能没有实际的意义,但是却可以通过不同的optionKey属性设置保证输入格式与输出格式的一致性。对于全部EPC来说,仅允许在输入与输出格式的两个具有相同optionKey属性值的<option>标记之间进行转换	任意文本 对 GS1 标识,值为 6-12,代表厂商识别代码的长度
Pattern	一个用来与输入数据进行匹配并提取相应变量字段的正则表达式	00110000([01]{3}) 010 ([01]{34}) ([01]{10}) ([01]{38})
Grammar	一个扩展巴科斯范式 ABNF 的语法定义来规定如何格式化输出结果	'00110000' filter '010' gs1companyprefix itemref serial 注意： 单引号包含的是常数值、不加单引号的是变量字段值需要用实际数据进行替换



7.3.4.3 选项元素的子元素

选项元素的子元素名称及描述见表 13。

表 13 选项元素的子元素

子元素名称	描述
Field	提供关于每一个变量字段的具体信息,如最大最小取值范围、允许的字符、长度、填充方式等

7.3.5 字段元素

TDT 定义文件的字段元素为<field>。字段元素的属性名称、说明及示例值见表 14。

表 14 字段元素的属性

属性名称	属性说明	示例值
Seq	使用特定的正则表达式匹配出来的子模式所对应的序号,子模式是从 1 开始的	1, 2, 3...
Name	数据字段(或变量)的名称,该名称用于构造输出结果或者产生派生字段时使用	filter, companyprefix, itemref, serial, ...
decimalMinimum	该字段允许的最小十进制数值	0
decimalMaximum	该字段允许的最大十进制数值	9999999
Length	该字段需要满足的字符串长度	7
bitLength	该字段需要的位长度,仅在二进制格式中使用	24
bitPadDir	在二进制编码中添加填充位 0 的方向	'LEFT', 'RIGHT'
characterSet	该字段允许的字符集合,采用正则表达式表示	[0-9]*, [01]*, [0-9A-HJ-NP-Z]*
padChar	为了使字段达到 length 属性规定的长度,填充进该字段结果字符串中的填充字符(对于二进制格式之外的格式,如果不需要填充可以忽略该属性)	'0', ' ' (ASCII 空格)
padDir	填充填充字符的方向	'LEFT', 'RIGHT'

7.3.6 规则元素

TDT 定义文件的规则元素为<rule>。规则元素的属性名称、说明及示例值见表 15。

表 15 规则元素的属性

属性名称	属性说明	示例值
Type	表明该规则需要在转换过程中的所属阶段执行”	'EXTRACT' (解析过程), 'FORMAT' (格式化过程)
inputFormat	表明输入变量(数据字段)的格式,是二进制编码格式还是非二进制编码的字符串格式	'STRING', 'BINARY'
Seq	表明一系列规则执行的顺序,按照 seq 属性的值从小到大依次执行	1,2,3,4,5...

表 15 (续)

属性名称	属性说明	示例值
newFieldName	在完成操作之后所得结果的新字段名称(变量名)	gtinprefixremainder、gtinprefix
Function	一个用来表示如何通过已知字段计算新字段的表达式(会使用核心函数或者核心函数的组合)	SUBSTR(itemref,0,1)
decimalMinimum	对于输出为数值型的字段,字段的输出所允许的最小十进制数值	0
decimalMaximum	对于输出为数值型的字段,字段的输出所允许的最大十进制数值	9999999
Length	输出的字符串所应达到的长度	7
padChar	为了满足 length 属性规定的输出字符串所应达到的长度而进行的填充所使用的填充字符(不需要填充时该属性可省略)	'0', ''
padDir	填充字符填充的方向	'LEFT', 'RIGHT'
bitLength	输出结果二进制编码需要达到的长度(对于非二进制格式可省略)	e.g.24
bitPadDir	为了达到 bitLength 规定的输出结果二进制编码长度而添加填充位 0 的方向	'LEFT', 'RIGHT'
characterSet	输出结果所允许的字符集合,用正则表达式表示	[0-9],[01]
tableURL	一个 URL 地址用来指明用于操作的表格的位置	http://www.onsepc.com/ManagerTranslation.xml
tableXPath	一个 XPath 表达式用来从 XML 表格中获得相应的标记值或者属性值 在该表达式中 \$1、\$2 等分别指明 tableParams 中的表格变量,\$1 是第一个变量,依此类推。在将本属性值传输到 XML 解析引擎之前需要对上述变量使用实际值进行替换	/GEPC64Table/entry[@index='\$1']/@companyPrefix
tableSQL	一个 SQL 表达式,用来从关系型数据库表中获取相应数据。 该表达式中 \$1、\$2 等分别指明 tableParams 中的表格变量,\$1 是第一个变量,依此类推。在将本属性值传输到关系型数据库查询引擎之前需要对上述变量使用实际值进行替换	 SELECT companyPrefix FROM GEPC64Table WHERE index='\$1'
tableParams	一个用逗号分割的表格变量列表,用于 tableXPath 属性或 tableSQL 属性中相应变量的替换。这些变量应为已知的字段或者常数	companyprefixindex

8 标签数据转换算法

8.1 概述

标签数据转换算法一方面通过读取最新的 TDT 标记文件得到各种编码结构的最新转换规则(称之为维护功能,详见 8.2.1),另一方面对用户具体的标签数据转换,即根据具体的输入数据、附加参数和输出格式将输入数据转换为输出数据(称之为标签数据转换功能,详见 8.2.2)。

标签数据转换算法可以是一个专门的软件,也可以是一个软件包、类库或者函数库。标签数据转换可以部署到 EPC 系统的任何部分,也可以作为 EPC 系统与其他应用相交互的软件接口中的一部分使用。另外,为了满足云端应用的需要,也可以将标签数据转换设计为一个独立的服务。这个服务可以采用 Web 表单或者是 Web 服务形式,可以一次完成一个输入数据的转换也可以一次完成一组输入数据的转换。

8.2 功能算法和应用程序接口(API)

8.2.1 维护功能算法和维护接口(TDT 标记文件读取和解析功能)

8.2.1.1 维护功能的算法

检查所有的 TDT 标记文件是否为最新版本,根据获得的版本重新加载 TDT 标记文件的各种规则。关于具体规则的读取参照第 7 章相关规定。

8.2.1.2 维护接口

对于采用不同的编程语言,维护接口的定义方法可能会有区别,表 16 给出了一些典型的编程语言维护接口的定义方式。

表 16 维护接口的定义

编程语言	建议的接口定义方式
C++	public: void refreshTranslations();
C++ .Net	public: void refreshTranslations();
Visual Basic	Public Sub refreshTranslations()
C#	public void refreshTranslations();
Java	public void refreshTranslations();
Perl	sub refreshTranslations
Python	def refreshTranslations()
说明	
检查所有的 TDT 标记文件是否为最新版本,根据获得的版本重新加载 TDT 标记文件的各种规则	
参数	
(无)	
返回值	
(无)	
抛出的异常或返回的错误	
(无)	

8.2.2 转换功能算法和转换接口

8.2.2.1 转换功能算法

标签数据转换功能是实现将标签数据从输入格式到同一编码结构的另一种输出格式的转换过程，具体算法流程如下：

第 1 步，设置：

第 1.1 步 读取输入数据及附加参数。

第 1.2 步 将附加参数转换为“键-值”列表形式（如 Dictionary、Hashtable）。“键-值”列表定义为本次转换过程的字段数据表。在此转换过程中，该字段数据表会不断扩展，增加从输入数据中解析出的原始数据字段的“字段-值”，以及根据解析阶段（'EXTRACT'）和格式化阶段（'FORMAT'）规则派生出的派生数据字段的“字段-值”。

第 1.3 步 根据用户的输入或者系统的参数，设置希望获得的输出格式。

第 2 步，判断输入数据的编码结构和输入格式：

第 2.1 步 遍历所有编码结构元素〈scheme〉以及其格式级别元素〈level〉；使用每一个格式级别元素〈level〉的 prefixMatch 属性¹⁾判断输入数据是否符合其规定，判断：

- a) 如果输入数据确实是以该〈level〉的 prefixMatch 属性规定的模式起始的：（先字母后数字，单括号）
 - 1) 如果该格式级别元素〈level〉上级的编码模式〈scheme〉没有设置 tagLength 属性，那么该格式级别元素〈level〉上级的编码模式〈scheme〉记为一个待选的编码模式，将该格式〈level〉记为待选的输入格式，用“待选的编码模式+格式级别”（记为：待选输入格式）来表示；
 - 2) 如果该格式级别元素〈level〉上级的编码模式〈scheme〉设置有 tagLength 属性，同时字段数据表中也有 tagLength 的定义，那么：如果 tagLength 属性和字段数据表中 tagLength 的值相一致，将该编码模式〈scheme〉以及该格式〈level〉记为一个待选输入格式；否则跳过继续进行下一个格式级别元素〈level〉的 prefixMatch 属性判断。
- b) 如果输入数据不符合该〈level〉的 prefixMatch 属性规定，跳过继续进行下一个格式级别标记〈level〉的 prefixMatch 属性判断。

第 2.2 步 找出所有的“待选输入格式”。

第 2.3 步 如果遍历后没有发现任何“待选输入格式”，报错退出。

第 3 步，查找匹配输入数据的选项〈option〉：

第 3.1 步 对上述找出的所有“待选输入格式”，判断：

- a) 如果该待选输入格式的编码结构元素〈scheme〉中指明了 optionKey 属性的含义（如 gslcompanyprefixlength），且字段数据表中附加参数部分包含相同含义的值（如 gslcompanyprefixlength），如果字段数据表中的值和 optionKey 属性的值相同，将该“待选

1) 请注意，在 TDT 标记文件中 prefixMatch 属性提供了一种优化的方法来判断输入格式，这种方法明显比正则表达式要效率更高。在设计标签数据转换软件时，如果一个格式既包含 prefixMatch 又包含正则表达式：如果输入数据不以 prefixMatch 开头，就可以认为该格式不匹配输入格式，继续下一个格式来验证；如果输入数据确实是以 prefixMatch 开头的，那还要验证正则表达式才能够判断具体是否匹配。

另外，TDT 标记文件中的 prefixMatch 属性应该是一个实际固定的字符串，而不应该是一个正则表达式，这样也是因为很多编程语言都具有字符串起始检查函数（如 startsWith），这个函数比正则表达式效率高很多。软件设计者如果愿意使用正则表达式来实现起始检查功能，完全可以根据 prefixMatch 的值自己构造正则表达式，但请注意其语法（如用 ‘.’ 表示开头）和字符转义（如用 ‘\\.’ 来转义 ‘.’）。

的编码模式+格式级别+匹配的选项”记为一个匹配输入选项(编码模式+格式级别+匹配的选项的组合);否则,继续往下寻找。

示例:

如果<scheme>元素有下列属性定义:optionKey="gs1companyprefixlength",并且字段数据表中有如如下“键-值”:gs1companyprefixlength=7,而只有属性值为 optionKey="7"的选项元素<option>,那么 optionKey="7"的选项(以及其编码模式和格式)就是一个匹配输入选项。

- b) 如果该待选输入格式的编码结构元素<scheme>中没有 optionKey 属性或 optionKey 属性的值不在字段数据表的名称列表中,则需要遍历该待选输入格式的所有选项,且使用每个选项元素<option>中 pattern 属性所规定的正则表达式检查其是否与输入数据相匹配,如果匹配,记为匹配输入选项;否则,继续往下寻找。

第 3.2 步 每个输入数据在每个待选的输入格式下应仅有一个匹配输入选项。如果找到多个匹配输入选项一般是由于存在多个待选输入格式,此情况下:如果输入格式和输出格式都是高于标签编码 URI 格式级别的格式,使用任一格式进行转换都可得到正确结果,即,可以选择第一个待选输入格式以及其匹配选项作为最终确定的输入格式和匹配选项;如果输入格式是标签编码 URI 格式或二进制格式,出现多个待选输入格式,优先使用编码模式<scheme>设置含 tagLength 属性且与输入数据 tagLength 相一致的作为最终确定的输入格式和匹配选项;如果输出格式是标签编码 URI 格式级别或者二进制格式级别,必须设置附加参数 tagLength。

第 3.3 步 记录最终确定的匹配选项元素<option>属性 optionKey 的值,该值在第 6 步中还会使用。

第 4 步 根据最终确定输入格式的匹配选项元素<option>的模式解析输入数据,并获得相应数据字段:

第 4.1 步 验证匹配选项元素<option>的 pattern 属性所规定的正则表达式是否与输入数据完全匹配(前面以及验证的可以使用前面检查的结果):如果完全匹配,开始考虑该选项元素<option>下所定义的各个字段元素<field>;否则,继续查找下一个待选的输入格式的匹配选项。

第 4.2 步 使用选项元素<option>中 pattern 属性所规定的正则表达式对输入数据进行匹配。在该匹配过程中,会有一堆使用正则表达式反向引用语法(backreference)解析出的匹配组,这些匹配组的值会逐个赋给各原始数据字段(由选项元素<option>下的各字段元素<field>定义):

- a) 使用上述获得各匹配组,每个原始数据字段用一个字段元素<field>定义,每个<field>元素包含了一个 seq 属性指明该数据字段对应的匹配组,其中 seq="1"代表匹配组 1,seq="2"代表匹配组 2,以此类推,完成赋值过程。
- b) 当完成上述赋值后:
 - 1) 如果<field>元素的 compaction 属性没有定义或定义值为“空”,将该字段的值作为整数处理(数值型字段),否则将其作为字符串处理(文本型字段)。
 - 2) 如果输入格式是二进制格式(<level>元素的 type 属性是 BINARY),将正则表达式匹配所获得的 0-1 字符串当作一个二进制字符串处理。如果该字段是数值型字段,将上述二进制字符串转换为整数;如果是文本型字段,将其按 compaction 属性的规定转换为字符串。二进制转换为整数或字符串时进行的必要剔除(反填充)和转换之后的必要填充见图 9。
- c) 如果是文本型字段且<field>元素定义了 characterSet 属性,检查该字段字符串是否符合该属性所规定的字符集合。

d) 如果是数值型字段:

- 1) 如果<field>元素设置了 decimalMinimum 属性,检查该字段整数值是否大于等于 decimalMinimum。
- 2) 如果<field>元素设置了 decimalMaximum 属性,检查该字段整数值是否小于等于 decimalMaximum。

第 4.3 步 将上述原始数据字段其 name(名称)属性和获得的值,按照名称-值对的方式存储到字段数据表以做后用。

第 5 步,执行解析阶段(EXTRACT)的各规则,获取派生数据字段的值:

第 5.1 步 在通过输入数据解析出相应的原始数据字段后,按处于解析阶段(EXTRACT)各规则元素<rule>定义的规则,依次(按照 seq 属性的值)执行各项操作,产生派生数据字段。

第 5.2 步 在计算派生字段过程中,进行必要的剔除(反填充)和转换之后必要的填充,并检查其结果是否符合 characterSet 属性、decimalMinimum 属性、decimalMaximum 属性的规定。

第 5.3 步 在完全无错的情况下,将这些派生数据字段的名称-值对存入字段数据表中。

第 6 步,查找输出格式中对应的选项

第 6.1 步 在与输入格式相同的编码结构<scheme>下,选择与第 1.3 步设置相同的格式级别元素<level>作为输出格式。

第 6.2 步 在该<level>元素下,找到与第 3 步确定的选项具有相同 optionKey 属性值的选项元素<option>作为输出格式所使用的输出选项。

第 7 步,执行格式化阶段(FORMAT)的各种规则,获取派生数据字段的值:

第 7.1 步 按处于格式化阶段(FORMAT)的各个规则元素<rule>所定义的规则,依次按 seq 属性的值执行各项操作,产生派生数据字段。为下面格式化输出结果做好准备。

第 7.2 步 在计算派生字段过程中,进行必要的剔除(反填充)和转换后必要的填充,并检查其结果是否符合定义的 characterSet 属性或 decimalMinimum 属性、decimalMaximum 属性的规定。

第 7.3 步 在完全无错的情况下,将这些派生数据字段的名称和值存入字段数据表中。

第 8 步,使用字段数据表和定义的输出格式语法产生输出结果:

第 8.1 步 根据输出格式元素<level>的输出选项<option>中 grammar 属性所规定的语法产生输出结果。该语法符合扩展巴科斯范式 ABNF 语法的规定,其中:单引号包含的是常数值、没有单引号包含的是数据字段名称(即字段数据表中的名称或键)。

第 8.2 步 在字段数据表中查找上述所有数据字段名称所对应的值。如果输出格式是二进制格式,应将上述值从十进制整数或者字符串转换为二进制,且进行必要的剔除(反填充)和转换后必要的填充。规则见图 8。

第 8.3 步 按照 grammar 属性所规定的语法,将上述查找到的(符合输出格式的)值和(去掉单引号的)常数值按 grammar 规定的顺序进行组合(组合过程中忽略 grammar 属性中的空格),最终将组合结果作为输出结果与输出数据输出。

8.2.2.2 转换接口

对于采用不同的编程语言,标签数据转换接口的定义方法可能会有区别,表 17 给出了一些典型的标签数据转换接口的定义方式。

表 17 标签数据转换接口的定义

编程语言	建议的接口定义方式
C++ (STL)	public: string translate(string epcIdentifier, string parameterList, string outputFormat);
C++ .Net	public: String^ translate(String^ epcIdentifier, String^ parameterList, String^ outputFormat);
Visual Basic	Public Function translate(Byval epcIdentifier As String, Byval parameterList As String, Byval Public outputFormat As String)
C#	public String translate(String epcIdentifier, String parameterList, String outputFormat);
Java	public String translate(String epcIdentifier, String parameterList, String outputFormat);
Perl	sub refreshTranslations { my (\$ epcIdentifier, \$ parameterList, \$ outputFormat) = @_; ... }
Python	def translate(epcIdentifier, parameterList , outputFormat)
说明	
将输入数据 epcIdentifier 从一种格式(输入格式,自动检测)转换为相同编码结构下的另一种格式(输出格式)	
参数	
参数名称	说明
epcIdentifier	输入数据。 字符串格式,该字符串应符合 TDT 元素文件的相应 pattern 属性的规定,是二进制格式、标签 编码 URI 格式、纯标识 URI 格式、GS1 数据传输字符串格式、GS1 应用标识符字符串格式、 GS1 文本数据字段格式之一,参见表 1
parameterList	附加参数。 字符串格式。采用名称-值对方式(名称=值),每个名称-值对之间用英文分号“;”分割,例如: filter=3;companyprefixlength=7;tagLength=96
outputFormat	输出格式。字符串,取值范围: 1.BINARY - 二进制格式 2.TAG_ENCODING - 标签编码 URI 格式 3.PURE_IDENTITY - 纯标识 URI 格式 4.ELEMENT_STRING - GS1 数据传输字符串格式 5.LEGACY_AI - GS1 应用标识符字符串格式 6.LEGACY - GS1 文本数据字段格式 7.ONS_HOSTNAME - ONS 域名格式
返回值	
经过标签数据转换过程返回的符合输出格式的输出结果字符串	
抛出的异常或返回的错误	
错误/异常代码	说明
101	TDTFileNotFound 没有找到相应的配置文件(TDT 元素文件)
102	TDTFieldBelowMinimum 数值型字段超出 decimalMinimum 的规定

表 17（续）

编程语言	建议的接口定义方式
错误/异常代码	说明
103	TDTFieldAboveMaximum 数值型字段超出 decimalMaximum 的规定
104	TDTFieldOutsideCharacterSet 字段字符不符合 characterSet 的规定
105	TDTUndefinedField 在转换过程中存在未定义或者无法获得的数据字段
106	TDTSchemeNotFound 没有匹配输入数据的编码结构(主要通过 prefixMatch 属性)
107	TDTLevelNotFound 没有匹配输入数据的格式(主要通过 prefixMatch 属性)
108	TDTOptionNotFound 没有匹配输入数据的选项(主要通过 optionKey 属性或者 pattern 属性)
109	TDTLookupFailed 外部表格未查找到相应的数据,并给出表的 URI 和路径
110	TDTNumericOverflow 数字溢出,比如在处理系列号等数值型发生溢出错误



附 录 A

(规范性附录)

TDT 标记文件

A.1 SGTIN-64

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="SGTIN-64" optionKey="gslcompanyprefixlength" tagLength="64">
    <level type="BINARY" prefixMatch="10" requiredFormattingParameters="filter,taglength">
      <option optionKey="12" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})" gram-
mar="'10' filter gslcompanyprefixindex itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet =
"[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]*"
bitPadDir="LEFT" bitLength="20" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[01]*" bitPadDir="LEFT" bitLength="25" name="serial"/>
      </option>
      <option optionKey="11" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})" gram-
mar="'10' filter gslcompanyprefixindex itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet =
"[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]*"
bitPadDir="LEFT" bitLength="20" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[01]*" bitPadDir="LEFT" bitLength="25" name="serial"/>
      </option>
      <option optionKey="10" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})" gram-
mar="'10' filter gslcompanyprefixindex itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet =
[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet =

```

```

[01]* " bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[01]* " bitPadDir="LEFT" bitLength="25" name="serial"/>
    </option>
    <option optionKey="9" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="" 10' filter gslcompanyprefixindex itemref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[01]* " bitPadDir="LEFT" bitLength="25" name="serial"/>
    </option>
    <option optionKey="8" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="" 10' filter gslcompanyprefixindex itemref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[01]* " bitPadDir="LEFT" bitLength="25" name="serial"/>
    </option>
    <option optionKey="7" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="" 10' filter gslcompanyprefixindex itemref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[01]* " bitPadDir="LEFT" bitLength="25" name="serial"/>
    </option>
    <option optionKey="6" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="" 10' filter gslcompanyprefixindex itemref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>

```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[01]* " bitPadDir="LEFT" bitLength="25" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="TABLELOOKUP(gslcompanyprefixindex,
tdt64bitcpi, gslcompanyprefixindex, gslcompanyprefix)" tableURL="http://www.onsepc.com/
ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@index=' $ 1 ']/@companyPrefix"
tableParams="gslcompanyprefixindex"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
gslcompanyprefixlength" characterSet="[0-9]*" function="LENGTH(gslcompanyprefix)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="
gslcompanyprefixindex" characterSet="[0-9]*" function="TABLELOOKUP(gslcompanyprefix,
tdt64bitcpi, gslcompanyprefix, gslcompanyprefixindex)" tableURL="http://www.onsepc.com/
ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@companyPrefix=' $ 1 ']/@index"
tableParams="gslcompanyprefix"/>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgtin-64" requiredFormattingPa-
rameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.[0-9]{12}\.[0-9]
{1})\.[0-9]{1,8})" grammar="urn:epc:tag:sgtin-64:' filter '.' gslcompanyprefix '.' itemref '.' seri-
al">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]* "
length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "
length="1" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="11" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.[0-9]{11}\.[0-9]
{2})\.[0-9]{1,8})" grammar="urn:epc:tag:sgtin-64:' filter '.' gslcompanyprefix '.' itemref '.' seri-
al">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]* " length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>

```



```

</option>
<option optionKey="10" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\,([0-9]{10})\,([0-9]{3})\,([0-9]{1,8})" grammar="" urn:epc:tag:sgtin-64:' filter '!' gslcompanyprefix '!' itemref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\,([0-9]{9})\,([0-9]{4})\,([0-9]{1,8})" grammar="" urn:epc:tag:sgtin-64:' filter '!' gslcompanyprefix '!' itemref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\,([0-9]{8})\,([0-9]{5})\,([0-9]{1,8})" grammar="" urn:epc:tag:sgtin-64:' filter '!' gslcompanyprefix '!' itemref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\,([0-9]{7})\,([0-9]{6})\,([0-9]{1,8})" grammar="" urn:epc:tag:sgtin-64:' filter '!' gslcompanyprefix '!' itemref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="

```



```

[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="6" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.[0-9]{6})\.[0-9]{7})\.[0-9]{1,8})" grammar="urn:epc:tag:sgtin-64:' filter!' gslcompanyprefix!' itemref!' serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgtin">
    <option optionKey="12" pattern="urn:epc:id:sgtin:([0-9]{12})\.[0-9]{1})\.[0-9]{1,8})" grammar="urn:epc:id:sgtin:' gslcompanyprefix!' itemref!' serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "
length="1" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:id:sgtin:([0-9]{11})\.[0-9]{2})\.[0-9]{1,8})" grammar="urn:epc:id:sgtin:' gslcompanyprefix!' itemref!' serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="
[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]* "
length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:id:sgtin:([0-9]{10})\.[0-9]{3})\.[0-9]{1,8})" grammar="urn:epc:id:sgtin:' gslcompanyprefix!' itemref!' serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="
[0-9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]* "

```

```

9]* " length="3" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="9" pattern="urn:epc:id:sgtin:([0-9]{9})\.[0-9]{4}\.([0-9]{1,
8})" grammar="urn:epc:id:sgtin:'gslcompanyprefix' itemref 'serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
="[0-9]* " length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]* " length="4" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="8" pattern="urn:epc:id:sgtin:([0-9]{8})\.[0-9]{5}\.([0-9]{1,
8})" grammar="urn:epc:id:sgtin:'gslcompanyprefix' itemref 'serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]* " length="5" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="7" pattern="urn:epc:id:sgtin:([0-9]{7})\.[0-9]{6}\.([0-9]{1,
8})" grammar="urn:epc:id:sgtin:'gslcompanyprefix' itemref 'serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="6" pattern="urn:epc:id:sgtin:([0-9]{6})\.[0-9]{7}\.([0-9]{1,
8})" grammar="urn:epc:id:sgtin:'gslcompanyprefix' itemref 'serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="1048575" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
    </option>
</level>
< level type = " LEGACY " prefixMatch = " gtin = " requiredParsingParameters = "
gslcompanyprefixlength">

```

```

    <option optionKey="12" pattern="gtin=([0-9]{14});serial=([0-9]{1,8})" grammar="
gtin='indicator digit gsl company prefix item reference remainder check digit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="gtin=([0-9]{14});serial=([0-9]{1,8})" grammar="
gtin='indicator digit gsl company prefix item reference remainder check digit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="gtin=([0-9]{14});serial=([0-9]{1,8})" grammar="
gtin='indicator digit gsl company prefix item reference remainder check digit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="gtin=([0-9]{14});serial=([0-9]{1,8})" grammar="
gtin='indicator digit gsl company prefix item reference remainder check digit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="gtin=([0-9]{14});serial=([0-9]{1,8})" grammar="
gtin='indicator digit gsl company prefix item reference remainder check digit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="gtin=([0-9]{14});serial=([0-9]{1,8})" grammar="
gtin='indicator digit gsl company prefix item reference remainder check digit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="gtin=([0-9]{14});serial=([0-9]{1,8})" grammar="

```

```

gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicatordigit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref" characterSet="[0-9]*" function="CONCAT(indicatordigit,itemrefremainder)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,0,gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicatordigit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix" characterSet="[0-9]*" length="13" function="CONCAT(indicatordigit,gslcompanyprefix,itemrefremainder)"/>
<rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit" characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
</level>
<level type="LEGACY_AI" prefixMatch="(01)" requiredParsingParameters="gslcompanyprefixlength">
    <option optionKey="12" pattern="\ (01\)([0-9]{14})\ (21\)([0-9]{1,8})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="\ (01\)([0-9]{14})\ (21\)([0-9]{1,8})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
    </option>

```

```

</option>
<option optionKey="10" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,8})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,8})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,8})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,8})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,8})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicatordigit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemrefremainder" characterSet="[0-9]*" length="1" function="SUBSTR(gtinprefixremainder,gslcompanyprefixlength)"/>

```

```

    <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref"
characterSet="[0-9]*" function="CONCAT(indicator digit,itemrefremainder)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemre-
fremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix"
characterSet="[0-9]*" length="13" function="CONCAT(indicator digit,gslcompanyprefix,itemre-
fremainder)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
  </level>
  <level type="LEGACY _ AI" prefixMatch="01" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="01([0-9]{14})21([0-9]{1,8})" grammar="'01' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="01([0-9]{14})21([0-9]{1,8})" grammar="'01' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="01([0-9]{14})21([0-9]{1,8})" grammar="'01' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="01([0-9]{14})21([0-9]{1,8})" grammar="'01' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="

```

```

[0-9]* " name="serial"/>
  </option>
  <option optionKey="8" pattern="01([0-9]{14})21([0-9]{1,8})" grammar="' 01 ' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="7" pattern="01([0-9]{14})21([0-9]{1,8})" grammar="' 01 ' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="6" pattern="01([0-9]{14})21([0-9]{1,8})" grammar="' 01 ' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="
[0-9]* " name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinpre-
fixremainder" characterSet="[0-9]* " length="12" function="SUBSTR(gtin,1,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicator-
digit" characterSet="[0-9]* " length="1" function="SUBSTR(gtin,0,1)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemre-
fremainder " characterSet = " [ 0-9 ] * " function = " SUBSTR ( gtinprefixremainder,
gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref"
characterSet="[0-9]* " function="CONCAT(indicatordigit,itemrefremainder)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="
gslcompanyprefix" characterSet = " [ 0-9 ] * " function = " SUBSTR ( gtinprefixremainder, 0,
gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator-
digit" characterSet="[0-9]* " length="1" function="SUBSTR(itemref,0,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemre-
fremainder" characterSet="[0-9]* " function="SUBSTR(itemref,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix"
characterSet="[0-9]* " length="13" function="CONCAT(indicatordigit,gslcompanyprefix,itemre-
fremainder)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"

```



```

characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
</level>
<level type="ONS_HOSTNAME">
  <option optionKey="12" grammar="itemref"! gs1companyprefix'.sgtin.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="11" grammar="itemref"! gs1companyprefix'.sgtin.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="10" grammar="itemref"! gs1companyprefix'.sgtin.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="9" grammar="itemref"! gs1companyprefix'.sgtin.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="8" grammar="itemref"! gs1companyprefix'.sgtin.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="7" grammar="itemref"! gs1companyprefix'.sgtin.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="6" grammar="itemref"! gs1companyprefix'.sgtin.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="

```



```
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
</level>
</scheme>
</tdt:epcTagDataTranslation>
```

A.2 SGTIN-96

```
<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="SGTIN-96" optionKey="gs1companyprefixlength" tagLength="96">
    <level type="BINARY" prefixMatch="00110000" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00110000([01]{3})000([01]{40})([01]{4})([01]
{38})" grammar="'00110000' filter '000' gs1companyprefix itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[01]* " bitPadDir="LEFT" bitLength="40" name="gs1companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
      </option>
      <option optionKey="11" pattern="00110000([01]{3})001([01]{37})([01]{7})([01]
{38})" grammar="'00110000' filter '001' gs1companyprefix itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="37" name="gs1companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="7" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
      </option>
      <option optionKey="10" pattern="00110000([01]{3})010([01]{34})([01]{10})([01]
{38})" grammar="'00110000' filter '010' gs1companyprefix itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="34" name="gs1companyprefix"/>
```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]*" bitPadDir="LEFT" bitLength="10" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*" bitPadDir="LEFT" bitLength="38" name="serial"/>
  </option>
  <option optionKey="9" pattern="00110000([01]{3})011([01]{30})([01]{14})([01]{38})" grammar="'00110000' filter '011' gslcompanyprefix itemref serial"/>
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*" bitPadDir="LEFT" bitLength="38" name="serial"/>
  </option>
  <option optionKey="8" pattern="00110000([01]{3})100([01]{27})([01]{17})([01]{38})" grammar="'00110000' filter '100' gslcompanyprefix itemref serial"/>
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*" bitPadDir="LEFT" bitLength="17" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*" bitPadDir="LEFT" bitLength="38" name="serial"/>
  </option>
  <option optionKey="7" pattern="00110000([01]{3})101([01]{24})([01]{20})([01]{38})" grammar="'00110000' filter '101' gslcompanyprefix itemref serial"/>
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*" bitPadDir="LEFT" bitLength="38" name="serial"/>
  </option>
  <option optionKey="6" pattern="00110000([01]{3})110([01]{20})([01]{24})([01]{38})" grammar="'00110000' filter '110' gslcompanyprefix itemref serial"/>
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*" bitPadDir="LEFT" bitLength="38" name="serial"/>
  </option>

```

```

[01]* " bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="24" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
    </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgtin-96" requiredFormattingPa-
rameters="filter,taglength">
    <option optionKey="12" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{12})\.[0-9]
{1})\.[0-9]{1,12})" grammar="urn:epc:tag:sgtin-96:' filter '!' gslcompanyprefix '!' itemref '!' seri-
al">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "
length="1" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]* " name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{11})\.[0-9]
{2})\.[0-9]{1,12})" grammar="urn:epc:tag:sgtin-96:' filter '!' gslcompanyprefix '!' itemref '!' seri-
al">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]* " length="2" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]* " name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{10})\.[0-9]
{3})\.[0-9]{1,12})" grammar="urn:epc:tag:sgtin-96:' filter '!' gslcompanyprefix '!' itemref '!' seri-
al">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]* " length="3" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"

```

```

characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{9}\.[0-9]{4}\.[0-9]{1,12}" grammar="" urn:epc:tag:sgtin-96:' filter '.' gslcompanyprefix '.' itemref '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{8}\.[0-9]{5}\.[0-9]{1,12}" grammar="" urn:epc:tag:sgtin-96:' filter '.' gslcompanyprefix '.' itemref '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{7}\.[0-9]{6}\.[0-9]{1,12}" grammar="" urn:epc:tag:sgtin-96:' filter '.' gslcompanyprefix '.' itemref '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{6}\.[0-9]{7}\.[0-9]{1,12}" grammar="" urn:epc:tag:sgtin-96:' filter '.' gslcompanyprefix '.' itemref '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgtin">
  <option optionKey="12" pattern="urn:epc:id:sgtin:([0-9]{12})\.[0-9]{1}\.[0-9]{1,
12})" grammar="urn:epc:id:sgtin:'gs1companyprefix' itemref 'serial'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="urn:epc:id:sgtin:([0-9]{11})\.[0-9]{2}\.[0-9]{1,
12})" grammar="urn:epc:id:sgtin:'gs1companyprefix' itemref 'serial'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="urn:epc:id:sgtin:([0-9]{10})\.[0-9]{3}\.[0-9]{1,
12})" grammar="urn:epc:id:sgtin:'gs1companyprefix' itemref 'serial'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="urn:epc:id:sgtin:([0-9]{9})\.[0-9]{4}\.[0-9]{1,
12})" grammar="urn:epc:id:sgtin:'gs1companyprefix' itemref 'serial'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943"

```

```

characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="urn:epc:id:sgtin:([0-9]{8})\.[0-9]{5}\.[0-9]{1,12})" grammar="urn:epc:id:sgtin:' gslcompanyprefix ' itemref ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="urn:epc:id:sgtin:([0-9]{7})\.[0-9]{6}\.[0-9]{1,12})" grammar="urn:epc:id:sgtin:' gslcompanyprefix ' itemref ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="urn:epc:id:sgtin:([0-9]{6})\.[0-9]{7}\.[0-9]{1,12})" grammar="urn:epc:id:sgtin:' gslcompanyprefix ' itemref ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
</option>
</level>
<level type="LEGACY" prefixMatch="gtin" requiredParsingParameters="gslcompanyprefixlength">
    <option optionKey="12" pattern="gtin=([0-9]{14});serial=([0-9]{1,12})" grammar="gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ' ;serial=' serial'"
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="gtin=([0-9]{14});serial=([0-9]{1,12})" grammar="gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ' ;serial=' serial'"
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="gtin=( $[0-9]\{14\}$ );serial=( $[0-9]\{1,12\}$ )" grammar=
="gtin='indicatordigit gslcompanyprefix itemrefremainder checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="gtin=( $[0-9]\{14\}$ );serial=( $[0-9]\{1,12\}$ )" grammar="
gtin='indicatordigit gslcompanyprefix itemrefremainder checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="gtin=( $[0-9]\{14\}$ );serial=( $[0-9]\{1,12\}$ )" grammar="
gtin='indicatordigit gslcompanyprefix itemrefremainder checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="gtin=( $[0-9]\{14\}$ );serial=( $[0-9]\{1,12\}$ )" grammar="
gtin='indicatordigit gslcompanyprefix itemrefremainder checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="gtin=( $[0-9]\{14\}$ );serial=( $[0-9]\{1,12\}$ )" grammar="
gtin='indicatordigit gslcompanyprefix itemrefremainder checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicator-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemre-

```



```

fremainder " characterSet = " [ 0-9 ]* " function = " SUBSTR ( gtinprefixremainder,
gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref"
characterSet="[0-9]*" function="CONCAT(indicatordigit,itemrefremainder)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="
gslcompanyprefix" characterSet=" [ 0-9 ]* " function = " SUBSTR ( gtinprefixremainder, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemre-
fremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix"
characterSet="[0-9]*" length="13" function="CONCAT(indicatordigit,gslcompanyprefix,itemre-
fremainder)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
</level>
    <level type="LEGACY _ AI" prefixMatch="(01)" requiredParsingParameters="
gslcompanyprefixlength">
        <option optionKey="12" pattern="\ (01\)([0-9]{14})\ (21\)([0-9]{1,12})" grammar="
(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
        </option>
        <option optionKey="11" pattern="\ (01\)([0-9]{14})\ (21\)([0-9]{1,12})" grammar="
(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
        </option>
        <option optionKey="10" pattern="\ (01\)([0-9]{14})\ (21\)([0-9]{1,12})" grammar="
(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
        </option>
        <option optionKey="9" pattern="\ (01\)([0-9]{14})\ (21\)([0-9]{1,12})" grammar="
(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-

```

```

terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,12})" grammar="(01)'
indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,12})" grammar="(01)'
indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="\(01\)([0-9]{14})\(21\)([0-9]{1,12})" grammar="(01)'
indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicator-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
itemrefremainder" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,
gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref"
characterSet="[0-9]*" function="CONCAT(indicatordigit,itemrefremainder)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder, 0,
gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemre-
fremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix"
characterSet="[0-9]*" length="13" function="CONCAT(indicatordigit,gslcompanyprefix,itemre-

```

```

fremainder)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
  </level>
  <level type="ELEMENT _ STRING" prefixMatch="01" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="01([0-9]{14})21([0-9]{1,12})" grammar="'01' indi-
catordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="01([0-9]{14})21([0-9]{1,12})" grammar="'01' indi-
catordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="01([0-9]{14})21([0-9]{1,12})" grammar="'01' indi-
catordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="01([0-9]{14})21([0-9]{1,12})" grammar="'01' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="8" pattern="01([0-9]{14})21([0-9]{1,12})" grammar="'01' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="7" pattern="01([0-9]{14})21([0-9]{1,12})" grammar="'01' indica-
tordigit gslcompanyprefix itemrefremainder checkdigit '21' serial">

```

```

    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="01([0-9]{14})21([0-9]{1,12})" grammar="01'indicator-digit gslcompanyprefix itemrefremainder checkdigit'21'serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicator-digit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref" characterSet="[0-9]*" function="CONCAT(indicator-digit,itemrefremainder)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,0,gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator-digit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix" characterSet="[0-9]*" length="13" function="CONCAT(indicator-digit,gslcompanyprefix,itemrefremainder)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit" characterSet="[0-9]*" length="1" function="GSLCHECKSUM(gtinprefix)"/>
</level>
<level type="ONS_HOSTNAME">
  <option optionKey="12" grammar="itemref'!gslcompanyprefix'.sgtin.id.onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*" length="1" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="11" grammar="itemref'!gslcompanyprefix'.sgtin.id.onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="10" grammar="itemref'!gslcompanyprefix'.sgtin.id.onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
</level>
</level>

```

```

9]* " length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet=
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="10" grammar="itemref '! gs1companyprefix '.sgtin.id.onsepc.com '">
        <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]* " length="3" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]* " length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        </option>
        <option optionKey="9" grammar="itemref '! gs1companyprefix '.sgtin.id.onsepc.com '">
            <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]* " length="4" padChar="0" padDir="LEFT" name="itemref"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]* " length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
            </option>
            <option optionKey="8" grammar="itemref '! gs1companyprefix '.sgtin.id.onsepc.com '">
                <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]* " length="5" padChar="0" padDir="LEFT" name="itemref"/>
                <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[0-9]* " length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
                </option>
                <option optionKey="7" grammar="itemref '! gs1companyprefix '.sgtin.id.onsepc.com '">
                    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="itemref"/>
                    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
                    </option>
                    <option optionKey="6" grammar="itemref '! gs1companyprefix '.sgtin.id.onsepc.com '">
                        <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="itemref"/>
                        <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
                        </option>
                    </level>
                </scheme>
            </tdt:epcTagDataTranslation>

```

A.3 SGTIN-198

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"

```

```

xmlns:tdt = "urn:epcglobal:tdt:xsd:1" xmlns:xsi = "http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="SGTIN-198" optionKey="gs1companyprefixlength" tagLength="198">
    <level type="BINARY" prefixMatch="00110110" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00110110([01]{3})000([01]{40})([01]{4})([01]
{140})" grammar="00110110' filter '000' gs1companyprefix itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="40" name="gs1companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="itemref"/>
        <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
      </option>
      <option optionKey="11" pattern="00110110([01]{3})001([01]{37})([01]{7})([01]
{140})" grammar="00110110' filter '001' gs1companyprefix itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="37" name="gs1companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="7" name="itemref"/>
        <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
      </option>
      <option optionKey="10" pattern="00110110([01]{3})010([01]{34})([01]{10})([01]
{140})" grammar="00110110' filter '010' gs1companyprefix itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="34" name="gs1companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="10" name="itemref"/>
        <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
      </option>
      <option optionKey="9" pattern="00110110([01]{3})011([01]{30})([01]{14})([01]
{140})" grammar="00110110' filter '011' gs1companyprefix itemref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet

```

```

="[01]*" bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="14" name="itemref"/>
    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
  <option optionKey="8" pattern="00110110([01]{3})100([01]{27})([01]{17})([01]
{140})" grammar="'00110110' filter '100' gslcompanyprefix itemref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="17" name="itemref"/>
    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
  <option optionKey="7" pattern="00110110([01]{3})101([01]{24})([01]{20})([01]
{140})" grammar="'00110110' filter '101' gslcompanyprefix itemref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="20" name="itemref"/>
    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
  <option optionKey="6" pattern="00110110([01]{3})110([01]{20})([01]{24})([01]
{140})" grammar="'00110110' filter '110' gslcompanyprefix itemref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="24" name="itemref"/>
    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgtin-198" requiredFormattingPa-
rameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:sgtin-198:([0-7]{1})\.[0-9]{12})\.[0-

```



```

9]{1})\.(?!%[-? A-Z _a-z \ x22]{1,20})" grammar = "' urn:epc:tag:sgtin-198:' filter '!'  

gs1companyprefix".itemref".serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "  

length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]* "  

length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "  

length="1" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" characterSet="(?!%[-? A-Z _a-z \ x22])* " name="serial"/>
</option>
<option optionKey="11" pattern="urn:epc:tag:sgtin-198:([0-7]{1})\.( [0-9]{11})\.( [0-9]{2})\.  

(?!%[-? A-Z _a-z \ x22]{1,20})" grammar = "' urn:epc:tag:sgtin-198:' filter '!'  

gs1companyprefix".itemref".serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "  

length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]* "  

length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]* "  

length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" characterSet="(?!%[-? A-Z _a-z \ x22])* " name="serial"/>
</option>
<option optionKey="10" pattern="urn:epc:tag:sgtin-198:([0-7]{1})\.( [0-9]{10})\.( [0-9]{3})\.  

(?!%[-? A-Z _a-z \ x22]{1,20})" grammar = "' urn:epc:tag:sgtin-198:' filter '!'  

gs1companyprefix".itemref".serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "  

length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]* "  

length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]* "  

length="3" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" characterSet="(?!%[-? A-Z _a-z \ x22])* " name="serial"/>
</option>
<option optionKey="9" pattern="urn:epc:tag:sgtin-198:([0-7]{1})\.( [0-9]{9})\.( [0-9]{4})\.  

(?!%[-? A-Z _a-z \ x22]{1,20})" grammar = "' urn:epc:tag:sgtin-198:' filter '!'  

gs1companyprefix".itemref".serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "  

length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]* "  

length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]* "  

length="4" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="4" characterSet="(?!%[-? A-Z _a-z \ x22])* " name="serial"/>
</option>

```

```

    <option optionKey="8" pattern="urn:epc:tag:sgtin-198:([0-7]{1})\.[0-9]{8})\.[0-9]{5})\.[0-9]{1,20}" grammar="urn:epc:tag:sgtin-198:'filter'.gs1companyprefix.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="4" characterSet="[0-9]{1,20}" name="serial"/>
    </option>
    <option optionKey="7" pattern="urn:epc:tag:sgtin-198:([0-7]{1})\.[0-9]{7})\.[0-9]{6})\.[0-9]{1,20}" grammar="urn:epc:tag:sgtin-198:'filter'.gs1companyprefix.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="4" characterSet="[0-9]{1,20}" name="serial"/>
    </option>
    <option optionKey="6" pattern="urn:epc:tag:sgtin-198:([0-7]{1})\.[0-9]{6})\.[0-9]{7})\.[0-9]{1,20}" grammar="urn:epc:tag:sgtin-198:'filter'.gs1companyprefix.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="4" characterSet="[0-9]{1,20}" name="serial"/>
    </option>
  </level>
  <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgtin">
    <option optionKey="12" pattern="urn:epc:id:sgtin:([0-9]{12})\.[0-9]{1})\.[0-9]{1,20}" grammar="urn:epc:id:sgtin:'gs1companyprefix'.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*" length="1" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="3" characterSet="[0-9]{1,20}" name="serial"/>
    </option>
  </level>

```

```

    <option optionKey="11" pattern="urn:epc:id:sgtin:([0-9]{11})\.[0-9]{2}\.([!%?-?
A-Z_a-z\x22]{1,20})" grammar="urn:epc:id:sgtin:'gslcompanyprefix'.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="([0-9])*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="([0-
9])*" length="2" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="3" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
    </option>

    <option optionKey="10" pattern="urn:epc:id:sgtin:([0-9]{10})\.[0-9]{3}\.([!%?-?
A-Z_a-z\x22]{1,20})" grammar="urn:epc:id:sgtin:'gslcompanyprefix'.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="([0-9])*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="([0-
9])*" length="3" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="3" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
    </option>

    <option optionKey="9" pattern="urn:epc:id:sgtin:([0-9]{9})\.[0-9]{4}\.([!%?-? A-Z
_a-z\x22]{1,20})" grammar="urn:epc:id:sgtin:'gslcompanyprefix'.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="([0-9])*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="([0-
9])*" length="4" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="3" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
    </option>

    <option optionKey="8" pattern="urn:epc:id:sgtin:([0-9]{8})\.[0-9]{5}\.([!%?-? A-Z
_a-z\x22]{1,20})" grammar="urn:epc:id:sgtin:'gslcompanyprefix'.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="
([0-9])*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="([0-
9])*" length="5" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="3" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
    </option>

    <option optionKey="7" pattern="urn:epc:id:sgtin:([0-9]{7})\.[0-9]{6}\.([!%?-? A-Z
_a-z\x22]{1,20})" grammar="urn:epc:id:sgtin:'gslcompanyprefix'.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
([0-9])*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
([0-9])*" length="6" padChar="0" padDir="LEFT" name="itemref"/>
      <field seq="3" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
    </option>

    <option optionKey="6" pattern="urn:epc:id:sgtin:([0-9]{6})\.[0-9]{7}\.([!%?-? A-Z
_a-z\x22]{1,20})" grammar="urn:epc:id:sgtin:'gslcompanyprefix'.itemref'.serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="

```

```

[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="3" characterSet="[!%-? A-Z_a-z\x22]* " name="serial"/>
    </option>
</level>
< level type = " LEGACY " prefixMatch = " gtin = " requiredParsingParameters = "
gslcompanyprefixlength">
    <option optionKey="12" pattern="gtin=([0-9]{14});serial=([!%-? A-Z_a-z\x22]{1,20})"
grammar=" ' gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="serial"/>
    </option>
    <option optionKey="11" pattern="gtin=([0-9]{14});serial=([!%-? A-Z_a-z\x22]{1,20})"
grammar=" ' gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="serial"/>
    </option>
    <option optionKey="10" pattern="gtin=([0-9]{14});serial=([!%-? A-Z_a-z\x22]{1,20})"
grammar=" ' gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="serial"/>
    </option>
    <option optionKey="9" pattern="gtin=([0-9]{14});serial=([!%-? A-Z_a-z\x22]{1,20})"
grammar=" ' gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="serial"/>
    </option>
    <option optionKey="8" pattern="gtin=([0-9]{14});serial=([!%-? A-Z_a-z\x22]{1,20})"
grammar=" ' gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="serial"/>
    </option>
    <option optionKey="7" pattern="gtin=([0-9]{14});serial=([!%-? A-Z_a-z\x22]{1,20})"
grammar=" ' gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" charac-
terSet="[0-9]* " length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="serial"/>

```

```

</option>
<option optionKey="6" pattern="gtin=([0-9]{14});serial=([!%~? A-Z_a-z\x22]{1,20})"
grammar="\" gtin=' indicatordigit gslcompanyprefix itemrefremainder checkdigit ';serial=' serial\""
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
  <field seq="2" characterSet="([!%~? A-Z_a-z\x22])*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicatordigit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref" characterSet="[0-9]*" function="CONCAT(indicatordigit,itemrefremainder)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,0,gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicatordigit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix" characterSet="[0-9]*" length="13" function="CONCAT(indicatordigit,gslcompanyprefix,itemrefremainder)"/>
<rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit" characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
</level>
<level type="LEGACY_AI" prefixMatch="(01)" requiredParsingParameters="gslcompanyprefixlength">
  <option optionKey="12" pattern="\ (01\)([0-9]{14})\ (21\)([!%~? A-Z_a-z\x22]{1,20})" grammar="\" (01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial\""
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="([!%~? A-Z_a-z\x22])*" name="serial"/>
  </option>
  <option optionKey="11" pattern="\ (01\)([0-9]{14})\ (21\)([!%~? A-Z_a-z\x22]{1,20})" grammar="\" (01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial\""
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="([!%~? A-Z_a-z\x22])*" name="serial"/>
  </option>

```

```

    <option optionKey="10" pattern="\ (01\)\ ([0-9]{14})\ (21\)\ ([!%?-? A-Z_a-z\x22]{1,
20})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="\ (01\)\ ([0-9]{14})\ (21\)\ ([!%?-? A-Z_a-z\x22]{1,
20})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="8" pattern="\ (01\)\ ([0-9]{14})\ (21\)\ ([!%?-? A-Z_a-z\x22]{1,
20})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="7" pattern="\ (01\)\ ([0-9]{14})\ (21\)\ ([!%?-? A-Z_a-z\x22]{1,
20})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="6" pattern="\ (01\)\ ([0-9]{14})\ (21\)\ ([!%?-? A-Z_a-z\x22]{1,
20})" grammar="(01)' indicatordigit gslcompanyprefix itemrefremainder checkdigit '(21)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicatordigit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref" characterSet="[0-9]*" function="CONCAT(indicatordigit,itemrefremainder)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,0,gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator-

```

```

digit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemre-
fremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix"
characterSet="[0-9]*" length="13" function="CONCAT(indicator digit,gs1company prefix,itemre-
fremainder)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
</level>
<level type="ELEMENT_STRING" prefixMatch="01" requiredParsingParameters="
gs1company prefix length">
  <option optionKey="12" pattern="01([0-9]{14})21([!%-? A-Z_a-z\\x22]{1,20})" gram-
mar="'01' indicator digit gs1company prefix itemref remainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="([!%-? A-Z_a-z\\x22])*" name="serial"/>
  </option>
  <option optionKey="11" pattern="01([0-9]{14})21([!%-? A-Z_a-z\\x22]{1,20})" gram-
mar="'01' indicator digit gs1company prefix itemref remainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="([!%-? A-Z_a-z\\x22])*" name="serial"/>
  </option>
  <option optionKey="10" pattern="01([0-9]{14})21([!%-? A-Z_a-z\\x22]{1,20})" gram-
mar="'01' indicator digit gs1company prefix itemref remainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="([!%-? A-Z_a-z\\x22])*" name="serial"/>
  </option>
  <option optionKey="9" pattern="01([0-9]{14})21([!%-? A-Z_a-z\\x22]{1,20})"
grammar="'01' indicator digit gs1company prefix itemref remainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="([!%-? A-Z_a-z\\x22])*" name="serial"/>
  </option>
  <option optionKey="8" pattern="01([0-9]{14})21([!%-? A-Z_a-z\\x22]{1,20})"
grammar="'01' indicator digit gs1company prefix itemref remainder checkdigit '21' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="([!%-? A-Z_a-z\\x22])*" name="serial"/>
  </option>
  <option optionKey="7" pattern="01([0-9]{14})21([!%-? A-Z_a-z\\x22]{1,20})"
grammar="'01' indicator digit gs1company prefix itemref remainder checkdigit '21' serial">

```



```

    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="01([0-9]{14})21([!%-? A-Z_a-z\x22]{1,20})"
grammar="01'indicator digit gs1company prefix itemref remainder checkdigit'21'serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicator digit" characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,gs1company prefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref" characterSet="[0-9]*" function="CONCAT(indicator digit,itemrefremainder)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="gs1company prefix" characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,0,gs1company prefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator digit" characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemrefremainder" characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix" characterSet="[0-9]*" length="13" function="CONCAT(indicator digit,gs1company prefix,itemrefremainder)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit" characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
</level>
<level type="ONS_HOSTNAME">
  <option optionKey="12" grammar="itemref'.gs1company prefix'.sgtin.id.onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*" length="1" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1company prefix"/>
  </option>
  <option optionKey="11" grammar="itemref'.gs1company prefix'.sgtin.id.onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="itemref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1company prefix"/>
  </option>

```

```

="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="10" grammar="itemref".gslcompanyprefix'.sgtin.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="itemref"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="9" grammar="itemref".gslcompanyprefix'.sgtin.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="itemref"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="8" grammar="itemref".gslcompanyprefix'.sgtin.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="itemref"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="7" grammar="itemref".gslcompanyprefix'.sgtin.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="itemref"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="6" grammar="itemref".gslcompanyprefix'.sgtin.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="itemref"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.4 SSCC-64

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="SSCC-64" optionKey="gslcompanyprefixlength" tagLength="64">

```

```

    <level type="BINARY" prefixMatch="00001000" requiredFormattingParameters="filter,tag-length">
        <option optionKey="12" pattern="00001000([01]{3})([01]{14})([01]{39})" grammar="00001000' filter gslcompanyprefixindex serialref">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*" bitPadDir="LEFT" bitLength="39" name="serialref"/>
        </option>
        <option optionKey="11" pattern="00001000([01]{3})([01]{14})([01]{39})" grammar="00001000' filter gslcompanyprefixindex serialref">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="39" name="serialref"/>
        </option>
        <option optionKey="10" pattern="00001000([01]{3})([01]{14})([01]{39})" grammar="00001000' filter gslcompanyprefixindex serialref">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="39" name="serialref"/>
        </option>
        <option optionKey="9" pattern="00001000([01]{3})([01]{14})([01]{39})" grammar="00001000' filter gslcompanyprefixindex serialref">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="39" name="serialref"/>
        </option>
        <option optionKey="8" pattern="00001000([01]{3})([01]{14})([01]{39})" grammar="00001000' filter gslcompanyprefixindex serialref">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"

```

```

[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet
   ="[01]* " bitPadDir="LEFT" bitLength="39" name="serialref"/>
    </option>
    <option optionKey="7" pattern="00001000([01]{3})([01]{14})([01]{39})" grammar="
    '00001000' filter gslcompanyprefixindex serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
        bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
        [01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet
       ="[01]* " bitPadDir="LEFT" bitLength="39" name="serialref"/>
        </option>
        <option optionKey="6" pattern="00001000([01]{3})([01]{14})([01]{39})" grammar="
        '00001000' filter gslcompanyprefixindex serialref">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
            bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
            [01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet
           ="[01]* " bitPadDir="LEFT" bitLength="39" name="serialref"/>
            </option>
            <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
            gslcompanyprefix" characterSet="[0-9]* " function="TABLELOOKUP(gslcompanyprefixindex,
            tdt64bitcpi, gslcompanyprefixindex, gslcompanyprefix)" tableURL="http://www.onsepc.com/
            ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
            tableParams="gslcompanyprefixindex"/>
            <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
            gslcompanyprefixlength" characterSet="[0-9]* " function="LENGTH(gslcompanyprefix)/>
            <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="
            gslcompanyprefixindex" characterSet="[0-9]* " function="TABLELOOKUP(gslcompanyprefix,
            tdt64bitcpi, gslcompanyprefix, gslcompanyprefixindex)" tableURL="http://www.onsepc.com/
            ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index"
            tableParams="gslcompanyprefix"/>
        </level>
        <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sscc-64" requiredFormattingPa-
        rameters="filter,taglength">
            <option optionKey="12" pattern="urn:epc:tag:sscc-64:([0-7]{1})\\.([0-9]{12})\\.([0-9]
            {5})" grammar="urn:epc:tag:sscc-64:' filter ".gslcompanyprefix ".serialref">
                <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
                length="1" padChar="0" padDir="LEFT" name="filter"/>
                <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
                characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>

```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="11" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{11})\.[0-9]{6})" grammar="urn:epc:tag:sscc-64:' filter ". gslcompanyprefix ". serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="10" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{10})\.[0-9]{7})" grammar="urn:epc:tag:sscc-64:' filter ". gslcompanyprefix ". serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="9" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{9})\.[0-9]{8})" grammar="urn:epc:tag:sscc-64:' filter ". gslcompanyprefix ". serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{8})\.[0-9]{9})" grammar="urn:epc:tag:sscc-64:' filter ". gslcompanyprefix ". serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{7})\.[0-9]{10})" grammar="urn:epc:tag:sscc-64:' filter ". gslcompanyprefix ". serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{6})\.[0-9]{11}" grammar="urn:epc:tag:sscc-64:' filter ".gslcompanyprefix ".serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sscc">
  <option optionKey="12" pattern="urn:epc:id:sscc:([0-9]{12})\.[0-9]{5}" grammar="urn:epc:id:sscc:' gslcompanyprefix ".serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="11" pattern="urn:epc:id:sscc:([0-9]{11})\.[0-9]{6}" grammar="urn:epc:id:sscc:' gslcompanyprefix ".serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="10" pattern="urn:epc:id:sscc:([0-9]{10})\.[0-9]{7}" grammar="urn:epc:id:sscc:' gslcompanyprefix ".serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="9" pattern="urn:epc:id:sscc:([0-9]{9})\.[0-9]{8}" grammar="urn:epc:id:sscc:' gslcompanyprefix ".serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="serialref"/>
  </option>

```

```

    </option>
    <option optionKey="8" pattern="urn:epc:id:sscc:([0-9]{8})\.([0-9]{9})" grammar="
urn:epc:id:sscc:'gslcompanyprefix'.serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="7" pattern="urn:epc:id:sscc:([0-9]{7})\.([0-9]{10})" grammar="
urn:epc:id:sscc:'gslcompanyprefix'.serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="6" pattern="urn:epc:id:sscc:([0-9]{6})\.([0-9]{11})" grammar="
urn:epc:id:sscc:'gslcompanyprefix'.serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
</level>
<level type="LEGACY" prefixMatch="sscc" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="sscc=([0-9]{18})" grammar="sscc='extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="11" pattern="sscc=([0-9]{18})" grammar="sscc='extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="10" pattern="sscc=([0-9]{18})" grammar="sscc='extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="9" pattern="sscc=([0-9]{18})" grammar="sscc='extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"

```



```

characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
</option>
<option optionKey="8" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
</option>
<option optionKey="7" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
</option>
<option optionKey="6" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="ssccpre-
fixremainder" characterSet="[0-9]*" length="16" function="SUBSTR(sscc,1,16)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(sscc,0,1)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serialre-
fremainder" characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,
gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="serialref"
characterSet="[0-9]*" function="CONCAT(extensiondigit,serialrefremainder)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder, 0,
gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(serialref,0,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="serialre-
fremainder" characterSet="[0-9]*" function="SUBSTR(serialref,1)"/>
<rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="ssccprefix"
characterSet="[0-9]*" length="17" function="CONCAT(extensiondigit,gslcompanyprefix,serial-
refremainder)"/>
<rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GSI_CHECKSUM(ssccprefix)"/>
</level>
<level type="LEGACY_AI" prefixMatch="(00)" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="\ (00\)([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">

```

```

    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="11" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="10" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="9" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="8" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="7" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="6" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="ssccpre-
fixremainder" characterSet="[0-9]*" length="16" function="SUBSTR(sscc,1,16)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(sscc,0,1)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serialre-
fremainder" characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,
gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="serialref"
characterSet="[0-9]*" function="CONCAT(extensiondigit,serialrefremainder)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="

```

```

gslcompanyprefix" characterSet = "[0-9]*" function = " SUBSTR ( ssccprefixremainder, 0,
gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(serialref,0,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="serialre-
fremainder" characterSet="[0-9]*" function="SUBSTR(serialref,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="ssccprefix"
characterSet="[0-9]*" length="17" function="CONCAT(extensiondigit,gslcompanyprefix,serial-
refremainder)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GSI CHECKSUM(ssccprefix)"/>
</level>
<level type="ELEMENT _ STRING" prefixMatch="00" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="00([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="11" pattern="00([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="10" pattern="00([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="9" pattern="00([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="8" pattern="00([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>
  <option optionKey="7" pattern="00([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
  </option>

```

```

</option>
  <option optionKey = "6" pattern = "00 ([0-9]{18})" grammar = "' 00 ' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "9999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "sscc"/>
  </option>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "1" newFieldName = "ssccpre-
fixremainder" characterSet = "[0-9]*" length = "16" function = "SUBSTR(sscc,1,16)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "2" newFieldName = "extension-
digit" characterSet = "[0-9]*" length = "1" function = "SUBSTR(sscc,0,1)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "3" newFieldName = "serialre-
fremainder" characterSet = "[0-9]*" function = "SUBSTR(ssccprefixremainder,
gslcompanyprefixlength)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "4" newFieldName = "serialref"
characterSet = "[0-9]*" function = "CONCAT(extensiondigit,serialrefremainder)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "5" newFieldName = "
gslcompanyprefix" characterSet = "[0-9]*" function = "SUBSTR(ssccprefixremainder, 0,
gslcompanyprefixlength)"/>
  <rule type = "FORMAT" inputFormat = "STRING" seq = "1" newFieldName = "extension-
digit" characterSet = "[0-9]*" length = "1" function = "SUBSTR(serialref,0,1)"/>
  <rule type = "FORMAT" inputFormat = "STRING" seq = "2" newFieldName = "serialre-
fremainder" characterSet = "[0-9]*" function = "SUBSTR(serialref,1)"/>
  <rule type = "FORMAT" inputFormat = "STRING" seq = "3" newFieldName = "ssccprefix"
characterSet = "[0-9]*" length = "17" function = "CONCAT(extensiondigit,gslcompanyprefix,serial-
refremainder)"/>
  <rule type = "FORMAT" inputFormat = "STRING" seq = "4" newFieldName = "checkdigit"
characterSet = "[0-9]*" length = "1" function = "GS1CHECKSUM(ssccprefix)"/>
</level>
<level type = "ONS_HOSTNAME">
  <option optionKey = "12" grammar = "gslcompanyprefix'.sscc.id.onsepc.com'">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999"
characterSet = "[0-9]*" length = "12" padChar = "0" padDir = "LEFT" name = "gslcompanyprefix"/>
  </option>
  <option optionKey = "11" grammar = "gslcompanyprefix'.sscc.id.onsepc.com'">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999" characterSet
= "[0-9]*" length = "11" padChar = "0" padDir = "LEFT" name = "gslcompanyprefix"/>
  </option>
  <option optionKey = "10" grammar = "gslcompanyprefix'.sscc.id.onsepc.com'">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999" characterSet
= "[0-9]*" length = "10" padChar = "0" padDir = "LEFT" name = "gslcompanyprefix"/>
  </option>
  <option optionKey = "9" grammar = "gslcompanyprefix'.sscc.id.onsepc.com'">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999" characterSet

```

```

="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="8" grammar="gslcompanyprefix'.sscc.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="7" grammar="gslcompanyprefix'.sscc.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="6" grammar="gslcompanyprefix'.sscc.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.5 SSCC-96

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="SSCC-96" optionKey="gslcompanyprefixlength" tagLength="96">
    <level type="BINARY" prefixMatch="00110001" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00110001([01]{3})000([01]{40})([01]{18})
000000000000000000000000000000000000" grammar="'00110001' filter '000' gslcompanyprefix serialref '
000000000000000000000000000000000000'">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[01]*" bitPadDir="LEFT" bitLength="40" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="18" name="serialref"/>
      </option>
      <option optionKey="11" pattern="00110001([01]{3})001([01]{37})([01]{21})
000000000000000000000000000000000000" grammar="'00110001' filter '001' gslcompanyprefix serialref '
000000000000000000000000000000000000'">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet

```

```

="[01]*" bitPadDir="LEFT" bitLength="37" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="21" name="serialref"/>
</option>
    <option optionKey="10" pattern="00110001 ([01]{3}) 010 ([01]{34}) ([01]{24})
000000000000000000000000" grammar="' 00110001 ' filter ' 010 ' gslcompanyprefix serialref '
000000000000000000000000'">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[01]*" bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="24" name="serialref"/>
</option>
    <option optionKey="9" pattern="00110001 ([01]{3}) 011 ([01]{30}) ([01]{28})
000000000000000000000000" grammar="' 00110001 ' filter ' 011 ' gslcompanyprefix serialref '
000000000000000000000000'">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[01]*" bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="28" name="serialref"/>
</option>
    <option optionKey="8" pattern="00110001 ([01]{3}) 100 ([01]{27}) ([01]{31})
000000000000000000000000" grammar="' 00110001 ' filter ' 100 ' gslcompanyprefix serialref '
000000000000000000000000'">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[01]*" bitPadDir="LEFT" bitLength="31" name="serialref"/>
</option>
    <option optionKey="7" pattern="00110001 ([01]{3}) 101 ([01]{24}) ([01]{34})
000000000000000000000000" grammar="' 00110001 ' filter ' 101 ' gslcompanyprefix serialref '
000000000000000000000000'">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[01]*" bitPadDir="LEFT" bitLength="34" name="serialref"/>

```

```

    </option>
    <option optionKey = "6" pattern = "00110001 ([01]{3}) 110 ([01]{20}) ([01]{38})
00000000000000000000000000000000" grammar = "' 00110001 ' filter ' 110 ' gslcompanyprefix serialref '
00000000000000000000000000000000" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[01]*" bitPadDir="LEFT" bitLength="38" name="serialref"/>
    </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sscc-96" requiredFormattingPa-
rameters="filter,taglength">
    <option optionKey="12" pattern="urn:epc:tag:sscc-96:([0-7]{1})\,([0-9]{12})\,([0-9]
{5})" grammar="urn:epc:tag:sscc-96:' filter ".gslcompanyprefix ".serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="11" pattern="urn:epc:tag:sscc-96:([0-7]{1})\,([0-9]{11})\,([0-9]
{6})" grammar="urn:epc:tag:sscc-96:' filter ".gslcompanyprefix ".serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="10" pattern="urn:epc:tag:sscc-96:([0-7]{1})\,([0-9]{10})\,([0-9]
{7})" grammar="urn:epc:tag:sscc-96:' filter ".gslcompanyprefix ".serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="9" pattern="urn:epc:tag:sscc-96:([0-7]{1})\,([0-9]{9})\,([0-9]
{8})" grammar="urn:epc:tag:sscc-96:' filter ".gslcompanyprefix ".serialref">

```



```

    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:sscc-96:([0-7]{1})\,([0-9]{8})\,([0-9]
{9})" grammar="urn:epc:tag:sscc-96:'filter'.gslcompanyprefix'.serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:sscc-96:([0-7]{1})\,([0-9]{7})\,([0-9]
{10})" grammar="urn:epc:tag:sscc-96:'filter'.gslcompanyprefix'.serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:sscc-96:([0-7]{1})\,([0-9]{6})\,([0-9]
{11})" grammar="urn:epc:tag:sscc-96:'filter'.gslcompanyprefix'.serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="serialref"/>
  </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sscc">
  <option optionKey="12" pattern="urn:epc:id:sscc:([0-9]{12})\,([0-9]{5})" grammar
="urn:epc:id:sscc:'gslcompanyprefix'.serialref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="serialref"/>
  </option>

```

```

    <option optionKey="11" pattern="urn:epc:id:sscc:([0-9]{11})\.[0-9]{6}" grammar=
    ="urn:epc:id:sscc:' gslcompanyprefix ". serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
    ="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
    [0-9]*" length="6" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="10" pattern="urn:epc:id:sscc:([0-9]{10})\.[0-9]{7}" grammar
    ="urn:epc:id:sscc:' gslcompanyprefix ". serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
    ="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
    [0-9]*" length="7" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="9" pattern="urn:epc:id:sscc:([0-9]{9})\.[0-9]{8}" grammar="
    urn:epc:id:sscc:' gslcompanyprefix ". serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
    ="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
    [0-9]*" length="8" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="8" pattern="urn:epc:id:sscc:([0-9]{8})\.[0-9]{9}" grammar="
    urn:epc:id:sscc:' gslcompanyprefix ". serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="
    [0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
    ="[0-9]*" length="9" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="7" pattern="urn:epc:id:sscc:([0-9]{7})\.[0-9]{10}" grammar="
    urn:epc:id:sscc:' gslcompanyprefix ". serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
    [0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
    ="[0-9]*" length="10" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
    <option optionKey="6" pattern="urn:epc:id:sscc:([0-9]{6})\.[0-9]{11}" grammar="
    urn:epc:id:sscc:' gslcompanyprefix ". serialref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
    [0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
    ="[0-9]*" length="11" padChar="0" padDir="LEFT" name="serialref"/>
    </option>
</level>

```

```

    < level type = " LEGACY " prefixMatch = " ssc = " requiredParsingParameters = "
gslcompanyprefixlength">
    <option optionKey="12" pattern="sscc=([0-9]{18})" grammar="' ssc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="11" pattern="sscc=([0-9]{18})" grammar="' ssc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="10" pattern="sscc=([0-9]{18})" grammar="' ssc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="9" pattern="sscc=([0-9]{18})" grammar="' ssc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="8" pattern="sscc=([0-9]{18})" grammar="' ssc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="7" pattern="sscc=([0-9]{18})" grammar="' ssc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="6" pattern="sscc=([0-9]{18})" grammar="' ssc=' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="ssccpre-
fixremainder" characterSet="[0-9]*" length="16" function="SUBSTR(sscc,1,16)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(sscc,0,1)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serialre-
fremainder" characterSet = " [ 0-9 ] * " function = " SUBSTR ( ssccprefixremainder,

```

```

gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="serialref"
characterSet="[0-9]*" function="CONCAT(extensiondigit,serialrefremainder)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,0,
gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(serialref,0,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="serialre-
fremainder" characterSet="[0-9]*" function="SUBSTR(serialref,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="ssccprefix"
characterSet="[0-9]*" length="17" function="CONCAT(extensiondigit,gslcompanyprefix,serial-
refremainder)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(ssccprefix)"/>
</level>
  <level type="LEGACY_AI" prefixMatch="(00)" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="\(00\)([0-9]{18})" grammar="(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="11" pattern="\(00\)([0-9]{18})" grammar="(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="10" pattern="\(00\)([0-9]{18})" grammar="(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="9" pattern="\(00\)([0-9]{18})" grammar="(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="8" pattern="\(00\)([0-9]{18})" grammar="(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>

```

```

    <option optionKey="7" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="6" pattern="\ (00\ ) ([0-9]{18})" grammar="'(00)' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="ssccpre-
fixremainder" characterSet="[0-9]*" length="16" function="SUBSTR(sscc,1,16)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(sscc,0,1)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serialre-
fremainder" characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,
gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="serialref"
characterSet="[0-9]*" function="CONCAT(extensiondigit,serialrefremainder)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="extension-
digit" characterSet="[0-9]*" length="1" function="SUBSTR(serialref,0,1)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="serialre-
fremainder" characterSet="[0-9]*" function="SUBSTR(serialref,1)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="ssccprefix"
characterSet="[0-9]*" length="17" function="CONCAT(extensiondigit,gslcompanyprefix,serial-
refremainder)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(ssccprefix)"/>
</level>
<level type="ELEMENT _ STRING" prefixMatch="00" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="00 ([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
    </option>
    <option optionKey="11" pattern="00 ([0-9]{18})" grammar="'00' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>

```

```

</option>
  <option optionKey = "10" pattern = "00 ([0-9]{18})" grammar = "' 00 ' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "sscc"/>
  </option>
  <option optionKey = "9" pattern = "00 ([0-9]{18})" grammar = "' 00 ' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "sscc"/>
  </option>
  <option optionKey = "8" pattern = "00 ([0-9]{18})" grammar = "' 00 ' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "sscc"/>
  </option>
  <option optionKey = "7" pattern = "00 ([0-9]{18})" grammar = "' 00 ' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "sscc"/>
  </option>
  <option optionKey = "6" pattern = "00 ([0-9]{18})" grammar = "' 00 ' extensiondigit
gslcompanyprefix serialrefremainder checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "sscc"/>
  </option>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "1" newFieldName = "ssccpre-
fixremainder" characterSet = "[0-9]*" length = "16" function = "SUBSTR(sscc,1,16)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "2" newFieldName = "extension-
digit" characterSet = "[0-9]*" length = "1" function = "SUBSTR(sscc,0,1)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "3" newFieldName = "serialre-
fremainder" characterSet = "[0-9]*" function = "SUBSTR(ssccprefixremainder,
gslcompanyprefixlength)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "4" newFieldName = "serialref"
characterSet = "[0-9]*" function = "CONCAT(extensiondigit,serialrefremainder)"/>
  <rule type = "EXTRACT" inputFormat = "STRING" seq = "5" newFieldName = "
gslcompanyprefix" characterSet = "[0-9]*" function = "SUBSTR(ssccprefixremainder, 0,
gslcompanyprefixlength)"/>
  <rule type = "FORMAT" inputFormat = "STRING" seq = "1" newFieldName = "extension-
digit" characterSet = "[0-9]*" length = "1" function = "SUBSTR(serialref,0,1)"/>
  <rule type = "FORMAT" inputFormat = "STRING" seq = "2" newFieldName = "serialre-
fremainder" characterSet = "[0-9]*" function = "SUBSTR(serialref,1)"/>
  <rule type = "FORMAT" inputFormat = "STRING" seq = "3" newFieldName = "ssccprefix"

```

```

characterSet="[0-9]*" length="17" function="CONCAT(extensiondigit,gs1companyprefix,serial-
refremainder)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(ssccprefix)"/>
</level>
<level type="ONS_HOSTNAME">
    <option optionKey="12" grammar="gs1companyprefix'.sscc.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="11" grammar="gs1companyprefix'.sscc.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet=
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="10" grammar="gs1companyprefix'.sscc.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet=
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="9" grammar="gs1companyprefix'.sscc.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet=
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="8" grammar="gs1companyprefix'.sscc.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="7" grammar="gs1companyprefix'.sscc.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="6" grammar="gs1companyprefix'.sscc.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.6 SGLN-64

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-

```



```

instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="SGLN-64" optionKey="gs1companyprefixlength" tagLength="64">
    <level type="BINARY" prefixMatch="00001001" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001001' filter gs1companyprefixindex locationref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gs1companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]* "
bitPadDir="LEFT" bitLength="20" name="locationref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[01]* " bitPadDir="LEFT" bitLength="19" name="serial"/>
      </option>
      <option optionKey="11" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001001' filter gs1companyprefixindex locationref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gs1companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="20" name="locationref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[01]* " bitPadDir="LEFT" bitLength="19" name="serial"/>
      </option>
      <option optionKey="10" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001001' filter gs1companyprefixindex locationref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gs1companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="20" name="locationref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[01]* " bitPadDir="LEFT" bitLength="19" name="serial"/>
      </option>
      <option optionKey="9" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001001' filter gs1companyprefixindex locationref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gs1companyprefixindex"/>

```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[01]* " bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <option optionKey="8" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001001' filter gslcompanyprefixindex locationref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[01]* " bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <option optionKey="7" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001001' filter gslcompanyprefixindex locationref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[01]* " bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <option optionKey="6" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001001' filter gslcompanyprefixindex locationref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[01]* " bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
gslcompanyprefix" characterSet="[0-9]* " function="TABLELOOKUP(gslcompanyprefixindex,
tdt64bitcpi, gslcompanyprefixindex, gslcompanyprefix)" tableURL="http://www.onsepc.com/
ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
tableParams="gslcompanyprefixindex"/>

```

```

    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
gslcompanyprefixlength" characterSet="[0-9]*" function="LENGTH(gslcompanyprefix)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="
gslcompanyprefixindex" characterSet="[0-9]*" function="TABLELOOKUP(gslcompanyprefix,
tdt64bitcpi, gslcompanyprefix, gslcompanyprefixindex)" tableURL="http://www. onsepc. com/
ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@companyPrefix=' $ 1 ']/@index"
tableParams="gslcompanyprefix"/>
  </level>
  <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgln-64" requiredFormattingPa-
rameters="filter,length">
    <option optionKey="12" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{12}\.[0-9]
{0}\.[0-9]{1,6}" grammar="urn:epc:tag:sgln-64:' filter '. gslcompanyprefix '. locationref '. seri-
al">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]* "
length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]* "
length="0" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{11}\.[0-9]
{1}\.[0-9]{1,6}" grammar="urn:epc:tag:sgln-64:' filter '. gslcompanyprefix '. locationref '. seri-
al">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "
length="1" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{10}\.[0-9]
{2}\.[0-9]{1,6}" grammar="urn:epc:tag:sgln-64:' filter '. gslcompanyprefix '. locationref '. seri-
al">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]* " length="2" padChar="0" padDir="LEFT" name="locationref"/>

```

```

    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{9}\.[0-9]{3}\.[0-9]{1,6}" grammar="urn:epc:tag:sgln-64:'filter'.gslcompanyprefix'.locationref'.serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{8}\.[0-9]{4}\.[0-9]{1,6}" grammar="urn:epc:tag:sgln-64:'filter'.gslcompanyprefix'.locationref'.serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{7}\.[0-9]{5}\.[0-9]{1,6}" grammar="urn:epc:tag:sgln-64:'filter'.gslcompanyprefix'.locationref'.serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{6}\.[0-9]{6}\.[0-9]{1,6}" grammar="urn:epc:tag:sgln-64:'filter'.gslcompanyprefix'.locationref'.serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"

```

```

length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgln">
    <option optionKey="12" pattern="urn:epc:id:sgln:([0-9]{12})\.[0-9]{0,6}\.([0-9]{1,
6})" grammar="urn:epc:id:sgln:'gs1companyprefix'.locationref'.serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="locationref"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
        </option>
    <option optionKey="11" pattern="urn:epc:id:sgln:([0-9]{11})\.[0-9]{1,6}\.([0-9]{1,
6})" grammar="urn:epc:id:sgln:'gs1companyprefix'.locationref'.serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="locationref"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
        </option>
    <option optionKey="10" pattern="urn:epc:id:sgln:([0-9]{10})\.[0-9]{2,6}\.([0-9]{1,
6})" grammar="urn:epc:id:sgln:'gs1companyprefix'.locationref'.serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="locationref"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
        </option>
    <option optionKey="9" pattern="urn:epc:id:sgln:([0-9]{9})\.[0-9]{3,6}\.([0-9]
{1,6})" grammar="urn:epc:id:sgln:'gs1companyprefix'.locationref'.serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="
[0-9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>

```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:id:sgln:([0-9]{8})\.[0-9]{4}\.([0-9]
{1,6})" grammar="urn:epc:id:sgln:'gslcompanyprefix'.locationref'.serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]* " length="4" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:id:sgln:([0-9]{7})\.[0-9]{5}\.([0-9]{1,
6})" grammar="urn:epc:id:sgln:'gslcompanyprefix'.locationref'.serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]* " length="5" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:id:sgln:([0-9]{6})\.[0-9]{6}\.([0-9]
{1,6})" grammar="urn:epc:id:sgln:'gslcompanyprefix'.locationref'.serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
</level>
<level type="LEGACY" prefixMatch="gln=" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="gln=([0-9]{13});serial=([0-9]{1,6})" grammar="
gln='gslcompanyprefix locationref checkdigit';serial='serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999"
characterSet="[0-9]* " length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="11" pattern="gln=([0-9]{13});serial=([0-9]{1,6})" grammar="
gln='gslcompanyprefix locationref checkdigit';serial='serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999"

```

```

characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="10" pattern="gln=([0-9]{13});serial=([0-9]{1,6})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="gln=([0-9]{13});serial=([0-9]{1,6})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="gln=([0-9]{13});serial=([0-9]{1,6})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="gln=([0-9]{13});serial=([0-9]{1,6})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="gln=([0-9]{13});serial=([0-9]{1,6})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref " characterSet = " [ 0-9 ]*" function = " SUBSTR ( glnprefixremainder,

```



```

gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR (glnprefixremainder, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(glnprefix)"/>
  </level>
  <level type="LEGACY _ AI" prefixMatch="(414)" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]* )" grammar="
(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]* )" grammar="
(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]* )" grammar="
(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]* )" grammar="
(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="8" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]* )" grammar="
(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="\ (414\)\ ([0-9]{13})\ (254\)\ ([0-9]* )" grammar="'
(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="\ (414\)\ ([0-9]{13})\ (254\)\ ([0-9]* )" grammar="'
(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref" characterSet=" [ 0-9 ]*" function=" SUBSTR ( glnprefixremainder,
gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet=" [ 0-9 ]*" function=" SUBSTR ( glnprefixremainder, 0,
gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GSI_CHECKSUM(glnprefix)"/>
</level>
<level type="ELEMENT _ STRING" prefixMatch="414" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="414 ([0-9]{13}) 254 ([0-9]* )" grammar="' 414 '
gslcompanyprefix locationref checkdigit ' 254 ' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="414 ([0-9]{13}) 254 ([0-9]* )" grammar="' 414 '
gslcompanyprefix locationref checkdigit ' 254 ' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="10" pattern="414([0-9]{13})254([0-9]*)" grammar="'414'
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="9" pattern="414([0-9]{13})254([0-9]*)" grammar="'414'
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="8" pattern="414([0-9]{13})254([0-9]*)" grammar="'414'
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="7" pattern="414([0-9]{13})254([0-9]*)" grammar="'414'
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <option optionKey="6" pattern="414([0-9]{13})254([0-9]*)" grammar="'414'
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]* " name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-
fixremainder" characterSet="[0-9]* " length="12" function="SUBSTR(gln,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref " characterSet = " [ 0-9 ]* " function = " SUBSTR ( glnprefixremainder,
gslcompanyprefixlength)"/>

```

```

    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
    gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder, 0,
    gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
    characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
    characterSet="[0-9]*" length="1" function="GSLCHECKSUM(glnprefix)"/>
  </level>
  <level type="ONS_HOSTNAME">
    <option optionKey="12" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
      length="0" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999"
      characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="11" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
      length="1" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
     ="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="10" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
      9]*" length="2" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
     ="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="9" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
      9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
     ="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="8" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
      9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
      [0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="7" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
      9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="

```

```

[0-9]* " length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="6" grammar="locationref'.sgln.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="locationref"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        </option>
    </level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.7 SGLN-96

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
    <scheme name="SGLN-96" optionKey="gs1companyprefixlength" tagLength="96">
        <level type="BINARY" prefixMatch="00110010" requiredFormattingParameters="filter,tag-
length">
            <option optionKey="12" pattern="00110010([01]{3})000([01]{40})([01]{1})([01]
{41})" grammar="'00110010' filter '000' gs1companyprefix locationref serial">
                <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
                <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[01]* " bitPadDir="LEFT" bitLength="40" name="gs1companyprefix"/>
                <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]* "
bitPadDir="LEFT" bitLength="1" name="locationref"/>
                <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[01]* " bitPadDir="LEFT" bitLength="41" name="serial"/>
            </option>
            <option optionKey="11" pattern="00110010([01]{3})001([01]{37})([01]{4})([01]
{41})" grammar="'00110010' filter '001' gs1companyprefix locationref serial">
                <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
                <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="37" name="gs1companyprefix"/>
                <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="locationref"/>
                <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[01]* " bitPadDir="LEFT" bitLength="41" name="serial"/>
            </option>

```

```

    <option optionKey="10" pattern="00110010([01]{3})010([01]{34})([01]{7})([01]{41})" grammar="00110010' filter '010' gslcompanyprefix locationref serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]*" bitPadDir="LEFT" bitLength="7" name="locationref"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>
    </option>

    <option optionKey="9" pattern="00110010([01]{3})011([01]{30})([01]{11})([01]{41})" grammar="00110010' filter '011' gslcompanyprefix locationref serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]*" bitPadDir="LEFT" bitLength="11" name="locationref"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>
    </option>

    <option optionKey="8" pattern="00110010([01]{3})100([01]{27})([01]{14})([01]{41})" grammar="00110010' filter '100' gslcompanyprefix locationref serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="locationref"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>
    </option>

    <option optionKey="7" pattern="00110010([01]{3})101([01]{24})([01]{17})([01]{41})" grammar="00110010' filter '101' gslcompanyprefix locationref serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*" bitPadDir="LEFT" bitLength="17" name="locationref"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>

```

```

    </option>
    <option optionKey="6" pattern="00110010([01]{3})110([01]{20})([01]{21})([01]{41})" grammar="'00110010' filter '110' gslcompanyprefix locationref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="21" name="locationref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>
    </option>
</level>

<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgln-96" requiredFormattingParameters="filter,taglength">
    <option optionKey="12" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{12}\.[0-9]{0})\.[0-9]{1,13}" grammar="urn:epc:tag:sgln-96:' filter '.gslcompanyprefix'.locationref'.serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*" length="0" padChar="0" padDir="LEFT" name="locationref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{11}\.[0-9]{1})\.[0-9]{1,13}" grammar="urn:epc:tag:sgln-96:' filter '.gslcompanyprefix'.locationref'.serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*" length="1" padChar="0" padDir="LEFT" name="locationref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{10}\.[0-9]{2})\.[0-9]{1,13}" grammar="urn:epc:tag:sgln-96:' filter '.gslcompanyprefix'.locationref'.serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"

```



```

length="1" padChar="0" padDir="LEFT" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet=
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
    </option>
  <option optionKey="9" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{9}\.[0-9]
{3}\.[0-9]{1,13}" grammar="urn:epc:tag:sgln-96:'filter'.gs1companyprefix'.locationref'.ser-
ial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet=
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[
0-9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{8}\.[0-9]
{4}\.[0-9]{1,13}" grammar="urn:epc:tag:sgln-96:'filter'.gs1companyprefix'.locationref'.ser-
ial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[
0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{7}\.[0-9]
{5}\.[0-9]{1,13}" grammar="urn:epc:tag:sgln-96:'filter'.gs1companyprefix'.locationref'.ser-
ial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[
0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
  </option>

```

```

    </option>
    <option optionKey="6" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{6})\.[0-9]{1,13}" grammar="urn:epc:tag:sgln-96:'filter'.gslcompanyprefix'.locationref'.serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
    </option>
  </level>
  <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgln">
    <option optionKey="12" pattern="urn:epc:id:sgln:([0-9]{12})\.[0-9]{0})\.[0-9]{1,13}" grammar="urn:epc:id:sgln:'gslcompanyprefix'.locationref'.serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*" length="0" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:id:sgln:([0-9]{11})\.[0-9]{1})\.[0-9]{1,13}" grammar="urn:epc:id:sgln:'gslcompanyprefix'.locationref'.serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*" length="1" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:id:sgln:([0-9]{10})\.[0-9]{2})\.[0-9]{1,13}" grammar="urn:epc:id:sgln:'gslcompanyprefix'.locationref'.serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="urn:epc:id:sgln:([0-9]{9})\.[0-9]{3})\.[0-9]{1,

```

```

13}}" grammar="urn:epc:id:sgln:' gslcompanyprefix ". locationref ". serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet=
   ="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
    9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551"
    characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:id:sgln:([0-9]{8})\.[0-9]{4})\.[0-9]{1,
  13}}" grammar="urn:epc:id:sgln:' gslcompanyprefix ". locationref ". serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
    [0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="
    [0-9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551"
    characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:id:sgln:([0-9]{7})\.[0-9]{5})\.[0-9]
  {1,13}}" grammar="urn:epc:id:sgln:' gslcompanyprefix ". locationref ". serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
    [0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
    9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551"
    characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:id:sgln:([0-9]{6})\.[0-9]{6})\.[0-9]{1,
  13}}" grammar="urn:epc:id:sgln:' gslcompanyprefix ". locationref ". serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
    [0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
    [0-9]*" length="6" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551"
    characterSet="[0-9]*" name="serial"/>
  </option>
</level>
<level type="LEGACY" prefixMatch="gln=" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="gln=([0-9]{13});serial=([0-9]{1,13})" grammar="
  gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999"
    characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"

```

```

characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="11" pattern="gln=([0-9]{13});serial=([0-9]{1,13})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="10" pattern="gln=([0-9]{13});serial=([0-9]{1,13})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="gln=([0-9]{13});serial=([0-9]{1,13})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="gln=([0-9]{13});serial=([0-9]{1,13})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="gln=([0-9]{13});serial=([0-9]{1,13})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="gln=([0-9]{13});serial=([0-9]{1,13})" grammar="
gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>

```

```

</option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="locationref" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder, gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder, 0, gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix" characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit" characterSet="[0-9]*" length="1" function="GSLCHECKSUM(glnprefix)"/>
</level>
<level type="LEGACY_AI" prefixMatch="(414)" requiredParsingParameters="gslcompanyprefixlength">
  <option optionKey="12" pattern="\ (414\) ([0-9]{13}) \(254\) ([0-9]{1,13})" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="\ (414\) ([0-9]{13}) \(254\) ([0-9]{1,13})" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="\ (414\) ([0-9]{13}) \(254\) ([0-9]{1,13})" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="\ (414\) ([0-9]{13}) \(254\) ([0-9]{1,13})" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-9]*" name="serial"/>
  </option>

```

```

</option>
<option optionKey="8" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]{1,13})" grammar
="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]{1,13})" grammar
="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="\ (414\)([0-9]{13})\ (254\)([0-9]{1,13})" grammar
="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref" characterSet="[0-9]*" function="SUBSTR ( glnprefixremainder,
gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR ( glnprefixremainder, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(glnprefix)"/>
</level>
<level type="ELEMENT_STRING" prefixMatch="414" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="414([0-9]{13})254([0-9]{1,13})" grammar="414'
gslcompanyprefix locationref checkdigit '254' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>

```

```

</option>
<option optionKey="11" pattern="414([0-9]{13})254([0-9]{1,13})" grammar="'414 '
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="10" pattern="414([0-9]{13})254([0-9]{1,13})" grammar="'414 '
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="414([0-9]{13})254([0-9]{1,13})" grammar="'414 '
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="414([0-9]{13})254([0-9]{1,13})" grammar="'414 '
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="414([0-9]{13})254([0-9]{1,13})" grammar="'414 '
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="414([0-9]{13})254([0-9]{1,13})" grammar="'414 '
gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>

```



```

    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,
gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(glnprefix)"/>
  </level>
  <level type="ONS_HOSTNAME">
    <option optionKey="12" grammar="locationref'.gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="11" grammar="locationref'.gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="10" grammar="locationref'.gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="9" grammar="locationref'.gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="8" grammar="locationref'.gslcompanyprefix'.sgln.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>

```

```

</option>
<option optionKey="7" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="6" grammar="locationref".gslcompanyprefix'.sgln.id.onsepc.com">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="locationref"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.8 SGLN-195

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="SGLN-195" optionKey="gslcompanyprefixlength" tagLength="195">
    <level type="BINARY" prefixMatch="00111001" requiredFormattingParameters="filter,tag-length">
      <option optionKey="12" pattern="00111001([01]{3})000([01]{40})([01]{1})([01]{140})" grammar="'00111001' filter '000' gslcompanyprefix locationref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="40" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]*" bitPadDir="LEFT" bitLength="1" name="locationref"/>
        <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="140" compaction="7-bit" name="serial"/>
      </option>
      <option optionKey="11" pattern="00111001([01]{3})001([01]{37})([01]{4})([01]{140})" grammar="'00111001' filter '001' gslcompanyprefix locationref serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="37" name="gslcompanyprefix"/>
      </option>
    </level>
  </scheme>
</tdt:epcTagDataTranslation>

```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="locationref"/>
    <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
  <option optionKey="10" pattern="00111001([01]{3})010([01]{34})([01]{7})([01]
{140})" grammar="'00111001' filter '010' gslcompanyprefix locationref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="7" name="locationref"/>
    <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
  <option optionKey="9" pattern="00111001([01]{3})011([01]{30})([01]{11})([01]
{140})" grammar="'00111001' filter '011' gslcompanyprefix locationref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="11" name="locationref"/>
    <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
  <option optionKey="8" pattern="00111001([01]{3})100([01]{27})([01]{14})([01]
{140})" grammar="'00111001' filter '100' gslcompanyprefix locationref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="locationref"/>
    <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
  </option>
  <option optionKey="7" pattern="00111001([01]{3})101([01]{24})([01]{17})([01]
{140})" grammar="'00111001' filter '101' gslcompanyprefix locationref serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="

```

```

[01]* " bitPadDir="LEFT" bitLength="24" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="17" name="locationref"/>
    <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
    </option>
    <option optionKey="6" pattern="00111001([01]{3})110([01]{20})([01]{21})([01]
{140})" grammar="00111001'filter'110'gs1companyprefix locationref serial"
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="21" name="locationref"/>
    <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="140" com-
paction="7-bit" name="serial"/>
    </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgln-195" requiredFormattingPa-
rameters="filter,taglength">
    <option optionKey="12" pattern="urn:epc:tag:sgln-195:([0-7]{1})\.[([0-9]{12})\.[([0-
9]{0})]\.([!%-? A-Z _ a-z \ x22]{1,20})" grammar="'urn:epc:tag:sgln-195:' filter '!'
gs1companyprefix!' locationref!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]* "
length="0" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" characterSet="[!%-? A-Z _ a-z \ x22]* " name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:tag:sgln-195:([0-7]{1})\.[([0-9]{11})\.[([0-
9]{1})]\.([!%-? A-Z _ a-z \ x22]{1,20})" grammar="'urn:epc:tag:sgln-195:' filter '!'
gs1companyprefix!' locationref!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "
length="1" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" characterSet="[!%-? A-Z _ a-z \ x22]* " name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:tag:sgln-195:([0-7]{1})\.[([0-9]{10})\.[([0-

```

```

9]{2})\.(?!%?-? A-Z_a-z\x22]{1,20})" grammar="urn:epc:tag:sgln-195:' filter '!'
gs1companyprefix '!' locationref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" characterSet="(!%?-? A-Z_a-z\x22)*" name="serial"/>
</option>
<option optionKey="9" pattern="urn:epc:tag:sgln-195:([0-7]{1})\.([0-9]{9})\.(
[0-9]{3})\.(?!%?-? A-Z_a-z\x22){1,20})" grammar="urn:epc:tag:sgln-195:' filter '!' gs1companyprefix
'!' locationref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" characterSet="(!%?-? A-Z_a-z\x22)*" name="serial"/>
</option>
<option optionKey="8" pattern="urn:epc:tag:sgln-195:([0-7]{1})\.(
[0-9]{8})\.(
[0-9]{4})\.(?!%?-? A-Z_a-z\x22){1,20})" grammar="urn:epc:tag:sgln-195:' filter '!' gs1companyprefix
'!' locationref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" characterSet="(!%?-? A-Z_a-z\x22)*" name="serial"/>
</option>
<option optionKey="7" pattern="urn:epc:tag:sgln-195:([0-7]{1})\.(
[0-9]{7})\.(
[0-9]{5})\.(?!%?-? A-Z_a-z\x22){1,20})" grammar="urn:epc:tag:sgln-195:' filter '!' gs1companyprefix
'!' locationref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="4" characterSet="(!%?-? A-Z_a-z\x22)*" name="serial"/>
</option>

```



```

    <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:id:sgln:([0-9]{8})\.[0-9]{4})\.[!%-? A-Z_a-z\x22]{1,20}" grammar="urn:epc:id:sgln:' gslcompanyprefix '!' locationref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:id:sgln:([0-9]{7})\.[0-9]{5})\.[!%-? A-Z_a-z\x22]{1,20}" grammar="urn:epc:id:sgln:' gslcompanyprefix '!' locationref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:id:sgln:([0-9]{6})\.[0-9]{6})\.[!%-? A-Z_a-z\x22]{1,20}" grammar="urn:epc:id:sgln:' gslcompanyprefix '!' locationref '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
</level>
< level type = " LEGACY " prefixMatch = " gln = " requiredParsingParameters = "
gslcompanyprefixlength">
  <option optionKey="12" pattern="gln=([0-9]{13});serial=([!%-? A-Z_a-z\x22]{0,20})" grammar="gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="gln=([0-9]{13});serial=([!%-? A-Z_a-z\x22]{0,20})" grammar="gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="gln=([0-9]{13});serial=([!%-? A-Z_a-z\x22]{0,20})" grammar="gln=' gslcompanyprefix locationref checkdigit ';serial=' serial">

```



```

    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
    <option optionKey="9" pattern="gln=([0-9]{13});serial=([!%-? A-Z_a-z\x22]{0,
20})" grammar="gln='gslcompanyprefix locationref checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
    <option optionKey="8" pattern="gln=([0-9]{13});serial=([!%-? A-Z_a-z\x22]{0,
20})" grammar="gln='gslcompanyprefix locationref checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
    <option optionKey="7" pattern="gln=([0-9]{13});serial=([!%-? A-Z_a-z\x22]{0,
20})" grammar="gln='gslcompanyprefix locationref checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
    <option optionKey="6" pattern="gln=([0-9]{13});serial=([!%-? A-Z_a-z\x22]{0,
20})" grammar="gln='gslcompanyprefix locationref checkdigit';serial='serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,
gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GSI_CHECKSUM(glnprefix)"/>
</level>
    <level type="LEGACY_AI" prefixMatch="(414)" requiredParsingParameters="
gslcompanyprefixlength">

```

```

    <option optionKey="12" pattern="\(414\)([0-9]{13})\((254\)([!%?-? A-Z_a-z\x22]{0,
20}))" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="\(414\)([0-9]{13})\((254\)([!%?-? A-Z_a-z\x22]{0,
20}))" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="\(414\)([0-9]{13})\((254\)([!%?-? A-Z_a-z\x22]{0,
20}))" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="\(414\)([0-9]{13})\((254\)([!%?-? A-Z_a-z\x22]{0,
20}))" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="8" pattern="\(414\)([0-9]{13})\((254\)([!%?-? A-Z_a-z\x22]{0,
20}))" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="7" pattern="\(414\)([0-9]{13})\((254\)([!%?-? A-Z_a-z\x22]{0,
20}))" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="6" pattern="\(414\)([0-9]{13})\((254\)([!%?-? A-Z_a-z\x22]{0,
20}))" grammar="(414)' gslcompanyprefix locationref checkdigit '(254)' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-

```

```

fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,
gslcompanylength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanylength" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder, 0,
gslcompanylength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanylength,locationref)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GSLCHECKSUM(glnprefix)"/>
</level>
<level type="ELEMENT_STRING" prefixMatch="414" requiredParsingParameters="
gslcompanylength">
  <option optionKey="12" pattern="414([0-9]{13})254([!%?-? A-Z_a-z\x22]{0,20})"
grammar="414' gslcompanylength locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
  </option>
  <option optionKey="11" pattern="414([0-9]{13})254([!%?-? A-Z_a-z\x22]{0,20})"
grammar="414' gslcompanylength locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
  </option>
  <option optionKey="10" pattern="414([0-9]{13})254([!%?-? A-Z_a-z\x22]{0,20})"
grammar="414' gslcompanylength locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
  </option>
  <option optionKey="9" pattern="414([0-9]{13})254([!%?-? A-Z_a-z\x22]{0,20})"
grammar="414' gslcompanylength locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
  </option>
  <option optionKey="8" pattern="414([0-9]{13})254([!%?-? A-Z_a-z\x22]{0,20})"
grammar="414' gslcompanylength locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="([!%?-? A-Z_a-z\x22])*" name="serial"/>
  </option>

```

```

</option>
<option optionKey="7" pattern="414([0-9]{13})254([!%-? A-Z_a-z\x22]{0,20})"
grammar="414' gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="6" pattern="414([0-9]{13})254([!%-? A-Z_a-z\x22]{0,20})"
grammar="414' gslcompanyprefix locationref checkdigit '254' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnpre-
fixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
locationref" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,
gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(glnprefixremainder, 0,
gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,locationref)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(glnprefix)"/>
</level>
<level type="ONS_HOSTNAME">
    <option optionKey="12" grammar="locationref!' gslcompanyprefix',sgln.id,onsepc.com
">
        <field seq="1" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="locationref"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="11" grammar="locationref!' gslcompanyprefix',sgln.id,onsepc.com
">
        <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="locationref"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="10" grammar="locationref!' gslcompanyprefix',sgln.id,onsepc.com
">

```

```

    <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="9" grammar="locationref '.' gs1companyprefix '.sgln.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="8" grammar="locationref '.' gs1companyprefix '.sgln.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="7" grammar="locationref '.' gs1companyprefix '.sgln.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="6" grammar="locationref '.' gs1companyprefix '.sgln.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="locationref"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.9 GDTI-96

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:
schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GDTI-96" optionKey="gs1companyprefixlength" tagLength="96">
    <level type="BINARY" prefixMatch="00101100" requiredFormattingParameters="filter, tag-
length">
      <option optionKey="12" pattern="00101100([01]{3})000([01]{40})(0)([01]{41})"

```

```

grammar="00101100' filter '000' gslcompanyprefix doctype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="40" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]* "
bitPadDir="LEFT" bitLength="1" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[01]* " bitPadDir="LEFT" bitLength="41" name="serial"/>
</option>
<option optionKey="11" pattern="00101100([01]{3})001([01]{37})([01]{4})([01]{41})"
grammar="00101100' filter '001' gslcompanyprefix doctype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="37" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[01]* " bitPadDir="LEFT" bitLength="41" name="serial"/>
</option>
<option optionKey="10" pattern="00101100([01]{3})010([01]{34})([01]{7})([01]{41})"
grammar="00101100' filter '010' gslcompanyprefix doctype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="7" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[01]* " bitPadDir="LEFT" bitLength="41" name="serial"/>
</option>
<option optionKey="9" pattern="00101100([01]{3})011([01]{30})([01]{11})([01]{41})"
grammar="00101100' filter '011' gslcompanyprefix doctype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="11" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[01]* " bitPadDir="LEFT" bitLength="41" name="serial"/>
</option>

```

```

    <option optionKey="8" pattern="00101100([01]{3})100([01]{27})([01]{14})([01]{41})" grammar="00101100'filter'100'gslcompanyprefix doctype serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="doctype"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>
    </option>

    <option optionKey="7" pattern="00101100([01]{3})101([01]{24})([01]{17})([01]{41})" grammar="00101100'filter'101'gslcompanyprefix doctype serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*" bitPadDir="LEFT" bitLength="17" name="doctype"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>
    </option>

    <option optionKey="6" pattern="00101100([01]{3})110([01]{20})([01]{21})([01]{41})" grammar="00101100'filter'110'gslcompanyprefix doctype serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="21" name="doctype"/>
      <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[01]*" bitPadDir="LEFT" bitLength="41" name="serial"/>
    </option>
  </level>

  <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:gdti-96" requiredFormattingParameters="filter,taglength">
    <option optionKey="12" pattern="urn:epc:tag:gdti-96:([0-7]{1})\.[0-9]{12}\.[0-9]{0}\.[0-9]{1,13}" grammar="urn:epc:tag:gdti-96:'filter'! 'gslcompanyprefix'! 'doctype'! 'serial'">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>

```



```

    <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]* "
length="0" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="11" pattern="urn:epc:tag:gdti-96:([0-7]{1})\.[0-9]{11}\.[0-9]{1}\.[0-9]{1,13}" grammar="urn:epc:tag:gdti-96:' filter ' gslcompanyprefix ' doctype ' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "
length="1" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="10" pattern="urn:epc:tag:gdti-96:([0-7]{1})\.[0-9]{10}\.[0-9]{2}\.[0-9]{1,13}" grammar="urn:epc:tag:gdti-96:' filter ' gslcompanyprefix ' doctype ' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]* " length="2" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="9" pattern="urn:epc:tag:gdti-96:([0-7]{1})\.[0-9]{9}\.[0-9]{3}\.[0-9]{1,13}" grammar="urn:epc:tag:gdti-96:' filter ' gslcompanyprefix ' doctype ' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]* " length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]* " length="3" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:gdti-96:([0-7]{1})\.[0-9]{8}\.[0-9]{4}\.[0-9]{1,13}" grammar="urn:epc:tag:gdti-96:' filter ' gslcompanyprefix ' doctype ' serial">

```

```

al">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="urn:epc:tag:gdti-96:([0-7]{1})\.[0-9]{7})\.[0-9]
{5})\.[0-9]{1,13})" grammar="urn:epc:tag:gdti-96:'filter'!gs1companyprefix'!doctype'!seri-
al">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="urn:epc:tag:gdti-96:([0-7]{1})\.[0-9]{6})\.[0-9]
{6})\.[0-9]{1,13})" grammar="urn:epc:tag:gdti-96:'filter'!gs1companyprefix'!doctype'!seri-
al">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:gdti">
    <option optionKey="12" pattern="urn:epc:id:gdti:([0-9]{12})\.[0-9]{0})\.[0-9]{1,
13})" grammar="urn:epc:id:gdti:'gs1companyprefix'!doctype'!serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="doctype"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551"

```

```

characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="urn:epc:id:gdtd:([0-9]{11})\.[0-9]{1}\.[0-9]{1,13})" grammar="urn:epc:id:gdtd:' gslcompanyprefix ' doctype ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*" length="1" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="urn:epc:id:gdtd:([0-9]{10})\.[0-9]{2}\.[0-9]{1,13})" grammar="urn:epc:id:gdtd:' gslcompanyprefix ' doctype ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="urn:epc:id:gdtd:([0-9]{9})\.[0-9]{3}\.[0-9]{1,13})" grammar="urn:epc:id:gdtd:' gslcompanyprefix ' doctype ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:id:gdtd:([0-9]{8})\.[0-9]{4}\.[0-9]{1,13})" grammar="urn:epc:id:gdtd:' gslcompanyprefix ' doctype ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:id:gdtd:([0-9]{7})\.[0-9]{5}\.[0-9]{1,13})" grammar="urn:epc:id:gdtd:' gslcompanyprefix ' doctype ' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-9]*" name="serial"/>
  </option>

```

```

9]" * " length="5" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]" * " name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:id:gdti:([0-9]{6})\.[0-9]{6})\.[0-9]{1,
13})" grammar="urn:epc:id:gdti:'gslcompanyprefix'!'doctype'!'serial"/>
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]" * " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]" * " length="6" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]" * " name="serial"/>
  </option>
</level>
<level type="LEGACY" prefixMatch="gdti" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="gdti=([0-9]{13})([0-9]{1,13})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999"
characterSet="[0-9]" * " length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]" * " name="serial"/>
  </option>
  <option optionKey="11" pattern="gdti=([0-9]{13})([0-9]{1,13})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999"
characterSet="[0-9]" * " length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]" * " name="serial"/>
  </option>
  <option optionKey="10" pattern="gdti=([0-9]{13})([0-9]{1,13})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999"
characterSet="[0-9]" * " length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]" * " name="serial"/>
  </option>
  <option optionKey="9" pattern="gdti=([0-9]{13})([0-9]{1,13})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999"
characterSet="[0-9]" * " length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]" * " name="serial"/>

```

```

</option>
<option optionKey="8" pattern="gdti=([0-9]{13})([0-9]{1,13})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="gdti=([0-9]{13})([0-9]{1,13})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="gdti=([0-9]{13})([0-9]{1,13})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" function="SUBSTR(gdtiprefix,0,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="doctype"
characterSet="[0-9]*" function="SUBSTR(gdtibody,gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix " characterSet = " [ 0-9 ] * " function = " SUBSTR ( gdtibody, 0,
gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,doctype)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gdtibody)"/>
</level>
<level type="LEGACY _ AI" prefixMatch="(253)" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="\ (253\)([0-9]{13})([0-9]{1,13})" grammar="(253)'
gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
    </option>

```

```

    <option optionKey="11" pattern="\ (253\)([0-9]{13})([0-9]{1,13})" grammar="(253)'
    gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
        characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
        characterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="10" pattern="\ (253\)([0-9]{13})([0-9]{1,13})" grammar="(253)'
    gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
        characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
        characterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="9" pattern="\ (253\)([0-9]{13})([0-9]{1,13})" grammar="(253)'
    gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
        characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
        characterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="8" pattern="\ (253\)([0-9]{13})([0-9]{1,13})" grammar="(253)'
    gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
        characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
        characterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="7" pattern="\ (253\)([0-9]{13})([0-9]{1,13})" grammar="(253)'
    gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
        characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
        characterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="6" pattern="\ (253\)([0-9]{13})([0-9]{1,13})" grammar="(253)'
    gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
        characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
        characterSet="[0-9]*" name="serial"/>
    </option>

    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gdtibody"

```

```

characterSet="[0-9]*" function="SUBSTR(gdtiprefix,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="doctype"
characterSet="[0-9]*" function="SUBSTR(gdtibody,gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gdtibody,0,
gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,doctype)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gdtibody)"/>
</level>
<level type="ELEMENT_STRING" prefixMatch="253" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="253([0-9]{13})([0-9]{1,13})" grammar="'253'
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="253([0-9]{13})([0-9]{1,13})" grammar="'253'
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="253([0-9]{13})([0-9]{1,13})" grammar="'253'
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="253([0-9]{13})([0-9]{1,13})" grammar="'253'
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="253([0-9]{13})([0-9]{1,13})" grammar="'253'
gslcompanyprefix doctype checkdigit serial">

```



```

    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="253([0-9]{13})([0-9]{1,13})" grammar="'253'
gs1companyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="253([0-9]{13})([0-9]{1,13})" grammar="'253'
gs1companyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551"
characterSet="[0-9]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" function="SUBSTR(gdtiprefix,0,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="doctype"
characterSet="[0-9]*" function="SUBSTR(gdtibody,gs1companyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gs1companyprefix " characterSet = " [ 0-9 ] * " function = " SUBSTR ( gdtibody , 0,
gs1companyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" length="12" function="CONCAT(gs1companyprefix,doctype)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gdtibody)"/>
</level>
<level type="ONS_HOSTNAME">
    <option optionKey="12" grammar="doctype '! gs1companyprefix '.gdti.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="doctype"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="11" grammar="doctype '! gs1companyprefix '.gdti.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="doctype"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>

```

```

</option>
<option optionKey="10" grammar="doctype '! gslcompanyprefix'.gdti.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="9" grammar="doctype '! gslcompanyprefix'.gdti.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="8" grammar="doctype '! gslcompanyprefix'.gdti.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="7" grammar="doctype '! gslcompanyprefix'.gdti.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="6" grammar="doctype '! gslcompanyprefix'.gdti.id.onsepc.com'">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.10 GDTI-113

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:21:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:
schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GDTI-113" optionKey="gslcompanyprefixlength" tagLength="113">
    <level type="BINARY" prefixMatch="00111010" requiredFormattingParameters="filter,tag-

```

```

length">
  <option optionKey="12" pattern="00111010([01]{3})000([01]{40})(0)([01]{58})"
grammar="00111010' filter '000' gslcompanyprefix doctype serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[01]* " bitPadDir="LEFT" bitLength="40" name="gslcompanyprefix"/>
  <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]* "
bitPadDir="LEFT" bitLength="1" name="doctype"/>
  <field seq="4" decimalMinimum="0" decimalMaximum="999999999999999999" char-
acterSet="[01]* " bitPadDir="LEFT" bitLength="58" name="serial"/>
</option>
  <option optionKey="11" pattern="00111010([01]{3})001([01]{37})([01]{4})([01]
{58})" grammar="00111010' filter '001' gslcompanyprefix doctype serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="37" name="gslcompanyprefix"/>
  <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="doctype"/>
  <field seq="4" decimalMinimum="0" decimalMaximum="999999999999999999" char-
acterSet="[01]* " bitPadDir="LEFT" bitLength="58" name="serial"/>
</option>
  <option optionKey="10" pattern="00111010([01]{3})010([01]{34})([01]{7})([01]
{58})" grammar="00111010' filter '010' gslcompanyprefix doctype serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
  <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="7" name="doctype"/>
  <field seq="4" decimalMinimum="0" decimalMaximum="999999999999999999" char-
acterSet="[01]* " bitPadDir="LEFT" bitLength="58" name="serial"/>
</option>
  <option optionKey="9" pattern="00111010([01]{3})011([01]{30})([01]{11})([01]
{58})" grammar="00111010' filter '011' gslcompanyprefix doctype serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
  <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="11" name="doctype"/>
  <field seq="4" decimalMinimum="0" decimalMaximum="999999999999999999" char-

```

```

acterSet="[01]*" bitPadDir="LEFT" bitLength="58" name="serial"/>
  </option>
  <option optionKey="8" pattern="00111010([01]{3})100([01]{27})([01]{14})([01]{58})" grammar="'00111010' filter '100' gslcompanyprefix doctype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="58" name="serial"/>
  </option>
  <option optionKey="7" pattern="00111010([01]{3})101([01]{24})([01]{17})([01]{58})" grammar="'00111010' filter '101' gslcompanyprefix doctype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*" bitPadDir="LEFT" bitLength="17" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="58" name="serial"/>
  </option>
  <option optionKey="6" pattern="00111010([01]{3})110([01]{20})([01]{21})([01]{58})" grammar="'00111010' filter '110' gslcompanyprefix doctype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="21" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="58" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="serial" characterSet="[0-9]*" function="SUBSTR(serial,1)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="serial" characterSet="[0-9]*" function="CONCAT(1,serial)"/>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:gdti-113" requiredFormattingParameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:gdti-113:([0-7]{1})\.[0-9]{12}\.([0-

```

```

9]{0})\.[0-9]{1,17})" grammar="" urn:epc:tag:gdti-113:' filter '.' gslcompanyprefix '.' doctype '.'
serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="11" pattern="urn:epc:tag:gdti-113:([0-7]{1})\.[0-9]{11})\.[0-
9]{1})\.[0-9]{1,17})" grammar="" urn:epc:tag:gdti-113:' filter '.' gslcompanyprefix '.' doctype '.'
serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="10" pattern="urn:epc:tag:gdti-113:([0-7]{1})\.[0-9]{10})\.[0-
9]{2})\.[0-9]{1,17})" grammar="" urn:epc:tag:gdti-113:' filter '.' gslcompanyprefix '.' doctype '.'
serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="urn:epc:tag:gdti-113:([0-7]{1})\.[0-9]{9})\.[0-9]
{3})\.[0-9]{1,17})" grammar="" urn:epc:tag:gdti-113:' filter '.' gslcompanyprefix '.' doctype '.' seri-
al">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-

```

```

9]* " length="3" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:gdti-113:([0-7]{1})\.[0-9]{8})\.[0-9]{4})\.[0-9]{1,17})" grammar="urn:epc:tag:gdti-113:' filter '! gslcompanyprefix!' doctype!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* " length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]* " length="4" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:gdti-113:([0-7]{1})\.[0-9]{7})\.[0-9]{5})\.[0-9]{1,17})" grammar="urn:epc:tag:gdti-113:' filter '! gslcompanyprefix!' doctype!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* " length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]* " length="5" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:gdti-113:([0-7]{1})\.[0-9]{6})\.[0-9]{6})\.[0-9]{1,17})" grammar="urn:epc:tag:gdti-113:' filter '! gslcompanyprefix!' doctype!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* " length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]* " length="6" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]* " name="serial"/>
  </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:gdti">
  <option optionKey="12" pattern="urn:epc:id:gdti:([0-9]{12})\.[0-9]{0})\.[0-9]{1,

```

```

17}}" grammar="urn:epc:id:gti:' gslcompanyprefix '!' doctype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="11" pattern="urn:epc:id:gti:([0-9]{11})\.[0-9]{1}\.[0-9]{1,
17}}" grammar="urn:epc:id:gti:' gslcompanyprefix '!' doctype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="10" pattern="urn:epc:id:gti:([0-9]{10})\.[0-9]{2}\.[0-9]{1,
17}}" grammar="urn:epc:id:gti:' gslcompanyprefix '!' doctype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="urn:epc:id:gti:([0-9]{9})\.[0-9]{3}\.[0-9]{1,
17}}" grammar="urn:epc:id:gti:' gslcompanyprefix '!' doctype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="urn:epc:id:gti:([0-9]{8})\.[0-9]{4}\.[0-9]{1,
17}}" grammar="urn:epc:id:gti:' gslcompanyprefix '!' doctype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>

```



```

    </option>
    <option optionKey="7" pattern="urn:epc:id:gdtd:([0-9]{7})\.[0-9]{5}\.[0-9]{1,17})" grammar="urn:epc:id:gdtd:'gslcompanyprefix' doctype 'serial'"
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="doctype"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="6" pattern="urn:epc:id:gdtd:([0-9]{6})\.[0-9]{6}\.[0-9]{1,17})" grammar="urn:epc:id:gdtd:'gslcompanyprefix' doctype 'serial'"
      <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="doctype"/>
      <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" name="serial"/>
    </option>
  </level>
  <level type="LEGACY" prefixMatch="gdtd" requiredParsingParameters="gslcompanyprefixlength">
    <option optionKey="12" pattern="gdtd=([0-9]{13})([0-9]{1,17})" grammar="gdtd='gslcompanyprefix doctype checkdigit serial'"
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtdprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="gdtd=([0-9]{13})([0-9]{1,17})" grammar="gdtd='gslcompanyprefix doctype checkdigit serial'"
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtdprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="gdtd=([0-9]{13})([0-9]{1,17})" grammar="gdtd='gslcompanyprefix doctype checkdigit serial'"
      <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtdprefix"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-9]*" name="serial"/>
    </option>
  </level>

```

```

    <option optionKey="9" pattern="gdti=([0-9]{13})([0-9]{1,17})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="8" pattern="gdti=([0-9]{13})([0-9]{1,17})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="7" pattern="gdti=([0-9]{13})([0-9]{1,17})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
    </option>

    <option optionKey="6" pattern="gdti=([0-9]{13})([0-9]{1,17})" grammar="gdti='
gslcompanyprefix doctype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
    </option>

    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" function="SUBSTR(gdtiprefix,0,12)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="doctype"
characterSet="[0-9]*" function="SUBSTR(gdtibody,gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix " characterSet = " [ 0-9 ] * " function = " SUBSTR ( gdtibody, 0,
gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,doctype)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gdtibody)"/>
</level>

<level type="LEGACY_AI" prefixMatch="(253)" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="\ (253\)([0-9]{13})([0-9]{1,17})" grammar="(253)'

```

```

gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="11" pattern="\ (253\)([0-9]{13})([0-9]{1,17})" grammar="(253)"
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="10" pattern="\ (253\)([0-9]{13})([0-9]{1,17})" grammar="(253)"
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="\ (253\)([0-9]{13})([0-9]{1,17})" grammar="(253)"
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="\ (253\)([0-9]{13})([0-9]{1,17})" grammar="(253)"
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="\ (253\)([0-9]{13})([0-9]{1,17})" grammar="(253)"
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" char-
acterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="6" pattern="\ (253\)([0-9]{13})([0-9]{1,17})" grammar="(253)"
gslcompanyprefix doctype checkdigit serial">

```

```

    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" function="SUBSTR(gdtiprefix,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="doctype"
characterSet="[0-9]*" function="SUBSTR(gdtibody,gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gdtibody,0,
gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="gdtibody"
characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,doctype)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GSLCHECKSUM(gdtibody)"/>
</level>
<level type="ELEMENT_STRING" prefixMatch="253" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="253([0-9]{13})([0-9]{1,17})" grammar="'253 '
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="253([0-9]{13})([0-9]{1,17})" grammar="'253 '
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="253([0-9]{13})([0-9]{1,17})" grammar="'253 '
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" char-
acterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="253([0-9]{13})([0-9]{1,17})" grammar="'253 '
gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"

```

```

characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" name="serial"/>
</option>
  <option optionKey="8" pattern="253([0-9]{13})([0-9]{1,17})" grammar="'253'gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="253([0-9]{13})([0-9]{1,17})" grammar="'253'gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="253([0-9]{13})([0-9]{1,17})" grammar="'253'gslcompanyprefix doctype checkdigit serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" length="13" padChar="0" padDir="LEFT" name="gdtiprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gdtibody" characterSet="[0-9]*" function="SUBSTR(gdtiprefix,0,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="doctype" characterSet="[0-9]*" function="SUBSTR(gdtibody,gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(gdtibody,0,gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="gdtibody" characterSet="[0-9]*" length="12" function="CONCAT(gslcompanyprefix,doctype)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit" characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gdtibody)"/>
</level>
<level type="ONS_HOSTNAME">
  <option optionKey="12" grammar="doctype!'gslcompanyprefix'.gdti.id.onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*" length="0" padChar="0" padDir="LEFT" name="doctype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>

```

```

</option>
<option optionKey="11" grammar="doctype '! gslcompanyprefix '.gdti.id.onsepc.com '">
  <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="10" grammar="doctype '! gslcompanyprefix '.gdti.id.onsepc.com '">
  <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="9" grammar="doctype '! gslcompanyprefix '.gdti.id.onsepc.com '">
  <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="8" grammar="doctype '! gslcompanyprefix '.gdti.id.onsepc.com '">
  <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="7" grammar="doctype '! gslcompanyprefix '.gdti.id.onsepc.com '">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
<option optionKey="6" grammar="doctype '! gslcompanyprefix '.gdti.id.onsepc.com '">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="doctype"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
</option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.11 GRAI-64

<? xml version="1.0" encoding="UTF-8"?>

```

<tdt:epcTagDataT ranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GRAI-64" optionKey="gslcompanyprefixlength" tagLength="64">
    <level type="BINARY" prefixMatch="00001010" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001010' filter gslcompanyprefixindex assettype serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]*"
bitPadDir="LEFT" bitLength="20" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
bitPadDir="LEFT" bitLength="19" name="serial"/>
      </option>
      <option optionKey="11" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001010' filter gslcompanyprefixindex assettype serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]*"
bitPadDir="LEFT" bitLength="20" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
bitPadDir="LEFT" bitLength="19" name="serial"/>
      </option>
      <option optionKey="10" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001010' filter gslcompanyprefixindex assettype serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]*"
bitPadDir="LEFT" bitLength="20" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
bitPadDir="LEFT" bitLength="19" name="serial"/>
      </option>
      <option optionKey="9" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001010' filter gslcompanyprefixindex assettype serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>

```



```

    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*" bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <option optionKey="8" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001010' filter gslcompanyprefixindex assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*" bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <option optionKey="7" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001010' filter gslcompanyprefixindex assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*" bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <option optionKey="6" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
grammar="00001010' filter gslcompanyprefixindex assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*" bitPadDir="LEFT" bitLength="19" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="TABLELOOKUP(gslcompanyprefixindex, tdt64bitcpi, gslcompanyprefixindex, gslcompanyprefix)" tableURL="http://www.onsepc.com/

```

```

ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
tableParams="gslcompanyprefixindex"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
gslcompanyprefixlength" characterSet="[0-9]*" function="LENGTH(gslcompanyprefix)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="
gslcompanyprefixindex" characterSet="[0-9]*" function="TABLELOOKUP(gslcompanyprefix,
tdt64bitcpi, gslcompanyprefix, gslcompanyprefixindex)" tableURL="http://www. onsepc. com/
ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index"
tableParams="gslcompanyprefix"/>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:grai-64" requiredFormattingPa-
rameters="filter,taglength">
    <option optionKey="12" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{12}\.[0-9]
{0})\.[0-9]{1,6}" grammar="urn:epc:tag:grai-64:' filter '!' gslcompanyprefix '!' assettype '!' seri-
al">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{11}\.[0-9]
{1})\.[0-9]{1,6}" grammar="urn:epc:tag:grai-64:' filter '!' gslcompanyprefix '!' assettype '!' seri-
al">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{10}\.[0-9]
{2})\.[0-9]{1,6}" grammar="urn:epc:tag:grai-64:' filter '!' gslcompanyprefix '!' assettype '!' seri-
al">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>

```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{9}\.[0-9]{3}\.[0-9]{1,6})" grammar="urn:epc:tag:grai-64:' filter '.' gslcompanyprefix '.' assettype '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{8}\.[0-9]{4}\.[0-9]{1,6})" grammar="urn:epc:tag:grai-64:' filter '.' gslcompanyprefix '.' assettype '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{7}\.[0-9]{5}\.[0-9]{1,6})" grammar="urn:epc:tag:grai-64:' filter '.' gslcompanyprefix '.' assettype '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{6}\.[0-9]{6}\.[0-9]{1,6})" grammar="urn:epc:tag:grai-64:' filter '.' gslcompanyprefix '.' assettype '.' serial">

```

```

al">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:grai">
    <option optionKey="12" pattern="urn:epc:id:grai:([0-9]{12})\.([0-9]{0})\.([0-9]{1,
6})" grammar="urn:epc:id:grai:' gs1companyprefix '.' assettype '.' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:id:grai:([0-9]{11})\.([0-9]{1})\.([0-9]{1,
6})" grammar="urn:epc:id:grai:' gs1companyprefix '.' assettype '.' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="urn:epc:id:grai:([0-9]{10})\.([0-9]{2})\.([0-9]{1,
6})" grammar="urn:epc:id:grai:' gs1companyprefix '.' assettype '.' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="
[0-9]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="urn:epc:id:grai:([0-9]{9})\.([0-9]{3})\.([0-9]{1,
6})" grammar="urn:epc:id:grai:' gs1companyprefix '.' assettype '.' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:id:grai:([0-9]{8})\.[0-9]{4}\.[0-9]{1,6})" grammar="urn:epc:id:grai:'gslcompanyprefix'!assettype!'serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:id:grai:([0-9]{7})\.[0-9]{5}\.[0-9]{1,6})" grammar="urn:epc:id:grai:'gslcompanyprefix'!assettype!'serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:id:grai:([0-9]{6})\.[0-9]{6}\.[0-9]{1,6})" grammar="urn:epc:id:grai:'gslcompanyprefix'!assettype!'serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*" name="serial"/>
  </option>
</level>
<level type="LEGACY" prefixMatch="grai=" requiredParsingParameters="gslcompanyprefixlength">
  <option optionKey="12" pattern="grai=([0-9]{15,20})" grammar="'grai=' '0' gslcompanyprefix assettype checkdigit serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999999999" characterSet="[0-9]*" name="grai"/>
  </option>
  <option optionKey="11" pattern="grai=([0-9]{15,20})" grammar="'grai=' '0' gslcompanyprefix assettype checkdigit serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999999999"

```

```

characterSet="[0-9]*" name="grai"/>
</option>
<option optionKey="10" pattern="grai = ([0-9]{15,20})" grammar="'grai=' '0'
gslcompanyprefix assettype checkdigit serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
</option>
<option optionKey="9" pattern="grai = ([0-9]{15,20})" grammar="'grai=' '0'
gslcompanyprefix assettype checkdigit serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
</option>
<option optionKey="8" pattern="grai = ([0-9]{15,20})" grammar="'grai=' '0'
gslcompanyprefix assettype checkdigit serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
</option>
<option optionKey="7" pattern="grai = ([0-9]{15,20})" grammar="'grai=' '0'
gslcompanyprefix assettype checkdigit serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
</option>
<option optionKey="6" pattern="grai = ([0-9]{15,20})" grammar="'grai=' '0'
gslcompanyprefix assettype checkdigit serial">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
graiprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(grai,1,12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="assettype"
characterSet="[0-9]*" function="SUBSTR(graiprefixremainder,gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serial"
characterSet="[0-9]*" function="SUBSTR(grai,14)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR (graiprefixremainder, 0,
gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiprefix"
characterSet="[0-9]*" length="13" function="CONCAT(0,gslcompanyprefix,assettype)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(graiprefix)"/>
</level>
<level type="LEGACY_AI" prefixMatch="(8003)" requiredParsingParameters="
gslcompanyprefixlength">

```

```

    <option optionKey="12" pattern="\ (8003\ ) ([0-9]{15,20})" grammar="'(8003)' '0 '
gslcompanyprefix assettype checkdigit serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="11" pattern="\ (8003\ ) ([0-9]{15,20})" grammar="'(8003)' '0 '
gslcompanyprefix assettype checkdigit serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="10" pattern="\ (8003\ ) ([0-9]{15,20})" grammar="'(8003)' '0 '
gslcompanyprefix assettype checkdigit serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="9" pattern="\ (8003\ ) ([0-9]{15,20})" grammar="'(8003)' '0 '
gslcompanyprefix assettype checkdigit serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="8" pattern="\ (8003\ ) ([0-9]{15,20})" grammar="'(8003)' '0 '
gslcompanyprefix assettype checkdigit serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="7" pattern="\ (8003\ ) ([0-9]{15,20})" grammar="'(8003)' '0 '
gslcompanyprefix assettype checkdigit serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="6" pattern="\ (8003\ ) ([0-9]{15,20})" grammar="'(8003)' '0 '
gslcompanyprefix assettype checkdigit serial">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
graiprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(grai,1,12)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="assettype"
characterSet="[0-9]*" function="SUBSTR(graiprefixremainder,gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serial"
characterSet="[0-9]*" function="SUBSTR(grai,14)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR (graiprefixremainder, 0,

```



```

gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiprefix"
characterSet="[0-9]*" length="13" function="CONCAT(0,gslcompanyprefix,assettype)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(graiprefix)"/>
</level>
<level type="ELEMENT_STRING" prefixMatch="8003" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="8003([0-9]{15,20})" grammar="'8003' '0'
gslcompanyprefix assettype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="11" pattern="8003([0-9]{15,20})" grammar="'8003' '0'
gslcompanyprefix assettype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="10" pattern="8003([0-9]{15,20})" grammar="'8003' '0'
gslcompanyprefix assettype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="9" pattern="8003([0-9]{15,20})" grammar="'8003' '0'
gslcompanyprefix assettype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="8" pattern="8003([0-9]{15,20})" grammar="'8003' '0'
gslcompanyprefix assettype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="7" pattern="8003([0-9]{15,20})" grammar="'8003' '0'
gslcompanyprefix assettype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>
    <option optionKey="6" pattern="8003([0-9]{15,20})" grammar="'8003' '0'
gslcompanyprefix assettype checkdigit serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" name="grai"/>
    </option>

```

```

    <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
    graiprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(grai,1,12)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="assettype"
    characterSet="[0-9]*" function="SUBSTR(graiprefixremainder,gs1companylength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serial"
    characterSet="[0-9]*" function="SUBSTR(grai,14)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="
    gs1companyprefix" characterSet="[0-9]*" function="SUBSTR(graiprefixremainder,0,
    gs1companylength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiprefix"
    characterSet="[0-9]*" length="13" function="CONCAT(0,gs1companyprefix,assettype)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
    characterSet="[0-9]*" length="1" function="GS1CHECKSUM(graiprefix)"/>
  </level>
  <level type="ONS_HOSTNAME">
    <option optionKey="12" grammar="assettype '.' gs1companyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
      length="0" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999"
      characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="11" grammar="assettype '.' gs1companyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
      length="1" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
     ="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="10" grammar="assettype '.' gs1companyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
      9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
     ="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="9" grammar="assettype '.' gs1companyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
      9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
     ="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    </option>
    <option optionKey="8" grammar="assettype '.' gs1companyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
      9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="

```

```

[0-9]* " length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="7" grammar="assettype!' gs1companyprefix'.grai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]* " length="5" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
  <option optionKey="6" grammar="assettype!' gs1companyprefix'.grai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  </option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.12 GRAI-96

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GRAI-96" optionKey="gs1companyprefixlength" tagLength="96">
    <level type="BINARY" prefixMatch="00110011" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00110011([01]{3})000([01]{40})([01]{4})([01]
{38})" grammar="'00110011' filter '000' gs1companyprefix assettype serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[01]* " bitPadDir="LEFT" bitLength="40" name="gs1companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
      </option>
      <option optionKey="11" pattern="00110011([01]{3})001([01]{37})([01]{7})([01]
{38})" grammar="'00110011' filter '001' gs1companyprefix assettype serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet

```

```

="[01]* " bitPadDir="LEFT" bitLength="37" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="7" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
</option>
<option optionKey="10" pattern="00110011([01]{3})010([01]{34})([01]{10})([01]
{38})" grammar="00110011' filter '010' gs1companyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet=
="[01]* " bitPadDir="LEFT" bitLength="34" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="10" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
</option>
<option optionKey="9" pattern="00110011([01]{3})011([01]{30})([01]{14})([01]
{38})" grammar="00110011' filter '011' gs1companyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet=
="[01]* " bitPadDir="LEFT" bitLength="30" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
</option>
<option optionKey="8" pattern="00110011([01]{3})100([01]{27})([01]{17})([01]
{38})" grammar="00110011' filter '100' gs1companyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="27" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="17" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
</option>
<option optionKey="7" pattern="00110011([01]{3})101([01]{24})([01]{20})([01]
{38})" grammar="00110011' filter '101' gs1companyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="24" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
  </option>
  <option optionKey="6" pattern="00110011([01]{3})110([01]{20})([01]{24})([01]
{38})" grammar="'00110011' filter '110' gs1companyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="24" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[01]* " bitPadDir="LEFT" bitLength="38" name="serial"/>
  </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:grai-96" requiredFormattingPa-
rameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[(0-9]{12})\.[(0-9]
{0})\.[(0-9]{1,12})" grammar="urn:epc:tag:grai-96:' filter '!' gs1companyprefix '!' ser-
ial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]* "
length="0" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"
characterSet="[0-9]* " name="serial"/>
  </option>
  <option optionKey="11" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[(0-9]{11})\.[(0-9]
{1})\.[(0-9]{1,12})" grammar="urn:epc:tag:grai-96:' filter '!' gs1companyprefix '!' ser-
ial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]* "
length="1" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943"

```

```

characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="10" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{10}\.[0-9]{2}\.[0-9]{1,12}" grammar="urn:epc:tag:grai-96:' filter '!' gslcompanyprefix '!' assettype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="9" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{9}\.[0-9]{3}\.[0-9]{1,12}" grammar="urn:epc:tag:grai-96:' filter '!' gslcompanyprefix '!' assettype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="8" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{8}\.[0-9]{4}\.[0-9]{1,12}" grammar="urn:epc:tag:grai-96:' filter '!' gslcompanyprefix '!' assettype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
</option>
<option optionKey="7" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{7}\.[0-9]{5}\.[0-9]{1,12}" grammar="urn:epc:tag:grai-96:' filter '!' gslcompanyprefix '!' assettype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{6}\.[0-9]{6}\.[0-9]{1,12}" grammar="urn:epc:tag:grai-96:' filter '.' gs1companyprefix '.' assettype '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:grai">
  <option optionKey="12" pattern="urn:epc:id:grai:([0-9]{12})\.[0-9]{0}\.[0-9]{1,12}" grammar="urn:epc:id:grai:' gs1companyprefix '.' assettype '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*" length="0" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="urn:epc:id:grai:([0-9]{11})\.[0-9]{1}\.[0-9]{1,12}" grammar="urn:epc:id:grai:' gs1companyprefix '.' assettype '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*" length="1" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="urn:epc:id:grai:([0-9]{10})\.[0-9]{2}\.[0-9]{1,12}" grammar="urn:epc:id:grai:' gs1companyprefix '.' assettype '.' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>

```



```

    <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="urn:epc:id:grai:([0-9]{9})\.[0-9]{3}\.[0-9]{1,12}" grammar="urn:epc:id:grai:'gslcompanyprefix'! assettype!' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="urn:epc:id:grai:([0-9]{8})\.[0-9]{4}\.[0-9]{1,12}" grammar="urn:epc:id:grai:'gslcompanyprefix'! assettype!' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="7" pattern="urn:epc:id:grai:([0-9]{7})\.[0-9]{5}\.[0-9]{1,12}" grammar="urn:epc:id:grai:'gslcompanyprefix'! assettype!' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
  <option optionKey="6" pattern="urn:epc:id:grai:([0-9]{6})\.[0-9]{6}\.[0-9]{1,12}" grammar="urn:epc:id:grai:'gslcompanyprefix'! assettype!' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-9]*" name="serial"/>
  </option>
</level>
<level type="LEGACY" prefixMatch="grai" requiredParsingParameters="

```


[illegible]


```

    </option>
    <option optionKey="10" grammar="assettype '!' gslcompanyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="9" grammar="assettype '!' gslcompanyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="8" grammar="assettype '!' gslcompanyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="7" grammar="assettype '!' gslcompanyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="6" grammar="assettype '!' gslcompanyprefix '.grai.id.onsepc.com'">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
  </level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.13 GRAI-170

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:21:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="urn:epcglobal:tdt:xsd:1
EpcTagDataTranslation.xsd">
  <scheme name="GRAI-170" optionKey="gslcompanyprefixlength" tagLength="170">

```



```

    <level type="BINARY" prefixMatch="00110111" requiredFormattingParameters="filter,tag-
length">
        <option optionKey="12" pattern="00110111([01]{3})000([01]{40})([01]{4})([01]
{112})" grammar="00110111' filter '000' gslcompanyprefix assettype serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]* "
bitPadDir="LEFT" bitLength="40" name="gslcompanyprefix"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[01]* "
bitPadDir="LEFT" bitLength="4" name="assettype"/>
            <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="112" com-
paction="7-bit" name="serial"/>
        </option>
        <option optionKey="11" pattern="00110111([01]{3})001([01]{37})([01]{7})([01]
{112})" grammar="00110111' filter '001' gslcompanyprefix assettype serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="37" name="gslcompanyprefix"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]* "
bitPadDir="LEFT" bitLength="7" name="assettype"/>
            <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="112" com-
paction="7-bit" name="serial"/>
        </option>
        <option optionKey="10" pattern="00110111([01]{3})010([01]{34})([01]{10})([01]
{112})" grammar="00110111' filter '010' gslcompanyprefix assettype serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]* "
bitPadDir="LEFT" bitLength="10" name="assettype"/>
            <field seq="4" characterSet="[01]* " bitPadDir="RIGHT" bitLength="112" com-
paction="7-bit" name="serial"/>
        </option>
        <option optionKey="9" pattern="00110111([01]{3})011([01]{30})([01]{14})([01]
{112})" grammar="00110111' filter '011' gslcompanyprefix assettype serial">
            <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
            <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
            <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="14" name="assettype"/>

```

```

    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="112" compaction="7-bit" name="serial"/>
  </option>
  <option optionKey="8" pattern="00110111([01]{3})100([01]{27})([01]{17})([01]{112})" grammar="00110111'filter'100'gslcompanyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*" bitPadDir="LEFT" bitLength="17" name="assettype"/>
    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="112" compaction="7-bit" name="serial"/>
  </option>
  <option optionKey="7" pattern="00110111([01]{3})101([01]{24})([01]{20})([01]{112})" grammar="00110111'filter'101'gslcompanyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="assettype"/>
    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="112" compaction="7-bit" name="serial"/>
  </option>
  <option optionKey="6" pattern="00110111([01]{3})110([01]{20})([01]{24})([01]{112})" grammar="00110111'filter'110'gslcompanyprefix assettype serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*" bitPadDir="LEFT" bitLength="24" name="assettype"/>
    <field seq="4" characterSet="[01]*" bitPadDir="RIGHT" bitLength="112" compaction="7-bit" name="serial"/>
  </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:grai-170" requiredFormattingParameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:grai-170:([0-7]{1})\.[0-9]{12}\.([0-9]{0})\.[0-9]{0}\.([!%-? A-Z _ a-z \ x22 ] { 1, 16 })" grammar="' urn: epc: tag: grai-170: ' filter ' .' gslcompanyprefix '!' assettype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"

```

```
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="11" pattern="urn:epc:tag:grai-170:([0-7]{1})\.([0-9]{11})\.([0-9]{1})\.(?!%?-? A-Z_a-z\x22){1,16}" grammar="' urn:epc:tag:grai-170:' filter '! gslcompanyprefix!' assettype!' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="10" pattern="urn:epc:tag:grai-170:([0-7]{1})\.([0-9]{10})\.([0-9]{2})\.(?!%?-? A-Z_a-z\x22){1,16}" grammar="' urn:epc:tag:grai-170:' filter '! gslcompanyprefix!' assettype!' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="9" pattern="urn:epc:tag:grai-170:([0-7]{1})\.([0-9]{9})\.([0-9]{3})\.(?!%?-? A-Z_a-z\x22){1,16}" grammar="' urn:epc:tag:grai-170:' filter '! gslcompanyprefix!' assettype!' serial'"
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="8" pattern="urn:epc:tag:grai-170:([0-7]{1})\.([0-9]{8})\.([0-9]{4})\.(?!%?-? A-Z_a-z\x22){1,16}" grammar="' urn:epc:tag:grai-170:' filter '! gslcompanyprefix!' assettype!' serial'"
```

```

    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-
9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="7" pattern="urn:epc:tag:grai-170:([0-7]{1})\.[(0-9]{7})\.[(0-9]
{5})\.[(0-9]{1,16})" grammar="urn:epc:tag:grai-170:' filter '!' gs1companyprefix
'!' assettype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="6" pattern="urn:epc:tag:grai-170:([0-7]{1})\.[(0-9]{6})\.[(0-9]
{6})\.[(0-9]{1,16})" grammar="urn:epc:tag:grai-170:' filter '!' gs1companyprefix
'!' assettype '!' serial">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="4" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:grai">
    <option optionKey="12" pattern="urn:epc:id:grai:([0-9]{12})\.[(0-9]{0})\.[(0-9]
{1,16})" grammar="urn:epc:id:grai:' gs1companyprefix '!' assettype '!' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
length="0" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="urn:epc:id:grai:([0-9]{11})\.[(0-9]{1})\.[(0-9]
{1,16})" grammar="urn:epc:id:grai:' gs1companyprefix '!' assettype '!' serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet

```

```

="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
length="1" padChar="0" padDir="LEFT" name="assettype"/>
  <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
  <option optionKey="10" pattern="urn:epc:id:grai:([0-9]{10})\.[0-9]{2})\.[!%-? A-
Z_a-z\x22]{1,16})" grammar="urn:epc:id:grai:' gs1companyprefix '!' assettype '!' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
  <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
  <option optionKey="9" pattern="urn:epc:id:grai:([0-9]{9})\.[0-9]{3})\.[!%-? A-Z_
a-z\x22]{1,16})" grammar="urn:epc:id:grai:' gs1companyprefix '!' assettype '!' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
  <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
  <option optionKey="8" pattern="urn:epc:id:grai:([0-9]{8})\.[0-9]{4})\.[!%-? A-Z_
a-z\x22]{1,16})" grammar="urn:epc:id:grai:' gs1companyprefix '!' assettype '!' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>
  <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
  <option optionKey="7" pattern="urn:epc:id:grai:([0-9]{7})\.[0-9]{5})\.[!%-? A-Z_
a-z\x22]{1,16})" grammar="urn:epc:id:grai:' gs1companyprefix '!' assettype '!' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
  <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
  <option optionKey="6" pattern="urn:epc:id:grai:([0-9]{6})\.[0-9]{6})\.[!%-? A-Z_
a-z\x22]{1,16})" grammar="urn:epc:id:grai:' gs1companyprefix '!' assettype '!' serial"
  <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>

```

```

    <field seq="3" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
  </option>
</level>
  <level type="LEGACY" prefixMatch="grai" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="grai=([0-9]{14})([!%-? A-Z_a-z\x22]{1,16})"
grammar="grai='0' gslcompanyprefix assettype checkdigit serial">
      <field seq="1" characterSet="[0-9]*" length="14" decimalMinimum="0" deci-
malMaximum="99999999999999" name="graibody"/>
      <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="11" pattern="grai=([0-9]{14})([!%-? A-Z_a-z\x22]{1,16})"
grammar="grai='0' gslcompanyprefix assettype checkdigit serial">
      <field seq="1" characterSet="[0-9]*" length="14" decimalMinimum="0" deci-
malMaximum="99999999999999" name="graibody"/>
      <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="10" pattern="grai=([0-9]{14})([!%-? A-Z_a-z\x22]{1,16})"
grammar="grai='0' gslcompanyprefix assettype checkdigit serial">
      <field seq="1" characterSet="[0-9]*" length="14" decimalMinimum="0" deci-
malMaximum="99999999999999" name="graibody"/>
      <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="9" pattern="grai=([0-9]{14})([!%-? A-Z_a-z\x22]{1,16})" gram-
mar="grai='0' gslcompanyprefix assettype checkdigit serial">
      <field seq="1" characterSet="[0-9]*" length="14" decimalMinimum="0" deci-
malMaximum="99999999999999" name="graibody"/>
      <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="8" pattern="grai=([0-9]{14})([!%-? A-Z_a-z\x22]{1,16})" gram-
mar="grai='0' gslcompanyprefix assettype checkdigit serial">
      <field seq="1" characterSet="[0-9]*" length="14" decimalMinimum="0" deci-
malMaximum="99999999999999" name="graibody"/>
      <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="7" pattern="grai=([0-9]{14})([!%-? A-Z_a-z\x22]{1,16})" gram-
mar="grai='0' gslcompanyprefix assettype checkdigit serial">
      <field seq="1" characterSet="[0-9]*" length="14" decimalMinimum="0" deci-
malMaximum="99999999999999" name="graibody"/>
      <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
    </option>
    <option optionKey="6" pattern="grai=([0-9]{14})([!%-? A-Z_a-z\x22]{1,16})" gram-
mar="grai='0' gslcompanyprefix assettype checkdigit serial">

```

```

    <field seq="1" characterSet="[0-9]*" length="14" decimalMinimum="0" decimalMaximum="9999999999999999" name="graibody"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="graiprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(graibody,1,12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="assettype" characterSet="[0-9]*" function="SUBSTR(graiprefixremainder,gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(graiprefixremainder,0,gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiprefix" characterSet="[0-9]*" length="13" function="CONCAT(0,gslcompanyprefix,assettype)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit" characterSet="[0-9]*" length="1" function="GS1CHECKSUM(graiprefix)"/>
</level>
<level type="LEGACY_ALT" prefixMatch="grai13=" requiredParsingParameters="gslcompanyprefixlength">
  <option optionKey="12" pattern="grai13=([0-9]{13})([!%?-? A-Z_a-z\x22]{1,16})" grammar="grai13='gslcompanyprefix assettype checkdigit serial'">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="9999999999999999" name="grai13body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="11" pattern="grai13=([0-9]{13})([!%?-? A-Z_a-z\x22]{1,16})" grammar="grai13='gslcompanyprefix assettype checkdigit serial'">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="9999999999999999" name="grai13body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="10" pattern="grai13=([0-9]{13})([!%?-? A-Z_a-z\x22]{1,16})" grammar="grai13='gslcompanyprefix assettype checkdigit serial'">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="9999999999999999" name="grai13body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="9" pattern="grai13=([0-9]{13})([!%?-? A-Z_a-z\x22]{1,16})" grammar="grai13='gslcompanyprefix assettype checkdigit serial'">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="9999999999999999" name="grai13body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <option optionKey="8" pattern="grai13=([0-9]{13})([!%?-? A-Z_a-z\x22]{1,16})"

```



```

grammar="\" grail3=' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="99999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\\x22]*" name="serial"/>
</option>
<option optionKey="7" pattern="grail3=([0-9]{13})([!%?-? A-Z_a-z\\x22]{1,16})"
grammar="\" grail3=' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="99999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\\x22]*" name="serial"/>
</option>
<option optionKey="6" pattern="grail3=([0-9]{13})([!%?-? A-Z_a-z\\x22]{1,16})"
grammar="\" grail3=' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="99999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\\x22]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
graiPREFIXremainder" characterSet="[0-9]*" length="12" function="SUBSTR(grail3body,0,
12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="assettype"
characterSet="[0-9]*" function="SUBSTR(graiPREFIXremainder,gslcompanyPREFIXlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="
gslcompanyPREFIX" characterSet="[0-9]*" function="SUBSTR(graiPREFIXremainder,0,
gslcompanyPREFIXlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiPREFIX"
characterSet="[0-9]*" length="13" function="CONCAT(0,gslcompanyPREFIX,assettype)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GSLCHECKSUM(graiPREFIX)"/>
</level>
<level type="LEGACY_AI" PREFIXMatch="(8003)" requiredParsingParameters="
gslcompanyPREFIXlength">
    <option optionKey="12" pattern="\\(8003\\)0([0-9]{13})([!%?-? A-Z_a-z\\x22]{1,16})"
grammar="\"(8003)'0' gslcompanyPREFIX assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="99999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\\x22]*" name="serial"/>
</option>
<option optionKey="11" pattern="\\(8003\\)0([0-9]{13})([!%?-? A-Z_a-z\\x22]{1,16})"
grammar="\"(8003)'0' gslcompanyPREFIX assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="99999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\\x22]*" name="serial"/>

```

```

</option>
<option optionKey="10" pattern="\ (8003\) 0([0-9]{13})([!%-? A-Z_a-z\x22]{1,16})"
grammar="(8003)''0' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="999999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="9" pattern="\ (8003\) 0([0-9]{13})([!%-? A-Z_a-z\x22]{1,16})"
grammar="(8003)''0' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="999999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="8" pattern="\ (8003\) 0([0-9]{13})([!%-? A-Z_a-z\x22]{1,16})"
grammar="(8003)''0' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="999999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="7" pattern="\ (8003\) 0([0-9]{13})([!%-? A-Z_a-z\x22]{1,16})"
grammar="(8003)''0' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="999999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<option optionKey="6" pattern="\ (8003\) 0([0-9]{13})([!%-? A-Z_a-z\x22]{1,16})"
grammar="(8003)''0' gslcompanyprefix assettype checkdigit serial">
    <field seq="1" characterSet="[0-9]*" length="13" decimalMinimum="0" decimalMaximum="999999999999999" name="grail3body"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="serial"/>
</option>
<rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
graiprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(grail3body, 0,
12)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="assettype"
characterSet="[0-9]*" function="SUBSTR(graiprefixremainder,gslcompanyprefixlength)"/>
<rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(graiprefixremainder, 0,
gslcompanyprefixlength)"/>
<rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiprefix"
characterSet="[0-9]*" length="13" function="CONCAT(0,gslcompanyprefix,assettype)"/>
<rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(graiprefix)"/>

```

```

</level>
<level type = " ELEMENT _ STRING " prefixMatch = " 8003 " requiredParsingParameters = "
gslcompanyprefixlength">
  <option optionKey = " 12 " pattern = " 80030 ([0-9]{13}) ([!%-? A-Z_a-z\x22]{1,16}) "
grammar = " 8003 ' ' 0 ' gslcompanyprefix assettype checkdigit serial ">
    <field seq = " 1 " characterSet = "[0-9]* " length = " 13 " decimalMinimum = " 0 " deci-
malMaximum = " 9999999999999 " name = "grai13body"/>
    <field seq = " 2 " characterSet = "[!%-? A-Z_a-z\x22]* " name = "serial"/>
  </option>
  <option optionKey = " 11 " pattern = " 80030 ([0-9]{13}) ([!%-? A-Z_a-z\x22]{1,16}) "
grammar = " 8003 ' ' 0 ' gslcompanyprefix assettype checkdigit serial ">
    <field seq = " 1 " characterSet = "[0-9]* " length = " 13 " decimalMinimum = " 0 " deci-
malMaximum = " 9999999999999 " name = "grai13body"/>
    <field seq = " 2 " characterSet = "[!%-? A-Z_a-z\x22]* " name = "serial"/>
  </option>
  <option optionKey = " 10 " pattern = " 80030 ([0-9]{13}) ([!%-? A-Z_a-z\x22]{1,16}) "
grammar = " 8003 ' ' 0 ' gslcompanyprefix assettype checkdigit serial ">
    <field seq = " 1 " characterSet = "[0-9]* " length = " 13 " decimalMinimum = " 0 " deci-
malMaximum = " 9999999999999 " name = "grai13body"/>
    <field seq = " 2 " characterSet = "[!%-? A-Z_a-z\x22]* " name = "serial"/>
  </option>
  <option optionKey = " 9 " pattern = " 80030 ([0-9]{13}) ([!%-? A-Z_a-z\x22]{1,16}) " gram-
mar = " 8003 ' ' 0 ' gslcompanyprefix assettype checkdigit serial ">
    <field seq = " 1 " characterSet = "[0-9]* " length = " 13 " decimalMinimum = " 0 " deci-
malMaximum = " 9999999999999 " name = "grai13body"/>
    <field seq = " 2 " characterSet = "[!%-? A-Z_a-z\x22]* " name = "serial"/>
  </option>
  <option optionKey = " 8 " pattern = " 80030 ([0-9]{13}) ([!%-? A-Z_a-z\x22]{1,16}) " gram-
mar = " 8003 ' ' 0 ' gslcompanyprefix assettype checkdigit serial ">
    <field seq = " 1 " characterSet = "[0-9]* " length = " 13 " decimalMinimum = " 0 " deci-
malMaximum = " 9999999999999 " name = "grai13body"/>
    <field seq = " 2 " characterSet = "[!%-? A-Z_a-z\x22]* " name = "serial"/>
  </option>
  <option optionKey = " 7 " pattern = " 80030 ([0-9]{13}) ([!%-? A-Z_a-z\x22]{1,16}) " gram-
mar = " 8003 ' ' 0 ' gslcompanyprefix assettype checkdigit serial ">
    <field seq = " 1 " characterSet = "[0-9]* " length = " 13 " decimalMinimum = " 0 " deci-
malMaximum = " 9999999999999 " name = "grai13body"/>
    <field seq = " 2 " characterSet = "[!%-? A-Z_a-z\x22]* " name = "serial"/>
  </option>
  <option optionKey = " 6 " pattern = " 80030 ([0-9]{13}) ([!%-? A-Z_a-z\x22]{1,16}) " gram-
mar = " 8003 ' ' 0 ' gslcompanyprefix assettype checkdigit serial ">
    <field seq = " 1 " characterSet = "[0-9]* " length = " 13 " decimalMinimum = " 0 " deci-
malMaximum = " 9999999999999 " name = "grai13body"/>

```

```

    <field seq="2" characterSet="[!%/? A-Z_a-z\x22]*" name="serial"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
    graiprefixremainder" characterSet="[0-9]*" length="12" function="SUBSTR(graip13body, 0,
    12)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="assettype"
    characterSet="[0-9]*" function="SUBSTR(graiprefixremainder, gslcompanyprefixlength)"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="
    gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(graiprefixremainder, 0,
    gslcompanyprefixlength)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiprefix"
    characterSet="[0-9]*" length="13" function="CONCAT(0, gslcompanyprefix, assettype)"/>
  <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
    characterSet="[0-9]*" length="1" function="GSLCHECKSUM(graiprefix)"/>
</level>
<level type="ONS_HOSTNAME">
  <option optionKey="12" grammar="assettype '! gslcompanyprefix ',grai.id,onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="0" characterSet="[0-9]*"
    length="0" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999"
    characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="11" grammar="assettype '! gslcompanyprefix ',grai.id,onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
    length="1" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
    ="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="10" grammar="assettype '! gslcompanyprefix ',grai.id,onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-
    9]*" length="2" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
    ="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="9" grammar="assettype '! gslcompanyprefix ',grai.id,onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-
    9]*" length="3" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet
    ="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="8" grammar="assettype '! gslcompanyprefix ',grai.id,onsepc.com">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-
    9]*" length="4" padChar="0" padDir="LEFT" name="assettype"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="7" grammar="assettype '.' gslcompanyprefix '.grai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*" length="5" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="6" grammar="assettype '.' gslcompanyprefix '.grai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="assettype"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.14 GIAI-64



```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GIAI-64" optionKey="gslcompanyprefixlength" tagLength="64">
    <level type="BINARY" prefixMatch="00001011" requiredFormattingParameters="filter,tag-length">
      <option optionKey="12" pattern="00001011([01]{3})([01]{14})([01]{39})" grammar="00001011' filter gslcompanyprefixindex indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]*" bitPadDir="LEFT" bitLength="39" name="indassetref"/>
      </option>
      <option optionKey="11" pattern="00001011([01]{3})([01]{14})([01]{39})" grammar="00001011' filter gslcompanyprefixindex indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*" bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
      </option>
    </level>
  </scheme>
</tdt:epcTagDataTranslation>

```

```

    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[01]" bitPadDir="LEFT" bitLength="39" name="indassetref"/>
  </option>
  <option optionKey="10" pattern="00001011([01]{3})([01]{14})([01]{39})" grammar="
00001011' filter gslcompanyprefixindex indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[01]" bitPadDir="LEFT" bitLength="39" name="indassetref"/>
  </option>
  <option optionKey="9" pattern="00001011([01]{3})([01]{14})([01]{39})" grammar="
00001011' filter gslcompanyprefixindex indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[01]" bitPadDir="LEFT" bitLength="39" name="indassetref"/>
  </option>
  <option optionKey="8" pattern="00001011([01]{3})([01]{14})([01]{39})" grammar="
00001011' filter gslcompanyprefixindex indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[01]" bitPadDir="LEFT" bitLength="39" name="indassetref"/>
  </option>
  <option optionKey="7" pattern="00001011([01]{3})([01]{14})([01]{39})" grammar="
00001011' filter gslcompanyprefixindex indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[01]" bitPadDir="LEFT" bitLength="39" name="indassetref"/>
  </option>
  <option optionKey="6" pattern="00001011([01]{3})([01]{14})([01]{39})" grammar="
00001011' filter gslcompanyprefixindex indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitPadDir="LEFT" bitLength="3" name="filter"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="
[01]*" bitPadDir="LEFT" bitLength="14" name="gslcompanyprefixindex"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[01]*" bitPadDir="LEFT" bitLength="39" name="indassetref"/>
  </option>
  <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="TABLELOOKUP(gslcompanyprefixindex,
tdt64bitcpi, gslcompanyprefixindex, gslcompanyprefix)" tableURL="http://www.onsepc.com/
ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
tableParams="gslcompanyprefixindex"/>
  <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
gslcompanyprefixlength" characterSet="[0-9]*" function="LENGTH(gslcompanyprefix)/>
  <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="
gslcompanyprefixindex" characterSet="[0-9]*" function="TABLELOOKUP(gslcompanyprefix,
tdt64bitcpi, gslcompanyprefix, gslcompanyprefixindex)" tableURL="http://www.onsepc.com/
ManagerTranslation.xml" tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index"
tableParams="gslcompanyprefix"/>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:giai-64" requiredFormattingPa-
rameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{12})\.[0-9]
{1,12})" grammar="urn:epc:tag:giai-64:'filter!' gslcompanyprefix!' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="11" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{11})\.[0-9]
{1,12})" grammar="urn:epc:tag:giai-64:'filter!' gslcompanyprefix!' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="10" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{10})\.[0-9]
{1,12})" grammar="urn:epc:tag:giai-64:'filter!' gslcompanyprefix!' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-

```



```

9] * " length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[0-9]*" name="indassetref"/>
    </option>
    <option optionKey="9" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{9}\.[0-9]{1,
12})" grammar="urn:epc:tag:giai-64:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[0-9]*" name="indassetref"/>
    </option>
    <option optionKey="8" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{8}\.[0-9]{1,
12})" grammar="urn:epc:tag:giai-64:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[0-9]*" name="indassetref"/>
    </option>
    <option optionKey="7" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{7}\.[0-9]{1,
12})" grammar="urn:epc:tag:giai-64:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[0-9]*" name="indassetref"/>
    </option>
    <option optionKey="6" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{6}\.[0-9]{1,
12})" grammar="urn:epc:tag:giai-64:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="549755813887"
characterSet="[0-9]*" name="indassetref"/>
    </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:giai">
    <option optionKey="12" pattern="urn:epc:id:giai:([0-9]{12})\.[0-9]{1,12})"

```

```

grammar="urn:epc:id:giai:'gslcompanyprefix'!indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[0-9]*" name="indassetref"/>
</option>
<option optionKey="11" pattern="urn:epc:id:giai:([0-9]{11})\.([0-9]{1,12})"
grammar="urn:epc:id:giai:'gslcompanyprefix'!indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999999" characterSet=
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[0-9]*" name="indassetref"/>
</option>
<option optionKey="10" pattern="urn:epc:id:giai:([0-9]{10})\.([0-9]{1,12})"
grammar="urn:epc:id:giai:'gslcompanyprefix'!indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999" charac-
terSet="[0-9]*" name="indassetref"/>
</option>
<option optionKey="9" pattern="urn:epc:id:giai:([0-9]{9})\.([0-9]{1,12})" grammar="
urn:epc:id:giai:'gslcompanyprefix'!indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" charac-
terSet="[0-9]*" name="indassetref"/>
</option>
<option optionKey="8" pattern="urn:epc:id:giai:([0-9]{8})\.([0-9]{1,12})" grammar="
urn:epc:id:giai:'gslcompanyprefix'!indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-
9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999999" charac-
terSet="[0-9]*" name="indassetref"/>
</option>
<option optionKey="7" pattern="urn:epc:id:giai:([0-9]{7})\.([0-9]{1,12})" grammar="
urn:epc:id:giai:'gslcompanyprefix'!indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999999" char-
acterSet="[0-9]*" name="indassetref"/>
</option>
<option optionKey="6" pattern="urn:epc:id:giai:([0-9]{6})\.([0-9]{1,12})" grammar="
urn:epc:id:giai:'gslcompanyprefix'!indassetref">

```

[illegible]

[illegible]

[illegible]

```

indassetref" characterSet="[0-9]*" function="SUBSTR(giai,gslcompanyprefixlength)"/>
    <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="
gslcompanyprefix" characterSet="[0-9]*" function="SUBSTR(giai, 0,
gslcompanyprefixlength)"/>
    </level>
    <level type="ONS_HOSTNAME">
        <option optionKey="12" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
            <field seq="1" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        </option>
        <option optionKey="11" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
            <field seq="1" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        </option>
        <option optionKey="10" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
            <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        </option>
        <option optionKey="9" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
            <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        </option>
        <option optionKey="8" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
            <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-
9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        </option>
        <option optionKey="7" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
            <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[
0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        </option>
        <option optionKey="6" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
            <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[
0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        </option>
    </level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.15 GIAI-96

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-

```

```

instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GIAI-96" optionKey="gslcompanyprefixlength" tagLength="96">
    <level type="BINARY" prefixMatch="00110100" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00110100([01]{3})000([01]{40})([01]{42})"
grammar="00110100'filter'000'gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[01]*" bitPadDir="LEFT" bitLength="40" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="4398046511103"
characterSet="[01]*" bitPadDir="LEFT" bitLength="42" name="indassetref"/>
      </option>
      <option optionKey="11" pattern="00110100([01]{3})001([01]{37})([01]{45})"
grammar="00110100'filter'001'gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[01]*" bitPadDir="LEFT" bitLength="37" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="35184372088831" charac-
terSet="[01]*" bitPadDir="LEFT" bitLength="45" name="indassetref"/>
      </option>
      <option optionKey="10" pattern="00110100([01]{3})010([01]{34})([01]{48})"
grammar="00110100'filter'010'gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="281474976710655" charac-
terSet="[01]*" bitPadDir="LEFT" bitLength="48" name="indassetref"/>
      </option>
      <option optionKey="9" pattern="00110100([01]{3})011([01]{30})([01]{52})" grammar
="00110100'filter'011'gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[01]*"
bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="4503599627370495" charac-
terSet="[01]*" bitPadDir="LEFT" bitLength="52" name="indassetref"/>
      </option>
      <option optionKey="8" pattern="00110100([01]{3})100([01]{27})([01]{55})" grammar
="00110100'filter'100'gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"

```



```

bitPadDir="LEFT" bitLength="3" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="36028797018963967" char-
acterSet="[01]*" bitPadDir="LEFT" bitLength="55" name="indassetref"/>
  </option>
  <option optionKey="7" pattern="00110100([01]{3})101([01]{24})([01]{58})" grammar
="00110100' filter '101' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="288230376151711743"
characterSet="[01]*" bitPadDir="LEFT" bitLength="58" name="indassetref"/>
  </option>
  <option optionKey="6" pattern="00110100([01]{3})110([01]{20})([01]{62})" grammar
="00110100' filter '110' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="4611686018427387903"
characterSet="[01]*" bitPadDir="LEFT" bitLength="62" name="indassetref"/>
  </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:giai-96" requiredFormattingPa-
rameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{12})\.([0-9]
{1,13})" grammar="urn:epc:tag:giai-96:' filter '!' gslcompanyprefix '!' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="4398046511103"
characterSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="11" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{11})\.([0-9]
{1,14})" grammar="urn:epc:tag:giai-96:' filter '!' gslcompanyprefix '!' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="35184372088831" charac-

```

```

terSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="10" pattern="urn:epc:tag:giai-96:([0-7]{1})\.[0-9]{10}\.[0-9]{1,15}" grammar="urn:epc:tag:giai-96:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="281474976710655" characterSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="9" pattern="urn:epc:tag:giai-96:([0-7]{1})\.[0-9]{9}\.[0-9]{1,16}" grammar="urn:epc:tag:giai-96:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="4503599627370495" characterSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="8" pattern="urn:epc:tag:giai-96:([0-7]{1})\.[0-9]{8}\.[0-9]{1,17}" grammar="urn:epc:tag:giai-96:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="36028797018963967" characterSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="7" pattern="urn:epc:tag:giai-96:([0-7]{1})\.[0-9]{7}\.[0-9]{1,18}" grammar="urn:epc:tag:giai-96:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="288230376151711743" characterSet="[0-9]*" name="indassetref"/>
  </option>
  <option optionKey="6" pattern="urn:epc:tag:giai-96:([0-7]{1})\.[0-9]{6}\.[0-9]{1,19}" grammar="urn:epc:tag:giai-96:' filter '.' gs1companyprefix '.' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*" length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"

```

```

[0-9]* " length="6" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="4611686018427387903"
characterSet="[0-9]* " name="indassetref"/>
    </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:giai">
    <option optionKey="12" pattern="urn:epc:id:giai:([0-9]{12})\.([0-9]{1,13})"
grammar="urn:epc:id:giai:'gs1companyprefix!' indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " name="indassetref"/>
    </option>
    <option optionKey="11" pattern="urn:epc:id:giai:([0-9]{11})\.([0-9]{1,14})"
grammar="urn:epc:id:giai:'gs1companyprefix!' indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet=
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" charac-
terSet="[0-9]* " name="indassetref"/>
    </option>
    <option optionKey="10" pattern="urn:epc:id:giai:([0-9]{10})\.([0-9]{1,15})"
grammar="urn:epc:id:giai:'gs1companyprefix!' indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" charac-
terSet="[0-9]* " name="indassetref"/>
    </option>
    <option optionKey="9" pattern="urn:epc:id:giai:([0-9]{9})\.([0-9]{1,16})" grammar="''
urn:epc:id:giai:'gs1companyprefix!' indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999999999" charac-
terSet="[0-9]* " name="indassetref"/>
    </option>
    <option optionKey="8" pattern="urn:epc:id:giai:([0-9]{8})\.([0-9]{1,17})" grammar="''
urn:epc:id:giai:'gs1companyprefix!' indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[0-9]* " length="8" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999999999" char-
acterSet="[0-9]* " name="indassetref"/>
    </option>
    <option optionKey="7" pattern="urn:epc:id:giai:([0-9]{7})\.([0-9]{1,18})" grammar="''
urn:epc:id:giai:'gs1companyprefix!' indassetref">

```

[illegible]

[illegible]

[illegible]


```

</scheme>
</tdt:epcTagDataTranslation>

```

A.16 GIAI-202

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:21:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GIAI-202" optionKey="gs1companyprefixlength" tagLength="202">
    <level type="BINARY" prefixMatch="00111000" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00111000([01]{3})000([01]{40})([01]{148})" gram-
mar="00111000'filter'000'gs1companyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[01]*" bitPadDir="LEFT" bitLength="40" name="gs1companyprefix"/>
        <field seq="3" characterSet="[01]*" bitPadDir="RIGHT" bitLength="148" com-
paction="7-bit" name="indassetref"/>
      </option>
      <option optionKey="11" pattern="00111000([01]{3})001([01]{37})([01]{151})" gram-
mar="00111000'filter'001'gs1companyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[01]*" bitPadDir="LEFT" bitLength="37" name="gs1companyprefix"/>
        <field seq="3" characterSet="[01]*" bitPadDir="RIGHT" bitLength="151" com-
paction="7-bit" name="indassetref"/>
      </option>
      <option optionKey="10" pattern="00111000([01]{3})010([01]{34})([01]{154})" gram-
mar="00111000'filter'010'gs1companyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="
[01]*" bitPadDir="LEFT" bitLength="34" name="gs1companyprefix"/>
        <field seq="3" characterSet="[01]*" bitPadDir="RIGHT" bitLength="154" com-
paction="7-bit" name="indassetref"/>
      </option>
      <option optionKey="9" pattern="00111000([01]{3})011([01]{30})([01]{158})"
grammar="00111000'filter'011'gs1companyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>

```

```

    <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*"
[01]* " bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
    <field seq="3" characterSet="[01]*" bitPadDir="RIGHT" bitLength="158" com-
paction="7-bit" name="indassetref"/>
  </option>
  <option optionKey="8" pattern="00111000 ([01]{3}) 100 ([01]{27}) ([01]{161})"
grammar="00111000 ' filter ' 100 ' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]*"
[01]* " bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" characterSet="[01]*" bitPadDir="RIGHT" bitLength="161" com-
paction="7-bit" name="indassetref"/>
  </option>
  <option optionKey="7" pattern="00111000 ([01]{3}) 101 ([01]{24}) ([01]{164})"
grammar="00111000 ' filter ' 101 ' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]*"
[01]* " bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" characterSet="[01]*" bitPadDir="RIGHT" bitLength="164" com-
paction="7-bit" name="indassetref"/>
  </option>
  <option optionKey="6" pattern="00111000 ([01]{3}) 110 ([01]{20}) ([01]{168})"
grammar="00111000 ' filter ' 110 ' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*"
[01]* " bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
    <field seq="3" characterSet="[01]*" bitPadDir="RIGHT" bitLength="168" com-
paction="7-bit" name="indassetref"/>
  </option>
</level>
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:giai-202" requiredFormattingPa-
rameters="filter,taglength">
  <option optionKey="12" pattern="urn:epc:tag:giai-202: ([0-7]{1})\.( [0-9]{12})\.(
[!%?-? A-Z_a-z\x22]{1,18})" grammar="urn:epc:tag:giai-202:' filter '!' gslcompanyprefix '!' in-
dassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" characterSet="[!%?-? A-Z_a-z\x22]*" name="indassetref"/>

```

```

</option>
<option optionKey="11" pattern="urn:epc:tag:giai-202:([0-7]{1})\.([0-9]{11})\.
([!%-? A-Z_a-z\x22]{1,19})" grammar="urn:epc:tag:giai-202:' filter '!' gslcompanyprefix '!' in-
dassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet=
"[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="3" characterSet="([!%-? A-Z_a-z\x22])* " name="indassetref"/>
</option>
<option optionKey="10" pattern="urn:epc:tag:giai-202:([0-7]{1})\.([0-9]{10})\.
([!%-? A-Z_a-z\x22]{1,20})" grammar="urn:epc:tag:giai-202:' filter '!' gslcompanyprefix '!' in-
dassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="3" characterSet="([!%-? A-Z_a-z\x22])* " name="indassetref"/>
</option>
<option optionKey="9" pattern="urn:epc:tag:giai-202:([0-7]{1})\.([0-9]{9})\.([!%-?
A-Z_a-z\x22]{1,21})" grammar="urn:epc:tag:giai-202:' filter '!' gslcompanyprefix '!' indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="3" characterSet="([!%-? A-Z_a-z\x22])* " name="indassetref"/>
</option>
<option optionKey="8" pattern="urn:epc:tag:giai-202:([0-7]{1})\.([0-9]{8})\.([!%-?
A-Z_a-z\x22]{1,22})" grammar="urn:epc:tag:giai-202:' filter '!' gslcompanyprefix '!' indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="3" characterSet="([!%-? A-Z_a-z\x22])* " name="indassetref"/>
</option>
<option optionKey="7" pattern="urn:epc:tag:giai-202:([0-7]{1})\.([0-9]{7})\.([!%-?
A-Z_a-z\x22]{1,23})" grammar="urn:epc:tag:giai-202:' filter '!' gslcompanyprefix '!' indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
  <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="3" characterSet="([!%-? A-Z_a-z\x22])* " name="indassetref"/>
</option>

```

```

    <option optionKey="6" pattern="urn:epc:tag:giai-202:([0-7]{1})\.[0-9]{6}\.(!%-?
A-Z_a-z\x22]{1,24})" grammar="urn:epc:tag:giai-202:' filter '!' gslcompanyprefix '!' indassetref">
      <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
      <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="3" characterSet="(!%-? A-Z_a-z\x22)* " name="indassetref"/>
    </option>
  </level>
  <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:giai">
    <option optionKey="12" pattern="urn:epc:id:giai:([0-9]{12})\.(!%-? A-Z_a-z\x22
{1,18})" grammar="urn:epc:id:giai:' gslcompanyprefix '!' indassetref">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" characterSet="(!%-? A-Z_a-z\x22)* " name="indassetref"/>
    </option>
    <option optionKey="11" pattern="urn:epc:id:giai:([0-9]{11})\.(!%-? A-Z_a-z\x22
{1,19})" grammar="urn:epc:id:giai:' gslcompanyprefix '!' indassetref">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" characterSet="(!%-? A-Z_a-z\x22)* " name="indassetref"/>
    </option>
    <option optionKey="10" pattern="urn:epc:id:giai:([0-9]{10})\.(!%-? A-Z_a-z\x22
{1,20})" grammar="urn:epc:id:giai:' gslcompanyprefix '!' indassetref">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" characterSet="(!%-? A-Z_a-z\x22)* " name="indassetref"/>
    </option>
    <option optionKey="9" pattern="urn:epc:id:giai:([0-9]{9})\.(!%-? A-Z_a-z\x22){1,
21})" grammar="urn:epc:id:giai:' gslcompanyprefix '!' indassetref">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" characterSet="(!%-? A-Z_a-z\x22)* " name="indassetref"/>
    </option>
    <option optionKey="8" pattern="urn:epc:id:giai:([0-9]{8})\.(!%-? A-Z_a-z\x22){1,
22})" grammar="urn:epc:id:giai:' gslcompanyprefix '!' indassetref">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
      <field seq="2" characterSet="(!%-? A-Z_a-z\x22)* " name="indassetref"/>
    </option>
    <option optionKey="7" pattern="urn:epc:id:giai:([0-9]{7})\.(!%-? A-Z_a-z\x22){1,
23})" grammar="urn:epc:id:giai:' gslcompanyprefix '!' indassetref">
      <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="

```

```

[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
</option>
    <option optionKey="6" pattern="urn:epc:id:giai:([0-9]{6})\.(?!%-? A-Z_a-z\x22){1,
24})" grammar="urn:epc:id:giai:' gslcompanyprefix ' indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
</option>
</level>
    <level type="LEGACY" prefixMatch="giai" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="giai=([0-9]{12})(?!%-? A-Z_a-z\x22){1,18})" gram-
mar="giai=' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
</option>
    <option optionKey="11" pattern="giai=([0-9]{11})(?!%-? A-Z_a-z\x22){1,19})" gram-
mar="giai=' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
</option>
    <option optionKey="10" pattern="giai=([0-9]{10})(?!%-? A-Z_a-z\x22){1,20})" gram-
mar="giai=' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
</option>
    <option optionKey="9" pattern="giai=([0-9]{9})(?!%-? A-Z_a-z\x22){1,21})" grammar
="giai=' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
</option>
    <option optionKey="8" pattern="giai=([0-9]{8})(?!%-? A-Z_a-z\x22){1,22})" grammar
="giai=' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
</option>
    <option optionKey="7" pattern="giai=([0-9]{7})(?!%-? A-Z_a-z\x22){1,23})" grammar

```

```

="giai='gslcompanyprefix indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="indassetref"/>
</option>
<option optionKey="6" pattern="giai=([0-9]{6})([!%-? A-Z_a-z\x22]{1,24})" grammar
="giai='gslcompanyprefix indassetref">
  <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="indassetref"/>
</option>
</level>
<level type="LEGACY_AI" prefixMatch="(8004)" requiredParsingParameters="
gslcompanyprefixlength">
  <option optionKey="12" pattern="\ (8004\)([0-9]{12})([!%-? A-Z_a-z\x22]{1,18})"
grammar="(8004)'gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="indassetref"/>
  </option>
  <option optionKey="11" pattern="\ (8004\)([0-9]{11})([!%-? A-Z_a-z\x22]{1,19})"
grammar="(8004)'gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="indassetref"/>
  </option>
  <option optionKey="10" pattern="\ (8004\)([0-9]{10})([!%-? A-Z_a-z\x22]{1,20})"
grammar="(8004)'gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="indassetref"/>
  </option>
  <option optionKey="9" pattern="\ (8004\)([0-9]{9})([!%-? A-Z_a-z\x22]{1,21})"
grammar="(8004)'gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="indassetref"/>
  </option>
  <option optionKey="8" pattern="\ (8004\)([0-9]{8})([!%-? A-Z_a-z\x22]{1,22})"
grammar="(8004)'gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]*" name="indassetref"/>
  </option>

```

```
</option>
<option optionKey = "7" pattern = "\ (8004\ ) ([0-9] {7} ([!%?-? A-Z_a-z\x22] {1,23}))"
grammar="(8004)' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="indassetref"/>
</option>
<option optionKey = "6" pattern = "\ (8004\ ) ([0-9] {6} ([!%?-? A-Z_a-z\x22] {1,24}))"
grammar="(8004)' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="indassetref"/>
</option>
</level>
<level type = " ELEMENT _ STRING " prefixMatch = " 8004 " requiredParsingParameters = "
gslcompanyprefixlength">
    <option optionKey = "12" pattern = "8004 ([0-9] {12} ([!%?-? A-Z_a-z\x22] {1,18}))"
grammar="(8004' gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="indassetref"/>
    </option>
    <option optionKey = "11" pattern = "8004 ([0-9] {11} ([!%?-? A-Z_a-z\x22] {1,19}))"
grammar="(8004' gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="indassetref"/>
    </option>
    <option optionKey = "10" pattern = "8004 ([0-9] {10} ([!%?-? A-Z_a-z\x22] {1,20}))"
grammar="(8004' gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="indassetref"/>
    </option>
    <option optionKey="9" pattern="8004([0-9]{9}([!%?-? A-Z_a-z\x22]{1,21}))" grammar="(8004' gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" characterSet="[!%?-? A-Z_a-z\x22]*" name="indassetref"/>
    </option>
    <option optionKey="8" pattern="8004([0-9]{8}([!%?-? A-Z_a-z\x22]{1,22}))" grammar="(8004' gslcompanyprefix indassetref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
```



```

9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
  </option>
  <option optionKey="7" pattern="8004([0-9]{7}([!%-? A-Z_a-z\x22]{1,23}))" grammar="
8004' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
  </option>
  <option optionKey="6" pattern="8004([0-9]{6}([!%-? A-Z_a-z\x22]{1,24}))" grammar="
8004' gslcompanyprefix indassetref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" characterSet="[!%-? A-Z_a-z\x22]* " name="indassetref"/>
  </option>
</level>
<level type="ONS_HOSTNAME">
  <option optionKey="12" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="11" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="10" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="9" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="8" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="7" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
  </option>
  <option optionKey="6" grammar="gslcompanyprefix'.giai.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="

```

```

[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
  </level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.17 GSRN-96

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:21:00Z" epcTDSVersion="1.6"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GSRN-96" optionKey="gslcompanyprefixlength" tagLength="96">
    <level type="BINARY" prefixMatch="00101101" requiredFormattingParameters="filter,tag-
length">
      <option optionKey="12" pattern="00101101 ([01]{3}) 000 ([01]{40}) ([01]{18})
00000000000000000000000000000000" grammar="' 00101101 ' filter ' 000 ' gslcompanyprefix serviceref '
00000000000000000000000000000000'">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[01]* " bitPadDir="LEFT" bitLength="40" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="18" name="serviceref"/>
      </option>
      <option optionKey="11" pattern="00101101 ([01]{3}) 001 ([01]{37}) ([01]{21})
00000000000000000000000000000000" grammar="' 00101101 ' filter ' 001 ' gslcompanyprefix serviceref '
00000000000000000000000000000000'">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[01]* " bitPadDir="LEFT" bitLength="37" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="21" name="serviceref"/>
      </option>
      <option optionKey="10" pattern="00101101 ([01]{3}) 010 ([01]{34}) ([01]{24})
00000000000000000000000000000000" grammar="' 00101101 ' filter ' 010 ' gslcompanyprefix serviceref '
00000000000000000000000000000000'">
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="34" name="gslcompanyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="

```

```

[01]* " bitPadDir="LEFT" bitLength="24" name="serviceref"/>
  </option>
  <option optionKey="9" pattern="00101101 ([01]{3}) 011 ([01]{30}) ([01]{28})
0000000000000000000000000000" grammar="' 00101101 ' filter ' 011 ' gslcompanyprefix serviceref '
0000000000000000000000000000">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="30" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="28" name="serviceref"/>
  </option>
  <option optionKey="8" pattern="00101101 ([01]{3}) 100 ([01]{27}) ([01]{31})
0000000000000000000000000000" grammar="' 00101101 ' filter ' 100 ' gslcompanyprefix serviceref '
0000000000000000000000000000">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="27" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="31" name="serviceref"/>
  </option>
  <option optionKey="7" pattern="00101101 ([01]{3}) 101 ([01]{24}) ([01]{34})
0000000000000000000000000000" grammar="' 00101101 ' filter ' 101 ' gslcompanyprefix serviceref '
0000000000000000000000000000">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="24" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="34" name="serviceref"/>
  </option>
  <option optionKey="6" pattern="00101101 ([01]{3}) 110 ([01]{20}) ([01]{38})
0000000000000000000000000000" grammar="' 00101101 ' filter ' 110 ' gslcompanyprefix serviceref '
0000000000000000000000000000">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]* "
bitPadDir="LEFT" bitLength="3" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="20" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="999999999999" characterSet="
[01]* " bitPadDir="LEFT" bitLength="38" name="serviceref"/>
  </option>
</level>

```

```
<level type="TAG_ENCODING" prefixMatch="urn:epc:tag:gsrn-96" requiredFormattingParameters="filter,taglength">
```

```
<option optionKey="12" pattern="urn:epc:tag:gsrn-96:([0-7]{1})\,([0-9]{12})\,([0-9]{5})" grammar="urn:epc:tag:gsrn-96:' filter ':' gslcompanyprefix ':' serviceref">
```

```
<field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
```

```
< field seq = " 2" decimalMinimum = " 0" decimalMaximum = " 999999999999"
characterSet="[-0-9]" length="12" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
```

```
<field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]" length="5" padChar="0" padDir="LEFT" name="serviceref"/>
```

</option>

```
<option optionKey="11" pattern="urn:epc:tag:gsrn-96:([0-7]{1})\,([0-9]{11})\,([0-9]{6})" grammar="urn:epc:tag:gsrn-96:'filter':'gs1companyprefix':'serviceref'>
```

```
<field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
```

```
<field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="0-9" length="11" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
```

```
<field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="serviceref"/>
```

</option>

```
<option optionKey="10" pattern="urn:epc:tag:gsrn-96:([0-7]{1})\,([0-9]{10})\,([0-9]{7})" grammar="urn:epc:tag:gsrn-96:' filter ':' gslcompanyprefix ':' serviceref">
```

```
<field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
```

```
<field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="0-9" length="10" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
```

```
<field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="serviceref"/>
```

</option>

```
<option optionKey="9" pattern="urn:epc:tag:gsrn-96;([0-7]{1})\.[0-9]{9})\.[0-9]{8})" grammar="" urn:epc:tag:gsrn-96;' filter '! gslcompanyprefix '! serviceref">
```

```
<field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
```

```
<field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*" length="9" padChar="0" padDir="LEFT" name="gs1companyprefix"/>
```

```
<field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*" length="8" padChar="0" padDir="LEFT" name="serviceref"/>
```

</option>

```
<option optionKey="8" pattern="urn:epc:tag:gsrn-96:([0-7]{1})\.[0-9]{8})\.[0-9]{9})" grammar="urn:epc:tag:gsrn-96:'filter':'gs1companyprefix':'serviceref">
```

```
<field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
length="1" padChar="0" padDir="LEFT" name="filter"/>
```

```
<field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-
```

```

9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
    <option optionKey="7" pattern="urn:epc:tag:gsrn-96:([0-7]{1})\.[0-9]{7}\.[0-9]
{10})" grammar="urn:epc:tag:gsrn-96:' filter '! gslcompanyprefix '! serviceref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
    <option optionKey="6" pattern="urn:epc:tag:gsrn-96:([0-7]{1})\.[0-9]{6}\.[0-9]
{11})" grammar="urn:epc:tag:gsrn-96:' filter '! gslcompanyprefix '! serviceref">
    <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]* "
length="1" padChar="0" padDir="LEFT" name="filter"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="99999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
</level>
<level type="PURE_IDENTITY" prefixMatch="urn:epc:id:gsrn">
    <option optionKey="12" pattern="urn:epc:id:gsrn:([0-9]{12})\.[0-9]{5}" grammar="
urn:epc:id:gsrn:' gslcompanyprefix '! serviceref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999"
characterSet="[0-9]* " length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-
9]* " length="5" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
    <option optionKey="11" pattern="urn:epc:id:gsrn:([0-9]{11})\.[0-9]{6}" grammar="
urn:epc:id:gsrn:' gslcompanyprefix '! serviceref">
    <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
    <option optionKey="10" pattern="urn:epc:id:gsrn:([0-9]{10})\.[0-9]{7}" grammar="
urn:epc:id:gsrn:' gslcompanyprefix '! serviceref">
    <field seq="1" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="

```

```

[0-9]* " length="7" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
    <option optionKey="9" pattern="urn:epc:id:gsrn:([0-9]{9})\.[0-9]{8}" grammar="urn:
epc:id:gsrn:' gslcompanyprefix '!' serviceref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]* " length="8" padChar="0" padDir="LEFT" name="serviceref"/>
 </option>
    <option optionKey="8" pattern="urn:epc:id:gsrn:([0-9]{8})\.[0-9]{9}" grammar="urn:
epc:id:gsrn:' gslcompanyprefix '!' serviceref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]* " length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-
9]* " length="9" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
    <option optionKey="7" pattern="urn:epc:id:gsrn:([0-9]{7})\.[0-9]{10}" grammar="urn:
epc:id:gsrn:' gslcompanyprefix '!' serviceref">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]* " length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-
9]* " length="10" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
    <option optionKey="6" pattern="urn:epc:id:gsrn:([0-9]{6})\.[0-9]{11}" grammar="urn:
epc:id:gsrn:' gslcompanyprefix '!' serviceref">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]* " length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet
="[0-9]* " length="11" padChar="0" padDir="LEFT" name="serviceref"/>
    </option>
</level>
<level type="LEGACY" prefixMatch="gsrn" requiredParsingParameters="
gslcompanyprefixlength">
    <option optionKey="12" pattern="gsrn=([0-9]{18})" grammar="' gsrn='
gslcompanyprefix serviceref checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999999"
characterSet="[0-9]* " length="18" padChar="0" padDir="LEFT" name="gsrn"/>
    </option>
    <option optionKey="11" pattern="gsrn=([0-9]{18})" grammar="' gsrn='
gslcompanyprefix serviceref checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999999"
characterSet="[0-9]* " length="18" padChar="0" padDir="LEFT" name="gsrn"/>
    </option>

```

```

    <option optionKey = "10" pattern = "gsrn = ([0-9]{18})" grammar = "' gsrn = '
gslcompanyprefix serviceref checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "9999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
    <option optionKey = "9" pattern = "gsrn = ([0-9]{18})" grammar = "' gsrn = '
gslcompanyprefix serviceref checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "9999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
    <option optionKey = "8" pattern = "gsrn = ([0-9]{18})" grammar = "' gsrn = '
gslcompanyprefix serviceref checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "9999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
    <option optionKey = "7" pattern = "gsrn = ([0-9]{18})" grammar = "' gsrn = '
gslcompanyprefix serviceref checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "9999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
    <option optionKey = "6" pattern = "gsrn = ([0-9]{18})" grammar = "' gsrn = '
gslcompanyprefix serviceref checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "9999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
    <rule type = "EXTRACT" inputFormat = "STRING" seq = "1" newFieldName = "gsrnprefix"
characterSet = "[0-9]*" length = "17" function = "SUBSTR(gsrn,0,17)"/>
    <rule type = "EXTRACT" inputFormat = "STRING" seq = "2" newFieldName = "serviceref"
characterSet = "[0-9]*" function = "SUBSTR(gsrnprefix,gslcompanyprefixlength)"/>
    <rule type = "EXTRACT" inputFormat = "STRING" seq = "3" newFieldName =
gslcompanyprefix " characterSet = " [ 0-9 ] * " function = " SUBSTR ( gsrnprefix, 0,
gslcompanyprefixlength)"/>
    <rule type = "FORMAT" inputFormat = "STRING" seq = "1" newFieldName = "gsrnprefix"
characterSet = "[0-9]*" length = "17" function = "CONCAT(gslcompanyprefix,serviceref)"/>
    <rule type = "FORMAT" inputFormat = "STRING" seq = "2" newFieldName = "checkdigit"
characterSet = "[0-9]*" length = "1" function = "GS1CHECKSUM(gsrnprefix)"/>
</level>
    <level type = "LEGACY _ AI" prefixMatch = "(8018)" requiredParsingParameters =
gslcompanyprefixlength">
    <option optionKey = "12" pattern = "\ (8018\) ([0-9]{18})" grammar = "'(8018)'
extensiondigit gslcompanyprefix servicerefremainder checkdigit">
    <field seq = "1" decimalMinimum = "0" decimalMaximum = "9999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>

```



```

</option>
<option optionKey = "11" pattern = "\ (8018 \) ([0-9] {18})" grammar = "' (8018) '
extensiondigit gslcompanyprefix servicerefremainder checkdigit">
  <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
<option optionKey = "10" pattern = "\ (8018 \) ([0-9] {18})" grammar = "' (8018) '
extensiondigit gslcompanyprefix servicerefremainder checkdigit">
  <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
<option optionKey = "9" pattern = "\ (8018 \) ([0-9] {18})" grammar = "' (8018) '
extensiondigit gslcompanyprefix servicerefremainder checkdigit">
  <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
<option optionKey = "8" pattern = "\ (8018 \) ([0-9] {18})" grammar = "' (8018) '
extensiondigit gslcompanyprefix servicerefremainder checkdigit">
  <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
<option optionKey = "7" pattern = "\ (8018 \) ([0-9] {18})" grammar = "' (8018) '
extensiondigit gslcompanyprefix servicerefremainder checkdigit">
  <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
<option optionKey = "6" pattern = "\ (8018 \) ([0-9] {18})" grammar = "' (8018) '
extensiondigit gslcompanyprefix servicerefremainder checkdigit">
  <field seq = "1" decimalMinimum = "0" decimalMaximum = "999999999999999999"
characterSet = "[0-9]*" length = "18" padChar = "0" padDir = "LEFT" name = "gsrn"/>
</option>
<rule type = "EXTRACT" inputFormat = "STRING" seq = "1" newFieldName = "gsrnprefix"
characterSet = "[0-9]*" length = "17" function = "SUBSTR(gsrn,0,17)"/>
<rule type = "EXTRACT" inputFormat = "STRING" seq = "2" newFieldName = "serviceref"
characterSet = "[0-9]*" function = "SUBSTR(gsrnprefix,gslcompanyprefixlength)"/>
<rule type = "EXTRACT" inputFormat = "STRING" seq = "3" newFieldName = "
gslcompanyprefix" characterSet = "[0-9]*" function = "SUBSTR ( gsrnprefix, 0,
gslcompanyprefixlength)"/>
<rule type = "FORMAT" inputFormat = "STRING" seq = "1" newFieldName = "gsrnprefix"
characterSet = "[0-9]*" length = "17" function = "CONCAT(gslcompanyprefix,serviceref)"/>
<rule type = "FORMAT" inputFormat = "STRING" seq = "2" newFieldName = "checkdigit"
characterSet = "[0-9]*" length = "1" function = "GS1CHECKSUM(gsrnprefix)"/>
</level>

```

```

    <level type="ELEMENT_STRING" prefixMatch="8018" requiredParsingParameters="
gslcompanyprefixlength">
      <option optionKey="12" pattern="8018([0-9]{18})" grammar="'8018' extensiondigit
gslcompanyprefix servicerefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="gsrn"/>
      </option>
      <option optionKey="11" pattern="8018([0-9]{18})" grammar="'8018' extensiondigit
gslcompanyprefix servicerefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="gsrn"/>
      </option>
      <option optionKey="10" pattern="8018([0-9]{18})" grammar="'8018' extensiondigit
gslcompanyprefix servicerefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="gsrn"/>
      </option>
      <option optionKey="9" pattern="8018([0-9]{18})" grammar="'8018' extensiondigit
gslcompanyprefix servicerefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="gsrn"/>
      </option>
      <option optionKey="8" pattern="8018([0-9]{18})" grammar="'8018' extensiondigit
gslcompanyprefix servicerefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="gsrn"/>
      </option>
      <option optionKey="7" pattern="8018([0-9]{18})" grammar="'8018' extensiondigit
gslcompanyprefix servicerefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="gsrn"/>
      </option>
      <option optionKey="6" pattern="8018([0-9]{18})" grammar="'8018' extensiondigit
gslcompanyprefix servicerefremainder checkdigit">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="gsrn"/>
      </option>
      <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gsrnprefix"
characterSet="[0-9]*" length="17" function="SUBSTR(gsrn,0,17)"/>
      <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="serviceref"
characterSet="[0-9]*" function="SUBSTR(gsrnprefix,gslcompanyprefixlength)"/>
      <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="
gslcompanyprefix " characterSet = " [ 0-9 ] * " function = " SUBSTR ( gsrnprefix, 0,

```

```

gslcompanyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="gsrnprefix"
characterSet="[0-9]*" length="17" function="CONCAT(gslcompanyprefix,serviceref)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gsrnprefix)"/>
</level>
<level type="ONS_HOSTNAME">
    <option optionKey="12" grammar="gslcompanyprefix'.gsrn.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999"
characterSet="[0-9]*" length="12" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="11" grammar="gslcompanyprefix'.gsrn.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999999" characterSet=
="[0-9]*" length="11" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="10" grammar="gslcompanyprefix'.gsrn.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]*" length="10" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="9" grammar="gslcompanyprefix'.gsrn.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-
9]*" length="9" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="8" grammar="gslcompanyprefix'.gsrn.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-
9]*" length="8" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="7" grammar="gslcompanyprefix'.gsrn.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="
[0-9]*" length="7" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
    <option optionKey="6" grammar="gslcompanyprefix'.gsrn.id.onsepc.com'">
        <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="
[0-9]*" length="6" padChar="0" padDir="LEFT" name="gslcompanyprefix"/>
    </option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```

A.18 GID-96

```

<? xml version="1.0" encoding="UTF-8"?>
<tdt:epcTagDataTranslation version="1.6" date="2011-01-20T12:20:00Z" epcTDSVersion="1.6"

```

```

xmlns:tdt = "urn:epcglobal:tdt:xsd:1" xmlns:xsi = "http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="urn:epcglobal:tdt:xsd:1 EpcTagDataTranslation.xsd">
  <scheme name="GID-96" optionKey="1" tagLength="96">
    <level type="BINARY" prefixMatch="00110101" requiredFormattingParameters="tag-
length">
      <option optionKey="1" pattern="00110101([01]{28})([01]{24})([01]{36})" grammar="
00110101'generalmanager objectclass serial">
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="
[01]*" bitPadDir="LEFT" bitLength="28" name="generalmanager"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="
[01]*" bitPadDir="LEFT" bitLength="24" name="objectclass"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet
="[01]*" bitPadDir="LEFT" bitLength="36" name="serial"/>
      </option>
    </level>
    <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:gid-96" requiredFormattingPa-
rameters="taglength">
      <option optionKey="1" pattern="urn:epc:tag:gid-96:([0-9]{1,9})\.[(0-9]{1,8})\.[(0-
9]{1,11})" grammar="urn:epc:tag:gid-96:'generalmanager' objectclass 'serial'">
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="[0-
9]*" name="generalmanager"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-
9]*" name="objectclass"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet
="[0-9]*" name="serial"/>
      </option>
    </level>
    <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:gid">
      <option optionKey="1" pattern="urn:epc:id:gid:([0-9]{1,9})\.[(0-9]{1,8})\.[(0-9]{1,
11})" grammar="urn:epc:id:gid:'generalmanager' objectclass 'serial'">
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="[0-
9]*" name="generalmanager"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-
9]*" name="objectclass"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet
="[0-9]*" name="serial"/>
      </option>
    </level>
    <level type="LEGACY" prefixMatch="generalmanager=">
      <option optionKey="1" pattern="generalmanager=([0-9]{1,9});objectclass=([0-9]{1,
8});serial=([0-9]{1,11})" grammar="'generalmanager=' generalmanager'; objectclass =
objectclass';serial=' serial'">
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="[0-

```

```

9] * " name="generalmanager"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-
9] * " name="objectclass"/>
    <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet
="[0-9] * " name="serial"/>
    </option>
</level>
<level type="ONS_HOSTNAME">
    <option optionKey="1" grammar="objectclass '.' generalmanager '.gid.id.onsepc.com'">
    <field seq="1" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-
9] * " name="objectclass"/>
    <field seq="2" decimalMinimum="0" decimalMaximum="268435455" characterSet="[0-
9] * " name="generalmanager"/>
    </option>
</level>
</scheme>
</tdt:epcTagDataTranslation>

```



参 考 文 献

- [1] EPCglobal 架构文档,可参考以下网址:(https://www.gs1.org/sites/default/files/docs/architecture/EPC_architecture_1_7-framework-May-2015.pdf)
 - [2] 标签数据标准(TDS-EPCglobal Tag Data Standard),可参考以下网址:(<http://www.gs1.org/gsmpt/kc/epcglobal/tds/>)
 - [3] 对象名称解析服务(ONS-Object Naming Service),可参考以下网址:(<http://www.gs1.org/gsmpt/kc/epcglobal/ons/>)
 - [4] GS1 校验位计算,可参考以下网址:(<http://www.gs1.org> 中 'The EAN.UCC System' 的 'Identification' 频道)
 - [5] NAPTR-名称权威指针,可参考以下网址:(<http://www.ietf.org/rfc/rfc2915.txt?number=2915>)
 - [6] PCRE-符合 Perl 规范的正则表达式,可参考以下网址:(<http://www.pcre.org>)
 - [7] ABNF-扩展巴科斯范式,可参考以下网址:(<http://www.ietf.org/rfc/rfc2234.txt?number=22349>)
 - [8] URI-统一资源标识符,可参考以下网址:(<http://www.ietf.org/rfc/rfc2396.txt?number=2234>)
 - [9] CGI-通用网关接口,可参考以下网址:(<https://tools.ietf.org/html/rfc3875>)
 - [10] UML-统一建模语言,可参考以下网址:(<http://www.uml.org/>)
 - [11] ISO AFI-应用类别标识符,可参考 ISO/IEC 15693 和 ISO/IEC 15961 和 15962
 - [12] UHF RFID 协议(UHF Generation 2 Protocol),可参考以下网址:(http://www.gs1.org/gsmpt/kc/epcglobal/uhfclg2/uhfclg2_1_2_0-standard-20080511.pdf.)
 - [13] XML DOM (文档对象模型) 和 XPath,可参考以下网址:(<http://www.w3.org/TR/xpath>)
 - [14] XSD 规范,可参考以下网址:(<http://www.w3.org/TR/xmlschema-2/>)
-